

Contribution to the Development of INWIN: A Smart Platform for Event Organization and Corporate Gift Customization

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Dedication

To my mother, To the most precious person in my life, no words will ever be enough to express what you mean to me. Thank you for your unconditional love, your patience, and your sacrifices. No words will ever be enough to express what you mean to me. Your endless patience, kindness, and strength have shaped who I am today. Whenever I felt weak, your love lifted me. Whenever I doubted myself, your faith in me gave me courage. You are the reason behind every step I take forward in life **To my father,**

Thank you for your constant support and your trust in me. You taught me perseverance and responsibility. I dedicate this work to you with all my gratitude. **To my dear**

family, Thank you for your love, encouragement, and support throughout my journey. You have always been my strength in both difficult and joyful moments. **To my dear**

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to have you in my life. You have been more than a friend—you have been my sister, my support system, and my safe place.

Mouna Oueldi

Dedication

To my family, who have supported me more than I could ever say. Through my toughest times, you have always been there for me, believing in me even when I did not believe in myself, and that truly means the world to me. Thank you for being my family.

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List of Abbreviations

UI	User Interface
UX	User Experience
API	Application Programming Interface

General Introduction

In today's digital era, event management has become an essential service for both individuals and businesses. The increasing need for efficient organization, personalized services, and smooth communication has led to the emergence of digital platforms that simplify the planning process and improve user experience.

Titled INWIN, this project aims to provide a comprehensive platform dedicated to facilitating event organization for both individual clients and businesses. The application allows users to request event organization services and communicate directly with service providers, ensuring a smooth, efficient, and transparent planning process.

In addition to event management services, INWIN offers advanced features designed for businesses. These include the customization and ordering of corporate gifts, with options such as logo integration, personalized text, and color selection. This functionality enables companies to strengthen their brand identity and enhance professional relationships through tailored and meaningful gifting solutions.

This report presents a comprehensive study of the development of the INWIN application, from its initial conception to its implementation. It begins by introducing the project context and objectives, followed by an analysis of user needs and market challenges. It then details the functional and technical specifications of the system, as well as its architecture and design choices. Finally, the report highlights the application's interface and workflow, demonstrating how usability, efficiency, and personalization are combined within a unified platform.

Ultimately, this project reflects the integration of digital innovation into event management, offering a practical and user-centered solution that simplifies organizational processes while meeting the evolving needs of both individuals and businesses.

Chapter 1

Project Context and Management Approach

Introduction

This chapter presents INWIN, the initial context that led to the project, an analysis of its environment, and the working methodology.

1.1 Host Company Presentation

1.1.1 INWIN Overview

HorizonLux is a Tunisian software development company located in Kelibia, Nabeul, that was established in 2021. Their mission is to help businesses grow their online presence and streamline their operations. They offer web and mobile app development as well as custom Customer Relationship Management (CRM) and Enterprise Resource Planning (ERP) system design. They also offer a variety of digital marketing solutions.

1.1.2 INWIN Services

The services offered by HorizonLux include:

- **Event Organization Services:** Provides comprehensive solutions for planning and managing various types of events, including personal events (weddings, birthdays, celebrations) and professional events (seminars, conferences, corporate events);
- **Service Provider Connection:** Connects clients with suitable service providers, allowing users to easily request services and choose options that match their needs;
- **Direct Communication:** Enables smooth and centralized communication between clients and service providers to ensure better coordination and follow-up;

- **Event Customization:** Allows users to personalize their events by selecting specific options and preferences through an interactive and user-friendly interface;
- **Corporate Gift Management:** Offers businesses the ability to customize and order corporate gifts with features such as logo integration, personalized text, and color selection;
- **User-Friendly Interface (UI/UX):** Ensures a simple, intuitive, and efficient user experience for both individuals and businesses using the platform.

1.1.3 INWIN Organigram

An organigram (or organizational chart) is a graphical representation of the structure of an organization, showing the different positions, departments, and the hierarchical relationships between them.

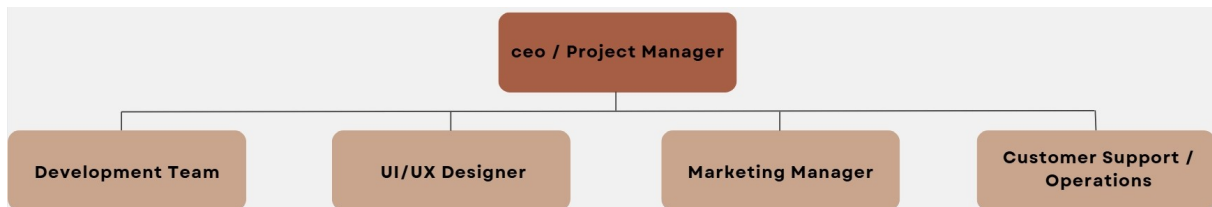


Figure 1.1: company organition chart

1.2 Project Presentation

1.2.1 Consumer Behavior

Consumer behavior in the field of event organization has evolved significantly in recent years. Both individuals and businesses are increasingly looking for simple, fast, and personalized solutions to organize their events, whether personal or professional. Instead of managing complex processes involving multiple service providers separately, users now prefer centralized platforms that offer better visibility and save time.

Users also place great importance on direct communication and service transparency. They expect to easily interact with service providers, track their requests, and ensure that the offered services meet their expectations. Personalization has become a key factor, especially in choosing event services, themes, and organizational details.

For businesses, these expectations are even more critical. In addition to organizing professional events such as seminars and conferences, companies seek to strengthen their brand image through complementary elements like customized corporate gifts. Therefore,

they prefer solutions that integrate both event management and corporate gift customization within a single platform. Problem Statement

1.2.2 Problem Statement

Event organization, whether personal or professional, requires significant effort, coordination, and time. Many clients struggle to find reliable service providers, manage communication, and ensure that all event elements are aligned with their expectations.

For businesses, the challenge is even more complex, as they must not only organize professional events but also manage corporate gifts. These gifts often require customization, such as logo integration, personalized text, and color selection. However, this process is usually handled separately from event planning, leading to inefficiencies, lack of coordination, and limited personalization.

Without a centralized digital solution, these processes remain manual, fragmented, and difficult to manage, which negatively impacts both efficiency and user experience.

This leads to the following question: How can we design a digital platform that simplifies event organization, centralizes communication, and integrates corporate gift management with personalized customization for both individuals and businesses?

1.2.3 Project Idea

The idea behind the INWIN project emerged from the lack of comprehensive digital solutions that effectively support event organization while integrating complementary services such as corporate gifting.

INWIN is designed as a centralized digital platform that enables users to plan and manage their events in a simple and efficient way. The application allows users to search for service providers, customize services according to their needs, and communicate directly with providers to ensure better coordination.

For businesses, INWIN also offers a dedicated corporate gift management feature, including customization options such as logo integration, color selection, and personalized text. This functionality helps companies strengthen their brand identity while simplifying the ordering process.

The main objective of INWIN is to provide an innovative, user-friendly, and complete solution that meets modern user needs in event organization, while enhancing the overall experience through centralization, personalization, and ease of use.

1.3 Benchmarking

Benchmarking is a management approach that involves a comparative analysis of the best practices used by leading organizations in a specific field. It aims to identify

performance gaps and adopt effective strategies in order to improve products, services, or internal processes.

The main objective of benchmarking is to enhance performance and competitiveness by relying on recognized industry standards, while promoting continuous learning and adaptation to market changes.

1.3.1 Competitor Analysis

In the context of an application development project, competitor analysis is an essential step. It helps to understand existing solutions in the market, identify their strengths, and highlight their limitations.

In Tunisia and similar markets, several platforms related to event planning, service booking, or business solutions already exist. However, most of these solutions are often limited to a single domain, lack personalization, or do not provide a fully integrated experience that combines event management and corporate services in one platform.

The purpose of this study is therefore to analyze existing applications in order to identify what works well, what is missing, and how INWIN can differentiate itself by offering a more comprehensive, centralized, and user-friendly experience that integrates both event organization and corporate gifting services in a single solution.

1.3.1.1 Event Organization applications in Tunisia

My Event



Figure 1.2: Logo My Event

My Event is one of the most complete event planning applications available in Tunisia. It provides users with a set of tools that simplify the organization and management of events.

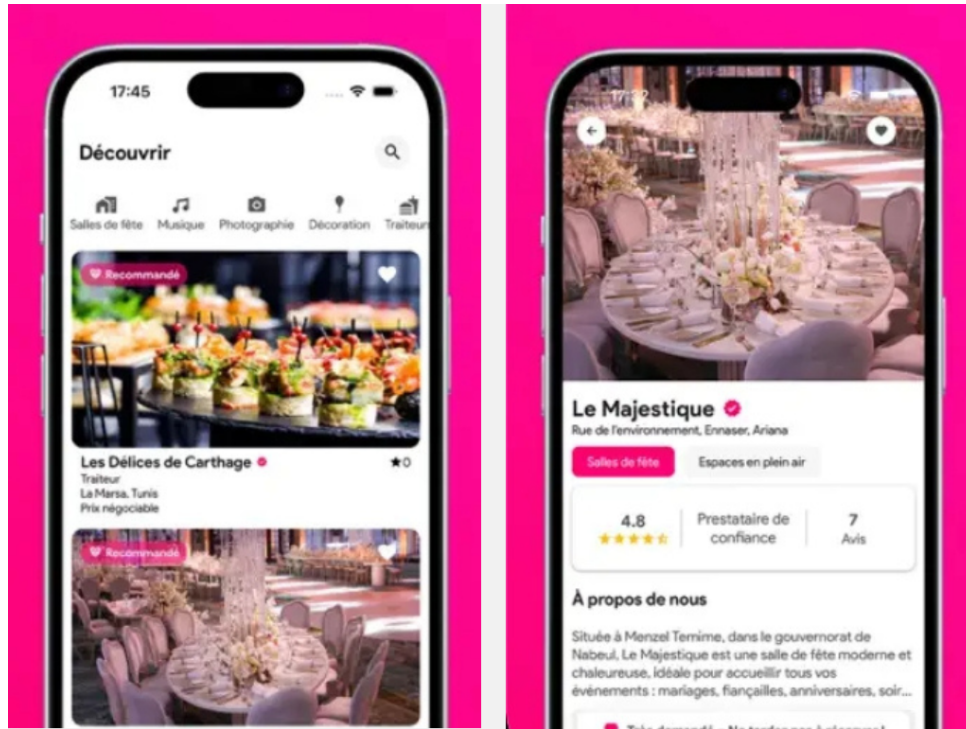


Figure 1.3: Page de découverte et détails du prestataire

The service discovery page in the My Event application serves as the main interface for exploring event organization services such as wedding halls, music, photography, decoration, and catering. It allows users to browse different service providers displayed in the form of cards containing images, provider names, locations, prices, recommendation badges, and favorite options, making the search and comparison process easier and more intuitive. In addition, the provider details page offers comprehensive information about a selected service provider, including photos, user ratings and reviews, service categories, detailed descriptions, and location information. The page also highlights trust and popularity indicators to help users make informed decisions before booking or contacting a provider, thereby improving the overall user experience through a clear and detailed presentation of available services.

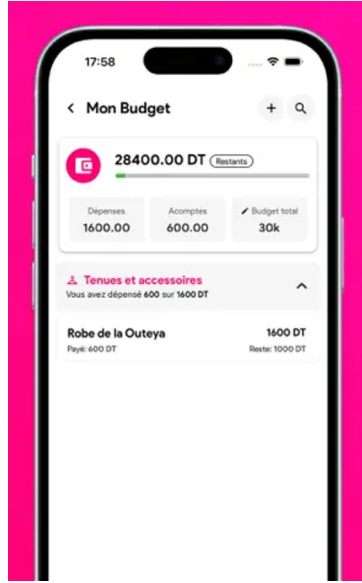


Figure 1.4: Page de découverte et détails du prestataire

The budget management page in the My Event application enables users to efficiently monitor and manage the financial aspects of their events. It provides an overview of the total budget, expenses made, paid deposits, and the remaining amount available. The page also organizes expenses into different categories, such as clothing and accessories, allowing users to track each event-related cost more easily. For every service or product, users can view the total price, the amount already paid, and the remaining balance. This functionality helps users better control their spending, avoid exceeding their budget, and ensure effective financial planning throughout the event organization process.



Figure 1.5: company organisation chart

This interface represents the messaging feature integrated into the INWIN application. It allows users to communicate directly with service providers in real time in order to discuss event details, check availability, request pricing information, and negotiate

services.

Through this functionality, users can easily exchange messages with providers without leaving the application, which improves communication and facilitates event planning. The chat system also enables users to receive quick responses, discuss payment methods, and confirm reservations efficiently.

This feature enhances the user experience by providing a simple, fast, and centralized communication channel between clients and service providers.



Figure 1.6: company organization chart

This interface represents the reservation request feature in the INWIN application. It allows users to submit a reservation request directly to a service provider by filling in important event details.

Through this page, users can select the event date and time, specify the event location, and write a personalized message describing their needs or requesting additional information. This functionality simplifies the reservation process and improves communication between clients and service providers.

The reservation request system helps users organize their events more efficiently by centralizing all booking information within a single platform, while enabling service providers to respond quickly and manage requests effectively.



Figure 1.7: company organization chart

This interface represents the review and rating feature integrated into the INWIN application. It allows users to evaluate service providers based on their experience and consult feedback from previous clients before making a reservation.

The page displays the provider's overall rating, customer reviews, and detailed comments regarding the quality of the service, professionalism, and event experience. Users can also access all reviews to better assess the reliability and reputation of the provider.

This functionality increases transparency and trust within the platform by helping users make informed decisions based on verified experiences and customer satisfaction.

1.3.1.2 corporate gifts for businesses applications in Tunisia

There is currently no well-known Tunisian mobile app specifically dedicated to customizable corporate gifts for businesses (such as pens, mugs, t-shirts, bags, goodies, notebooks, etc.).

Most Tunisian companies in this field operate mainly through websites and social media pages, not mobile applications. Platforms such as Teewinek and Princo provide customization services for promotional products and business gifts, but they do not offer a dedicated mobile application focused on corporate gift management and customization.

1.3.2 Critiquedel'existant

Through the analysis of existing solutions in Tunisia, it was observed that MyEvent is currently one of the few platforms available as a mobile application dedicated to event organization. The application offers several advantages, such as simplifying communication with service providers, facilitating event planning, and providing users with a more

accessible and practical digital experience. These features make it a useful solution for users seeking assistance in organizing personal events.

However, despite its strengths, MyEvent also presents certain limitations. The application mainly focuses on personal events and does not fully address the needs of businesses and professional event management.

The study also revealed that most other existing solutions in Tunisia are limited to websites rather than complete mobile applications. These platforms generally provide only basic services such as provider listings or quotation requests, without offering a fully integrated and centralized management experience.

There is currently no dedicated mobile application in Tunisia specifically designed for customizable corporate gifts for businesses (such as pens, mugs, t-shirts, bags, and promotional items). Existing solutions in this field are mainly web-based platforms or local service providers that operate through websites and direct orders rather than mobile apps. This highlights a clear gap in the Tunisian digital market.

⇒ There is currently no Tunisian application that combines both personal and professional event organization within a single platform while also integrating corporate gift customization services. This represents a significant gap in the market and highlights the innovative aspect of the INWIN project, which aims to provide a comprehensive, modern, and centralized solution adapted to the needs of both individuals and businesses.

1.4 L'analyse SWOT

SWOT analysis is a strategic planning tool widely used by organizations and businesses to evaluate their current situation and support decision-making processes. It helps identify the internal strengths and weaknesses of an organization, as well as the external opportunities and threats present in its environment.

SWOT analysis enables organizations to understand their competitive position and develop strategies that improve performance and achieve objectives. Strengths and weaknesses refer to internal factors such as resources, capabilities, and organizational limitations, while opportunities and threats relate to external factors like market trends, competition, and economic conditions.

Strengths (advantages of INWIN)	Weaknesses (limitations of INWIN)
• All-in-one platform combining two major services: event organization + corporate gift customization, which is rare in the market. • Centralized digital solution that reduces the need to use multiple platforms or providers. • High level of personalization (events + gifts with logos, colors, text, preferences). • Direct communication system between clients and service providers (chat + requests). • Simplified event management tools (budget tracking, reservations, reviews, provider discovery). • Addresses both individuals and businesses, increasing potential user base. • Strong user experience focus (simple interface, structured information, easy booking process).	• High development complexity, because the app integrates multiple systems (marketplace + messaging + customization + payments). • Need for strong supplier network (event providers + printing/custom gift suppliers must be onboarded). • Operational management challenge, especially for quality control of services and products. • Dependence on user adoption, as both clients and providers must actively use the platform. • Initial lack of brand recognition compared to existing local solutions like MyEvent.

Opportunities (market advantages INWIN can exploit)	Threats (risks from the environment)
• Clear market gap in Tunisia: no app combines event management + corporate gifting. • Growing digital transformation in event planning and SME marketing tools. • Increasing demand from businesses for personalized promotional items (branding, marketing). • Expansion potential beyond Tunisia to North African and MENA markets. • Opportunity to partner with local suppliers and printing companies. • Rise of mobile-first services, especially for booking and customization platforms. • Corporate branding trend (companies increasingly invest in customized gifts for marketing).	• Existing competitors in separate domains (event apps like MyEvent, and gift printing websites). • Traditional offline service providers (event planners and printing shops) resisting digitalization. • Low trust in online service quality in some user segments. • Price competition from informal or low-cost providers. • Rapid technological changes, requiring continuous updates and innovation. • Potential entry of international platforms offering similar integrated services in the future.

Table 1.1: Analyse SWOT

1.5 Proposed Solution and Objectives

By conducting a detailed and comparative analysis of the existing solutions available in the Tunisian market for event organization and corporate gift management, several limitations and gaps were identified that may negatively affect the user experience or fail to fully satisfy the needs of both individuals and businesses.

Through our project INWIN, and based on the results obtained from the SWOT analysis, we propose an innovative and centralized mobile solution capable of better addressing these needs by combining event management services and corporate gift cus-

tomization within a single platform.

The objective of INWIN is to simplify event organization, improve communication between clients and service providers, and provide businesses with an easy and modern way to manage and customize their corporate gifts.

Problems Identified	Proposed Solutions
Existing applications mainly focus only on personal event organization	Develop a platform that supports both personal and professional events
Lack of centralized services in one application	Integrate event planning, reservations, communication, and corporate gift management in a single platform
Difficulty communicating efficiently with service providers	Provide a real-time messaging system between users and providers
Limited personalization options in existing platforms	Offer customizable services and corporate gifts with logo integration, colors, and personalized text
Most corporate gift solutions exist only as websites and not mobile applications	Develop a dedicated mobile application specialized in corporate gift management and customization
Difficulty managing event budgets and reservations	Integrate budget tracking and reservation request functionalities
Lack of trust and transparency in some services	Include verified ratings, reviews, and customer feedback systems
Existing platforms do not fully meet business needs	Provide features specifically designed for companies and professional event management
Fragmented user experience across multiple platforms	Create a centralized, intuitive, and user-friendly interface that simplifies all processes

Table 1.2: Les solutions proposées

⇒ Following the analysis of competing solutions, our project consists of developing a mobile application dedicated to both event organization and corporate gift management, offering a modern, personalized, and centralized experience adapted to the needs of individuals and businesses. In addition to the core functionalities of the application, such as

service provider discovery, reservation requests, direct communication, budget management, and review systems, INWIN also integrates advanced corporate gifting features that allow businesses to customize and order promotional products directly through the platform. These functionalities aim to simplify event planning, improve service coordination, strengthen business branding through personalized gifts, and enhance the overall user experience by offering a complete digital ecosystem within a single mobile application.

1.6 Choice of Methodology

Several working methodologies are commonly used in software and mobile application development projects. Each methodology has its own principles, advantages, and limitations.

As part of our project, we conducted an in-depth analysis of the different project management methodologies in order to identify the one that best meets the project requirements and objectives.

1.6.1 definition of Methodologies

1.6.1.1 Traditional Method (Waterfall / Cascade)

The Waterfall Model methodology is a traditional software development approach based on a linear and sequential process. Each phase of the project must be completed before moving to the next one, such as requirements analysis, design, implementation, testing, deployment, and maintenance. This model emphasizes detailed planning and documentation at the beginning of the project

Characteristics

- Sequential and structured development process
- Detailed planning and documentation
- Clear definition of project phases
- Limited flexibility for changes
- User feedback integrated mainly at the end
- Easier management of deadlines and resources

1.6.1.2 Agile Methodology

Agile Software Development is a flexible software development methodology focused on iterative development, customer collaboration, and rapid adaptation to changes. Agile promotes continuous delivery of functional software through small development cycles called iterations or sprints.

Characteristics

- Iterative and incremental development
- Strong customer collaboration
- Flexibility toward requirement changes
- Continuous testing and feedback
- Fast delivery of working software
- Team communication and collaboration

1.6.1.3 Scrum Methodology

The Scrum methodology is an Agile framework used to manage and develop software projects through short iterations called sprints. Scrum encourages teamwork, regular feedback, and continuous improvement to deliver high-quality products quickly.

Characteristics

- Development divided into short sprints
- Daily meetings called Daily Scrum
- Defined roles: Product Owner, Scrum Master,
- Development Team
- Continuous product improvement
- Frequent delivery of functional increments
- Strong collaboration and transparency

1.6.1.4 RAD (Rapid Application Development)

Rapid Application Development is a software development methodology that focuses on rapid prototyping, iterative development, and continuous user feedback rather than extensive planning. RAD allows developers to quickly build and improve applications according to user needs.

Characteristics

- Rapid prototyping
- Iterative development process
- Continuous user involvement
- Fast application delivery
- Flexibility and adaptability
- Reusable software components
- Continuous testing and integration

1.6.2 Comparison of Methodologies

Methodology	Strengths	Weaknesses
Waterfall	• Simple and structured methodology • Clear documentation and planning • Easy to manage and understand • Well defined project phases	• Not flexible to changes • Problems are detected late • Difficult to modify requirements after development starts

Agile	• Flexible and adaptable to changes • Continuous improvement and testing • Better customer involvement • Faster delivery of features	• Requires continuous communication • Difficult to estimate total project cost and duration • Needs strong team collaboration
Scrum	• Well suited for mobile and software projects • Improves productivity and task management • Allows rapid development through sprints • Encourages teamwork and communication	• Requires experienced and organized team members • Daily meetings can consume time • Difficult to manage very large teams
RAD	• Fast development process • Quick prototyping and feedback • Good for user interface development	• Less suitable for complex projects • Requires highly skilled developers • Difficult to manage large-scale systems

Table 1.3: Comparaison entre la méthode traditionnelle et la méthode Agile

1.6.3 chosen Methodology for INWIN

For the development of the INWIN project, the most suitable methodology is: Agile Scrum Methodology

Reasons for Choosing Scrum

INWIN is a mobile application that combines several modules such as:

- Event organization
- Reservation management
- Real-time communication
- Budget management
- Corporate gift customization

These features require continuous improvements, testing, and flexibility during development. Scrum allows the project to be developed progressively while facilitating modifications and feature integration according to user needs.

In addition, Scrum improves task management, team collaboration, and project monitoring, making it highly suitable for innovative and evolving applications like INWIN.

1.6.4 Applying Scrum Methodology to the INWIN Project

To develop the INWIN application efficiently, the Scrum methodology can be applied by organizing the project into iterative development phases called sprints. Each sprint focuses on designing, developing, and testing a specific set of functionalities.



Figure 1.8: Scrum Process Diagram

1. Define the Scrum Team

The Scrum methodology is based on three main roles:

Product Owner

Responsible for defining the project requirements and prioritizing functionalities according to user and business needs. In the INWIN project, the Product Owner defines features such as:

- Event management
- Corporate gift customization
- Reservation requests
- Messaging system
- Budget management

Scrum Master

Responsible for ensuring that the Scrum process is correctly applied and helping the team solve organizational problems.

Development Team

Responsible for:

- Mobile application development
- UI/UX design
- Database management
- Testing and integration

2. Create the Product Backlog

The Product Backlog contains all project functionalities and requirements organized by priority.

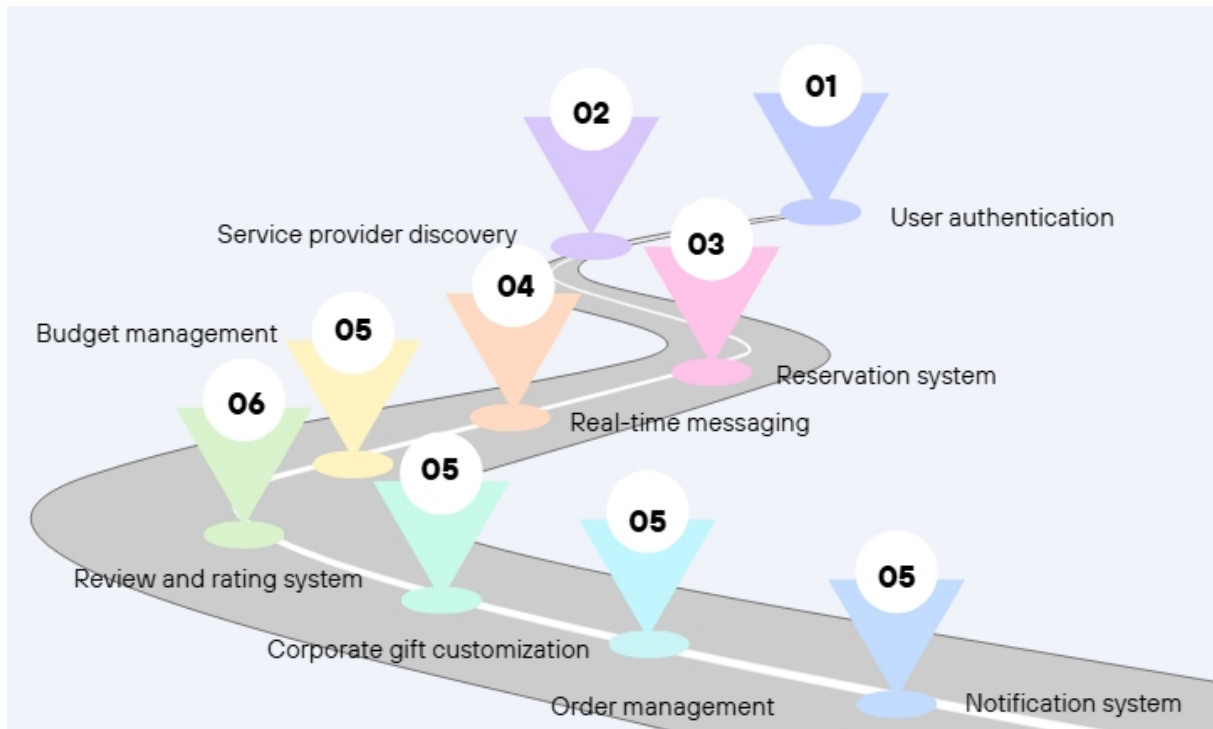


Figure 1.9: INWIN Product Backlog

3. Organize the Project into Sprints

The project is divided into short development cycles called sprints (generally 1 to 4 weeks).

Sprint	Main Objectives
Sprint 1	Project setup, authentication, database creations
Sprint 2	Service provider discovery and search system
Sprint 3	Reservation request functionality
Sprint 4	Messaging and communication system
Sprint 5	Budget management and reviews
Sprint 6	Corporate gift customization module
Sprint 7	Testing, optimization, and deployment

Table 1.4: Sprint Organization for INWIN

4. Daily Scrum Meetings

Short daily meetings are organized to discuss:

- Completed tasks

- Current tasks
- Problems encountered

5. Sprint Review

At the end of each sprint, the developed functionalities are presented and evaluated to ensure they meet project requirements.

6. Sprint Retrospective

The team analyzes:

- What worked well
- Problems encountered
- Possible improvements for the next sprint

1.7 Conclusion

This chapter presented the general framework of the INWIN project and analyzed existing event organization and corporate gift solutions in Tunisia. The study revealed several limitations in current platforms, especially the absence of a centralized mobile application combining event management and corporate gift customization. Based on this analysis, INWIN was proposed as an innovative and user-friendly solution adapted to both individuals and businesses. Finally, after comparing different project management methodologies, the Agile Scrum methodology was selected because of its flexibility and suitability for mobile application development.

Chapter 2

Theoretical Study and Marketing Plan

Introduction

This chapter presents the theoretical framework of the project, supported by market research conducted through a questionnaire, as well as a marketing plan developed based on the obtained results.

2.1 Theoretical Study

This section presents the key concepts related to the INWIN project, including the importance of digital event organization, service provider management, and corporate gift customization. It also highlights the role of mobile technologies and modern development approaches in improving user experience, simplifying event planning, and enhancing communication between clients and service providers.

2.1.1 Importance of Event Organization and Corporate Gift Management

Event organization has become an essential activity for both individuals and businesses, as it helps coordinate different services and ensure successful event planning. Modern users increasingly seek digital solutions that simplify the management of reservations, communication with service providers, budget tracking, and event customization. Mobile applications dedicated to event management provide users with a more practical, centralized, and efficient experience.

In addition, corporate gifting has become an important marketing and branding strategy for businesses. Customized promotional products such as mugs, pens, t-shirts, bags, and notebooks help companies strengthen their brand identity, improve customer relationships, and increase brand visibility. However, in Tunisia, most corporate gift

services are mainly offered through websites and social media pages rather than dedicated mobile applications.

Therefore, integrating event organization and corporate gift customization into a single mobile platform represents an innovative solution that simplifies management processes for both individuals and businesses while improving user experience and service accessibility.

2.1.2 Mobile Technologies and Modern Digital Solutions

The rapid evolution of mobile technologies has significantly transformed the way users interact with digital services. Mobile applications now play a major role in simplifying daily tasks by providing fast, accessible, and user-friendly solutions. In the field of event organization, mobile platforms facilitate communication, reservations, service discovery, and real-time management of events.

Modern application development technologies also contribute to improving system performance, user experience, and scalability. Features such as real-time messaging, notifications, geolocation, online reservations, and personalized customization options enhance interaction between clients and service providers while making event planning more efficient.

Furthermore, centralized digital platforms help reduce complexity by integrating multiple services into a single application. In the case of INWIN, combining event management and corporate gift customization within one mobile solution allows users and businesses to manage their needs more efficiently while benefiting from a modern, intuitive, and connected digital environment.

2.1.3 The Impact of Modern Digital Technologies on Event Management and Corporate Services

The organization and management of events can often be complex and time-consuming, especially when users need to coordinate with multiple service providers, manage reservations, monitor budgets, and ensure effective communication. Modern digital technologies have significantly transformed this process by providing centralized and automated solutions that simplify event planning and improve overall efficiency.

Mobile applications now allow users to access services quickly, communicate in real time, and manage event-related activities more effectively. Features such as instant messaging, online reservation requests, notifications, reviews, and budget tracking reduce manual effort and improve coordination between clients and service providers.

In addition, personalization technologies play an important role in enhancing user experience. Businesses increasingly require customized corporate gifts that reflect their brand identity through logos, colors, and personalized text. Digital customization tools

simplify this process by allowing companies to manage and personalize promotional products directly through a mobile application.

Furthermore, centralized platforms improve accessibility and save time by integrating multiple services into a single environment. Instead of relying on different websites or traditional communication methods, users can organize events, communicate with providers, and customize corporate gifts within one application. This approach improves efficiency, user satisfaction, and service management while providing a more modern and connected digital experience.

Through INWIN, these technologies are integrated into a single mobile platform designed to simplify event organization, strengthen communication, improve personalization, and provide businesses and individuals with a complete and user-friendly solution adapted to modern market needs.

2.1.4 The Role of Modern Development Technologies in Mobile Application Workflows

A workflow represents a sequence of organized tasks and processes that work together to achieve a specific objective. In mobile application development, efficient workflows are essential to ensure better task management, faster development, and improved user experience.

Modern development technologies and digital tools have significantly simplified the creation and management of mobile applications. These technologies help developers design user interfaces, manage databases, integrate APIs, and implement real-time functionalities more efficiently while reducing development complexity and time.

In the context of event management applications, modern technologies facilitate important processes such as reservation management, real-time communication, notifications, geolocation, payment integration, and service customization. These tools improve system performance and enable developers to create scalable and user-friendly applications adapted to evolving market needs.

Furthermore, integrated development approaches allow different services and functionalities to work together within a centralized platform. For the INWIN project, these technologies support the integration of event organization services and corporate gift customization into a single mobile application. This centralized workflow simplifies management processes for users and businesses while improving accessibility, communication, and overall service efficiency.

The use of modern mobile development frameworks and digital solutions also contributes to faster application development, easier maintenance, and better scalability, ensuring that the platform remains flexible and capable of adapting to future improvements and user requirements.

Technology	Definition
Flutter	Flutter is an open-source UI software development framework developed by Google that allows developers to create cross-platform mobile applications using a single codebase for Android and iOS platforms.
Firebase	Firebase is a backend development platform provided by Google that offers services such as authentication, real-time databases, cloud storage, notifications, and application hosting.
Figma	Figma is a cloud-based design and prototyping tool used for creating user interfaces, wireframes, and interactive application designs collaboratively.

Table 2.1: Mobile Development Technologies Overview

The comparison focuses on important criteria such as ease of development, scalability, cross-platform compatibility, backend integration, and suitability for mobile application development.

Feature/Technology	Flutter	Firebase	Figma
Type	Mobile UI Framework	Backend Development Platform	UI/UX Design Tool
Main Use Case	Cross-platform mobile app development	Backend services and database management	Interface and prototype design
Platform Support Android	iOS Mobile	Web Applications	Web-based collaborative platform
Real-Time Features	Supported through integration	Real-time database and notifications	Not applicable
Ease of Use	Moderate	Easy to Moderate	Easy
Scalability	High	High	Moderate
Development Speed	Fast development with single codebase	Simplifies backend development	Accelerates interface design
Collaboration Features	Team development support	Shared cloud services	Real-time collaborative design
Ideal For	Mobile developers	Backend and cloud services	UI/UX designers
Key Advantages	Cross-platform, responsive UI, high performance	Authentication, database, storage, notifications	Interactive prototypes and easy collaboration

Table 2.2: Mobile Development Technologies Comparison

For the development of the INWIN application, Flutter was selected as the main development framework due to its ability to create high-performance cross-platform mobile applications using a single codebase. This significantly reduces development time and ensures a consistent user experience across Android and iOS devices.

In addition, Firebase provides powerful backend services such as authentication, cloud database management, real-time communication, and notifications, which are essential for functionalities like messaging, reservation management, and user interaction within the INWIN platform.

Finally, Figma was used for designing and prototyping the application interfaces,

allowing the creation of a modern, intuitive, and user-friendly design adapted to both individuals and businesses.

2.2 Questionnaire Presentation and Analysis

To better understand our target audience, we conducted a questionnaire to identify the needs of potential users. This section presents the questionnaire's structure and analyses the results.

2.2.1 Presentation and Analysis of the Results for business

In this section, we will analyze the most important statistical graphs in order to draw meaningful conclusions. A total of 54 responses were collected.

2.2.1.1 Target Audience Specification Questions

The majority of participants (81.5%) belong to the 18–25 age group, followed by the 26–35 age group (11.1%), while older age groups remain marginal (36–50 years old and over 50 years old). This indicates a very young target audience that is connected and strongly integrated into the digital era.

Regarding their habits, 57.4% of respondents stated that they organize events rarely, while 31.5% organize events one to two times per year and 11.1% organize several events per year.

→ Conclusion: The typical user of our application is a young person between 18 and 25 years old who organizes events occasionally or periodically.

2.2.1.2 Past Experiences and Encountered Obstacles

Only 29.1% of respondents have already used an event organization application, while 70.9% have never used one.

For those who have never used such applications (50 responses collected for this sub-question), the main reasons mentioned are:

Lack of awareness (66%): The majority simply do not know that such tools exist. Perceived usefulness (24%): Nearly a quarter believe that these applications are not useful. Complexity (14%): A significant portion fears that the application may be too complicated to use.

→ Conclusion: There is a strong market opportunity, provided that users are made aware of the application and that the platform remains simple and accessible in order to eliminate concerns related to complexity.

2.2.1.3 Interests, Expectations, and Budget

When participants were asked about the types of events that interest them the most, private parties ranked first (35.2%), followed by birthdays (24.1%) and weddings (14.8%). Baby showers remained very limited (3.7%), while other types of celebrations represented 29.6%.

The most important criteria for organizing these events are:

Quality (44.4%) Creativity (42.6%)

These two factors largely dominate all other expectations, proving that users mainly seek memorable, original, and high-quality experiences.

Price (14.8%), speed (3.7%), and reputation (1.9%) were considered secondary factors.

This search for quality is also reflected in users' financial resources, as the majority of respondents (70.7%) allocate a medium budget for their events. Small budgets represent 24.1%, while high budgets remain very limited.

→ Conclusion: The application should highlight creative ideas and high-quality service providers adapted to users with medium budgets who mainly organize private events and birthdays.

2.2.1.4 Desired Features and Motivation Factors

Regarding the potential use of the application, the main factor encouraging respondents to use it is time-saving (53.7%). This is followed by service quality (25.9%) and simplicity (18.5%). Attractive pricing only represents 9.3% of motivations.

To meet these expectations related to simplicity and support, the most desired features within the application are:

Chat with an organizer (51.9%): Demonstrating the need for direct support and communication. Inspiration gallery (24.1%): Supporting users' need for creativity and visual inspiration. Quick quotation requests (16.7%): Reflecting the importance of saving time. Online payment (11.1%)

→ Conclusion: Users are looking for an efficient tool that saves time. Integrating a direct messaging system and an inspirational gallery are essential features for increasing application adoption.

2.2.1.5 Difficulties Encountered During Event Organization

To better define the platform's value proposition, the analysis of users' current difficulties revealed that general organization is the main issue (33.3%), closely followed by budget management (27.8%). Other difficulties include finding venues (18.5%), finding service providers (16.7%), and lack of time (11.1%).

→ Conclusion: The application should position itself as a true event coordination assistant capable of simplifying overall planning and helping users manage their budgets effectively.

2.3 Survey Findings

The analysis of the survey results highlights clear trends that confirm the relevance and usefulness of a modern event organization application. The collected responses reveal a predominantly young target audience (18–25 years old), connected to digital technologies, who occasionally organizes events such as private parties and birthdays while facing difficulties related to overall organization and budget management.

A major finding is that more than 70% of respondents have never used a dedicated event organization application, mainly due to a lack of awareness. This demonstrates a largely untapped market and a strong opportunity for our solution, provided that the application remains simple, intuitive, and easy to use in order to reduce concerns related to complexity.

Future users' expectations mainly focus on quality, creativity, and time-saving. Although financial considerations remain important through the dominance of medium budgets, users place greater emphasis on creating original and successful events. To attract this audience, the application should not function as a simple service directory, but rather as an interactive and supportive platform, as reflected by the strong interest in direct chat with organizers and visual inspiration galleries.

2.3.1 Presentation and Analysis of the Results for business

In this section, we will analyze the most important statistical graphs in order to draw meaningful conclusions. A total of 17 responses were collected.

2.3.1.1 Target Audience Specification Questions

The respondents' business sectors are mainly concentrated in the commercial field (50%), followed equally by the healthcare sector (25% for "Healthcare" and 25% for "healthcare").

Regarding the size of participating companies, the sample is perfectly balanced between small businesses (41.2%) and medium-sized enterprises (41.2%). Large enterprises represent 17.6% of respondents.

→ Conclusion: The main target audience consists of small and medium-sized enterprises (SMEs) operating mainly in the commercial and healthcare sectors.

2.3.1.2 Professional Event Organization Habits

Nearly half of the respondents (47.1%) organize professional events occasionally, while 41.2% organize them regularly. Only 11.8% of participants never organize professional events.

Concerning the most common types of events, conferences rank first with 64.7% of responses (11 responses). Seminars and meetings follow with 29.4% each (5 responses). Team-building events (23.5%) and product launches (11.8%) are less frequent.

Faced with event logistics, companies identified two major challenges:

Lack of reliable service providers (52.9%) Complex event organization (41.2%)

Budget management difficulties (29.4%) and lack of time (11.8%) complete the list of pain points.

→ Conclusion: An overwhelming majority of companies (88.3%) are active in corporate event organization, mainly conferences and seminars, but they face organizational complexity and difficulties finding trustworthy service providers.

2.3.1.3 Expectations and Preferences for Corporate Gifts

The use of corporate gifts is strongly established within the participating companies, with 89.5% of respondents confirming that they use them.

Among companies using corporate gifts, preferences are equally divided between:

Customized goodies (47.1% – 8 responses) Personalized products (47.1% – 8 responses)

Customized gift boxes attracted 23.5% of responses.

When asked about the usefulness of a digital solution (mobile application), companies expressed strong interest in two main functionalities:

Ordering customized corporate gifts (64.7% – 11 responses) Managing corporate events (52.9% – 9 responses)

→ Conclusion: The corporate gifting market is highly present among businesses. There is a clear market opportunity for a centralized application that combines personalized product ordering and event management functionalities.

2.3.1.4 Expected Features, Budget, and Selection Criteria

Regarding the expected features of such a platform, quick quotation requests appear as the top priority (47.1%), closely followed by product customization options (41.2%). Customer support also remains important for 29.4% of respondents.

From a financial perspective, the survey reveals diverse investment capacities:

41.2% allocate a budget below 2,000 TND 35.3% allocate between 2,000 and 5,000 TND 23.5% have a budget exceeding 5,000 TND

Finally, when selecting a service provider, price (47.1%) and quality (47.1%) are tied as the most important criteria, followed closely by service speed (41.2%). Provider experience is considered secondary (17.6%).

→ Conclusion: Future users expect responsiveness (quick quotations and fast execution), an excellent quality-price ratio, and easily customizable products.

2.4 Survey Findings

The analysis of the survey results highlights clear trends confirming the relevance and usefulness of a B2B application dedicated to event management and personalized corporate gift ordering. The collected responses reveal a target audience mainly composed of professionals from commercial and healthcare SMEs, who frequently organize conferences and seminars while also requiring corporate gifting solutions.

Participants place strong importance on the reliability of service providers and seek to simplify what they perceive as a complex organizational process. The significant interest in automated functionalities, such as rapid quotation generation and online customization modules, demonstrates a strong demand for time-saving and operational efficiency.

From a financial perspective, although a significant proportion of businesses operate with moderate budgets (below 2,000 TND), more than half of the respondents are willing to invest larger amounts (up to more than 5,000 TND), provided that the platform delivers an effective balance between price, quality, and service speed.

2.5 Marketing Plan

A marketing plan is a strategic document that outlines an organization's marketing objectives, target market, marketing strategies, and actions required to achieve business goals within a specific period. It serves as a guide for coordinating marketing activities, allocating resources, and evaluating performance. According to scholars, a marketing plan helps organizations identify customer needs, analyze competitors, and develop effective promotional and communication strategies.

2.5.1 Segmentation

Market segmentation is the process of dividing a broad market into smaller groups of consumers who share similar needs, characteristics, behaviors, or preferences. This approach allows companies to develop targeted marketing strategies and offer products or services that better satisfy specific customer groups.

irmographic Criteria: This refers to segmenting the market according to organizational characteristics such as company size, type of business, and industry (event

agencies, SMEs, startups, and large companies).

Geographic Criteria: This divides the market based on location, mainly focusing on Tunisia and potentially expanding to North African markets where event organization and corporate gifting demand is growing.

Behavioral Criteria: This focuses on how users interact with event planning and corporate gifting services, including their needs for customization, digital booking, communication tools, and centralized management solutions.

Why we did NOT choose other criteria Demographic → not useful for business clients and service providers Psychographic → difficult to measure in a technical B2B platform Technographic → optional, not essential for defining INWIN's core market

Based on these criteria, three main market segments were identified:

- **Individuals (Personal Event Clients)**

This segment includes individuals who organize personal events such as weddings, birthdays, engagements, graduations, and private parties. They often lack professional experience in event planning and therefore seek simple, centralized, and user-friendly solutions.

Their main challenges include managing multiple service providers, controlling budgets, and coordinating event details efficiently. For this segment, INWIN offers value through features such as service provider discovery, reservation requests, messaging, budget tracking, and reviews. The platform simplifies the entire planning process and helps users organize events more easily and efficiently.

- **Businesses (Corporate Clients)**

This segment includes

SMEs : companies that fall within a certain size limit in terms of number of employees, revenue, or assets, depending on the country.

Startups : newly created companies in the early stages of development, designed to search for a scalable and repeatable business model.

Large enterprises : well-established organizations with a high number of employees, significant financial resources, and complex organizational structures.

That organize professional events (conferences, seminars, corporate parties) and require customized corporate gifts for branding and marketing purposes.

Their main challenges include managing corporate events efficiently, ensuring brand consistency, and ordering personalized gifts in a simple and centralized way.

INWIN addresses these needs by offering corporate event management tools and a dedicated corporate gifting module, allowing businesses to customize products (logos, colors, text), place orders, and manage their event-related requirements within a single platform.

2.5.2 Targeting

Targeting is a strategic process that consists of selecting one or more market segments on which a company will focus its marketing and communication efforts. This step follows market segmentation and precedes the positioning strategy. Targeting allows companies to optimize resources by concentrating on the most relevant and profitable customer groups.

For the INWIN project, we adopted a niche marketing strategy focused mainly on individuals, startups, SMEs, and large enterprises that require event organization services and corporate gift customization solutions.

Niche marketing is the process of targeting a specific part of the market whose needs are not fully satisfied by existing solutions. In the case of INWIN, current platforms in Tunisia generally focus either on event organization or on promotional product customization, without offering a centralized mobile solution that combines both services.

Individuals organizing personal events often face challenges such as finding reliable service providers, managing reservations, controlling budgets, and coordinating event details. Startups and SMEs generally seek flexible and affordable digital solutions to organize professional events and manage branded corporate gifts while optimizing time and resources.

Large enterprises, on the other hand, require more advanced and scalable solutions capable of supporting large-scale corporate events, multiple service providers, and customized promotional products while ensuring consistency with their brand identity and professional standards.

By targeting these specific segments, INWIN aims to provide a modern, centralized, and user-friendly platform that simplifies event planning, improves communication between clients and service providers, and facilitates corporate gift customization. This strategy allows INWIN to differentiate itself in the Tunisian market by offering a complete digital ecosystem adapted to both personal and professional needs.

Positioning refers to the place a product, brand, or company occupies in the minds of consumers compared to its competitors. For the INWIN project, we adopted a differentiation strategy by offering a broader and more integrated range of features than existing solutions available in the Tunisian market.

This approach allows INWIN to provide a more comprehensive and centralized experience by combining event organization and corporate gift customization services within a single mobile application. Unlike most existing platforms that focus only on one specific domain, INWIN aims to simplify the entire event management process while addressing both personal and professional needs.

The main differentiating features of INWIN include:

- **Centralized Event Management:** A complete platform that allows users to search for

service providers, manage reservations, communicate directly with providers, and organize event details within a single application.

- **Corporate Gift Customization:** A dedicated feature enabling businesses to customize promotional products such as mugs, pens, t-shirts, bags, and notebooks through logo integration, color selection, and personalized text.
- **Real-Time Communication System:** An integrated messaging feature that facilitates direct communication between clients and service providers, improving coordination and service efficiency.
- **Budget Management and Reservation Tracking:** Tools that help users monitor expenses, manage payments, and track reservations efficiently throughout the event planning process.
- **Review and Rating System:** A transparent feedback system allowing users to consult reviews and ratings before selecting service providers, helping improve trust and service quality.
- **User-Friendly and Modern Interface:** A simple, intuitive, and responsive mobile interface designed to provide a smooth and accessible user experience for both individuals and businesses.

Through this differentiation strategy, INWIN positions itself as an innovative and modern digital solution that addresses a significant gap in the Tunisian market by integrating both event management and corporate gifting services into one centralized platform.

2.5.3 Marketing Mix

The Marketing Mix refers to the set of marketing tools and strategies that a company uses to promote its products or services and satisfy customer needs. Traditionally, the marketing mix is composed of the 4Ps: Product, Price, Place, and Promotion. These elements help organizations position their offerings effectively in the market and achieve their marketing objectives. The 4Ps of the Marketing Mix:

2.5.3.1 Product

INWIN is a mobile application designed to simplify event organization and corporate gift management by providing a centralized and user-friendly digital platform.

For businesses, the platform also provides tools to manage professional events and customize promotional products such as mugs, pens, bags, notebooks, and t-shirts in order to strengthen brand identity and improve corporate communication.

2.5.3.2 Price

The pricing strategy for INWIN has not yet been fully finalized. However, the objective is to create a flexible and accessible pricing model adapted to the needs and budgets of individuals, startups, SMEs, and large enterprises.

The application may offer:

- Free basic access for standard users
- Customized pricing for businesses and large enterprises
- Service commission fees for reservations or corporate gift orders

2.5.3.3 Place

INWIN will mainly be available as a mobile application accessible through Android and iOS platforms. Users will be able to download the application online and access its services anywhere and at any time.

In addition, the platform may also be supported by:

- An official website presenting the application and services
- Social media platforms for communication and customer support
- Partnerships with local event service providers and corporate gift suppliers

This digital distribution strategy ensures accessibility, convenience, and wider market reach within Tunisia and potentially other markets in the future.

2.5.3.4 Promotion

The promotional strategy of INWIN aims to increase brand awareness, attract users, and establish trust with both individuals and businesses.

The promotion strategy will mainly rely on digital marketing channels such as:

- Social Media Marketing: Promoting the application through platforms such as Facebook, Instagram, and LinkedIn by sharing event-related content, application features, customer experiences, and promotional campaigns.
- Website and Online Presence: Developing a website to present the platform, showcase services, and provide information about event organization and corporate gift customization.
- Partnership Marketing: Collaborating with event service providers, photographers, decorators, and corporate gift suppliers to increase platform visibility and attract more users.

- **Digital Advertising:** Using targeted online advertisements and sponsored social media campaigns to reach potential customers and businesses interested in event management services.

Through these promotional activities, INWIN aims to build a strong digital presence, increase user engagement, and position itself as an innovative and comprehensive solution in the Tunisian market.

2.5.4 Brand Identity

The brand identity of INWIN has been developed with a focus on conveying elegance, creativity, and professionalism. As a platform specialised in customizable corporate gifts and business solutions, its visual identity reflects the company's core values: innovation, quality, and customer satisfaction. This section presents the logic behind the logo design, colour palette, typography, and slogan used throughout the application and branding materials.

2.5.4.1 Application Name Choice

The term “IN” refers to integration and inclusion, representing a centralized platform where all services related to event organization and corporate gifting are brought together in one place. It also conveys the idea of being “in,” meaning modern, trendy, and aligned with current user needs.

The word “WIN” symbolizes success, achievement, and satisfaction. It reflects the main goal of the application: to ensure a successful and satisfying experience for users, whether they are organizing events or managing corporate gifts.

Therefore, INWIN represents a winning solution for both individuals and businesses by providing a smooth, efficient, and successful experience.

2.5.4.2 Application Logo Choice

Le logo InWin reflète une identité visuelle élégante, moderne et haut de gamme, en parfaite cohérence avec le positionnement de l'entreprise dans l'organisation d'événements et la création de cadeaux d'affaires personnalisés.



Figure 2.1: INWIN logo

- **Star Element:**

The golden star positioned above the logo symbolises innovation, inspiration, and excellence. It represents the idea of transforming simple concepts into successful achievements, perfectly aligning with the slogan “From Idea to Realisation.” The star also reinforces the premium image of the brand by adding a sense of brilliance and distinction.

2.5.4.3 Colour Palette

The colour palette of INWIN was selected to create a premium, modern, and visually appealing identity. Each colour was chosen carefully to reflect the company’s values and strengthen brand recognition.

Royal Blue (041C64): Used as the primary background colour, royal blue symbolises professionalism, trust, stability, and confidence. It creates a strong corporate identity while enhancing the luxurious appearance of the brand.

Elegant Gold (D4B484): Used for the logo typography and decorative elements, gold represents prestige, sophistication, success, and quality. It reinforces the premium positioning of the brand and draws attention to important visual elements.

Soft White (FFFFFF): Used for secondary backgrounds and clean spaces, soft white creates balance and improves readability while maintaining a minimalist and elegant design.

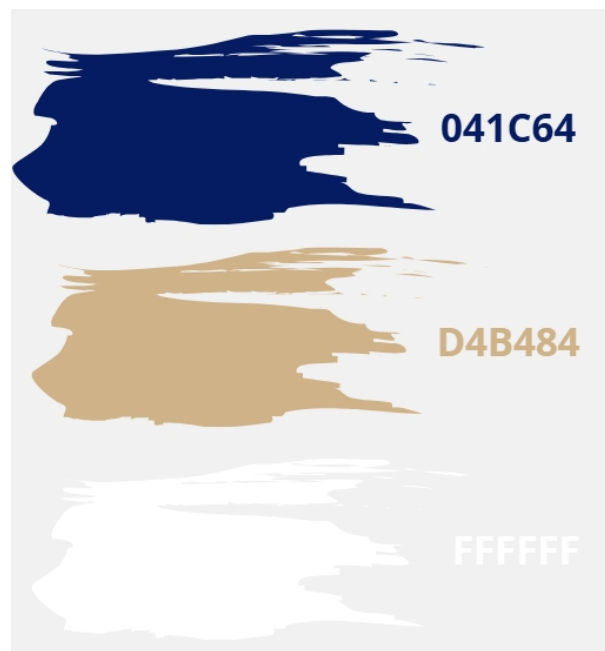


Figure 2.2: INWIN Colour Palette

2.5.4.4 Typography

Typography in INWIN was selected to reflect elegance, readability, and brand consistency. The following fonts are used throughout the application and branding materials:

Primary Typography: The main font used in the logo is Iridescent BC. This font was chosen for its luxurious and distinctive appearance, helping the brand stand out while conveying creativity and sophistication.

Secondary Typography: The secondary font used is Carrois Gothic SC. This typeface offers a clean and modern structure that complements the elegance of the primary font. It is mainly used for subtitles, interface elements, and supporting text to ensure readability and consistency across digital and printed materials.

Interface Typography:

Titles and Headings: Styled using bold variations of Carrois Gothic SC to create strong visual hierarchy and clear section organisation;

Body Text: Presented in a simple and readable sans serif style to ensure user comfort and maintain focus on the content.

Conclusion

This chapter presented the theoretical study and market research conducted for the INWIN application, including the survey analysis, market segmentation, benchmarking study, and the marketing plan established for the platform. It also highlighted the brand identity of INWIN, its target customers, and the marketing strategies proposed to promote the application effectively. The next chapter will outline the functional and non-functional specifications necessary for the design and development of the application.

Chapter 3

Functional Analysis and Requirements Engineering

Introduction

This chapter presents the requirements necessary for the development of our application, including the functional and non-functional requirements, use case diagrams, and technical specifications.

3.1 Functional and Non-Functional Requirements

This section presents the actors of the platform along with the functional and non-functional requirements necessary for the system's operation and development.

3.1.1 Actors Identification

The platform includes several actors, each with specific roles and responsibilities in interacting with the system and managing its different functionalities.

Primary Actors

Primary actors are those users who interact directly with the software in order to perform certain actions and enjoy the service of the software. The following are the main primary actors in the INWIN case:

- **Administrator:** Controls the operations of the system by reviewing requests coming in, studying details of the requests, preparing and sending quotes to the clients, updating status of the request, canceling requests and replying to the customers.

- **Business Customer:** Makes use of the software on behalf of the organization in browsing the gifts catalog, accessing the events area, making gift and event requests, checking progress on the projects, managing the profile details, responding to quotes, and interacting with the administrator.
- **Individual Customer:** Interacts with the system primarily for making requests related to events, monitoring the progress of his project submission, reviewing quotes, updating his personal information, and communicating with the administrator.

Secondary Actors

While secondary actors do not interact with the application in the same manner as primary actors, they offer the technical framework that enables proper functionality of the platform. The secondary actors include:

- **Firebase Authentication:** assists in managing the user signup and login process and enables access control. **Firebase Storage:** provides the ability to store media content and other documents, such as the logos associated with the companies registered. **Firebase Cloud Messaging:** facilitates the delivery of notifications in relation to quotations, requests, and messages sent.

3.1.2 Functional Requirements

Functional requirements describe the main functionalities and services provided by the system to its users. They define what the application should accomplish and how users interact with it. Below are the core features of our application.

Customer Functional Requirements:

- The customer must have the capability to register and login as well as edit personal information.
- The customer must be able to access the list of gifts and the list of events.
- The customer must have the capability to request for the gifts that include the product preference, number of products, logo location, attachment, and other notes.
- The customer must have the capability to request for events that include the event type, date, location, attendees, budget, services needed, colors, atmosphere, and notes.
- The customer must be able to track down his/her requested projects as well as review their information such as status, attachments, and quotations.
- The customer must be able to respond to quotations by accepting or rejecting them and download or share the purchase order when applicable.
- The customer must be able to exchange messages with the administrator and receive notifications related to requests, quotations, and project updates.

Admin Functional Requirements:

- The admin will be able to authenticate himself/herself and log into the administration section of the system.
- The admin will be able to browse, filter, and view all customer requests.
- The admin will be able to generate and send quotations to customers.
- The admin will be able to modify the status of all requests based on the project life cycle and even cancel requests where required.
- The admin will be able to interact with customers and track the progress of all ongoing projects.

System Functional Requirements:

- The system should be able to provide functionality for authentication and role-based access control.
- The system should be able to maintain records of user information, requests, projects, and message exchanges.
- The system should be able to upload files like logos and other supporting documentation.
- The system should be able to notify users about quotations, status updates, and new messages.

3.1.3 Non-Functional Requirements

Non-functional requirements specify the quality features and performance requirements that the INWIN program has to meet for its effective operation. The major non-functional requirements of the system are outlined below:

- **Performance:** The system will have to respond adequately to user operations, extract data about the projects, and present information about the requests.
- **Usability:** The system will have to possess an intuitive user interface that would be convenient for both the customers and administrators.
- **Compatibility:** The system will have to work effectively on the targeted platform, especially Android mobile devices.
- **Security:** The system will have to ensure the security of users' credentials, their messages, and uploaded documents.
- **Reliability:** The application will ensure consistent access to requests, quotations, messages, and notifications without loss of essential data.

3.2 Use Case Diagrams

The use case diagrams give a visual depiction of how the actors interact with the INWIN application. The use case diagrams for this project are listed below to demonstrate

the system's functionality and user interaction within the software.

3.2.1 Global Use Case Diagram

The global use case diagram presents an overview of the INWIN system by describing its main functions and their association with the various actors. The diagram focuses on the primary services provided by the system, such as submitting requests, monitoring projects, managing profiles, sending messages, handling quotes, and conducting administration activities. Below is the global use case diagram for the INWIN application.

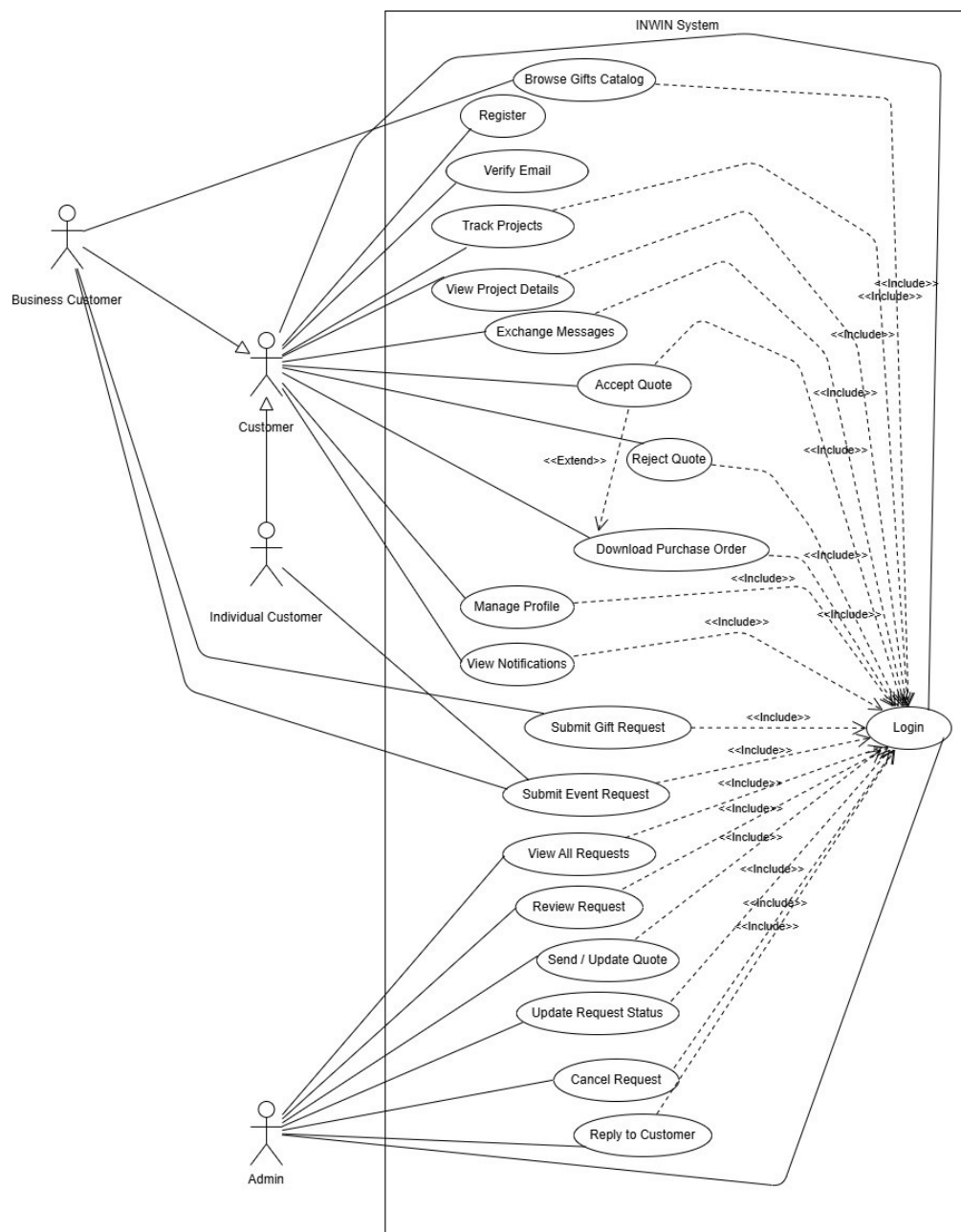


Figure 3.1: Global Use Case Diagram

3.2.2 Refined Use Case Diagrams

For a deeper analysis of the INWIN application, the global use case diagram is supplemented by a number of use case diagrams. Each diagram focuses on a specific functionality of the system and illustrates the interactions between the corresponding actor and the specific service provided by the platform. next section describes some of these use cases in detail.

3.2.2.1 Manage Profile

This use case diagram shows how the customer uses the system to access and modify their profile details such as personal information, company name, telephone number, and profile picture.

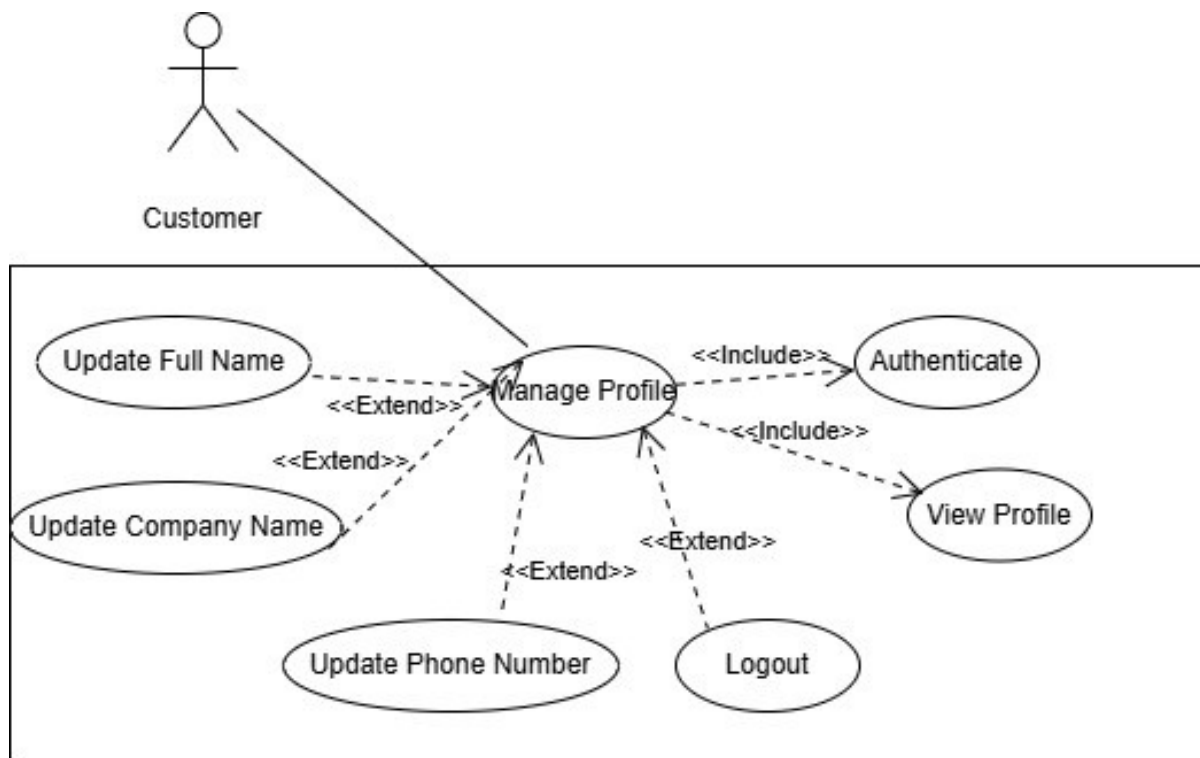


Figure 3.2: Manage Profile Refined Use Case Diagram

Below is a description of the use case "Manage Users".

Use Case Name	Manage Users
Description	This use case allows the customer to view and update their profile information.
Actors	Customer
Pre-conditions	The customer must be authenticated.
Normal Flow	1. The customer accesses the profile section. 2. The system displays the current profile information. 3. The customer updates one or more profile fields such as full name, company name, phone number. 4. The system saves the updated information in the database.
Alternative Flow	1. If the profile data cannot be loaded, an error message is displayed. 2. If the update operation fails, the system displays an error message and keeps the previous data unchanged.

Table 3.1: Description of the Use Case "Manage Profile"

3.2.2.2 Submit Gift Request

This use case diagram presents the interactions that allow the business customer to browse the gifts catalogue and submit a customised gift request according to their preferences and requirements.

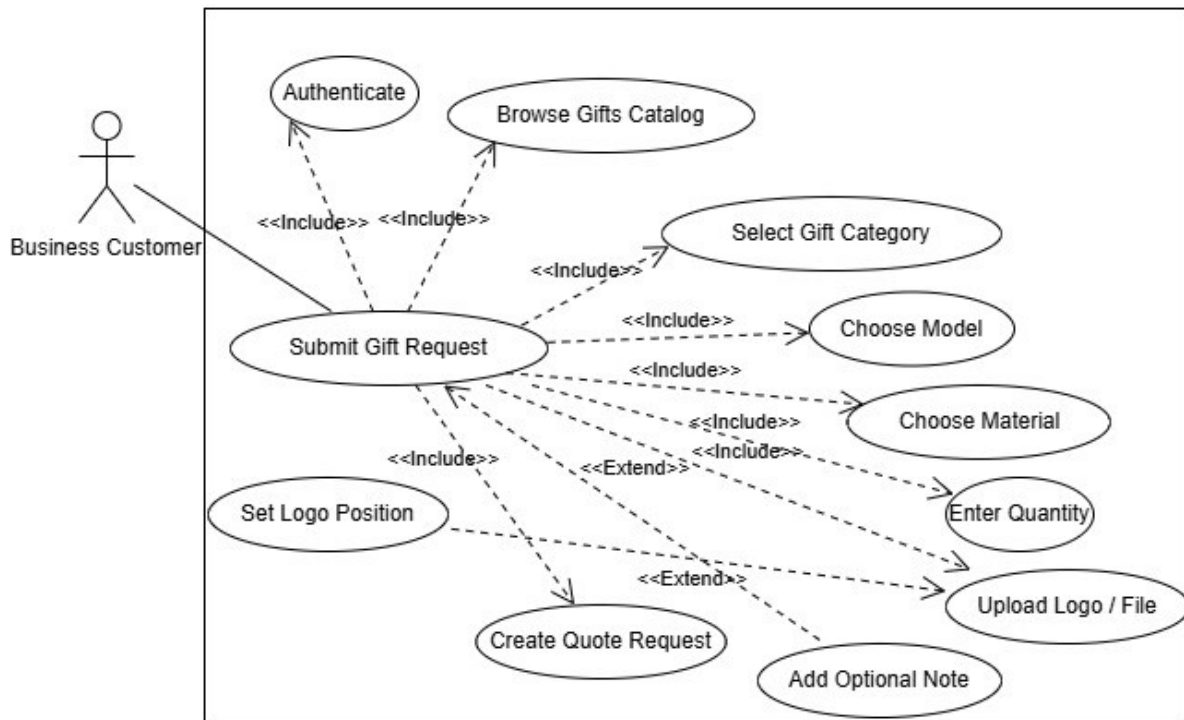


Figure 3.3: Submit Gift Request Refined Use Case Diagram

Below is a description of the use case " Submit Gift Request".

Use Case Name	Access IT Trends Dashboard
Description	This use case allows the business customer to create and submit a customised gift request.
Actors	Business Customer
Pre-conditions	The customer must be authenticated.
Normal Flow	1. The customer accesses the profile section. 2. The system displays the the gifts catalogue.. 3. The customer selects a category and configures the gift by choosing the model, material, quantity, logo position, and optional notes. 4. The customer submits the request 5. The system stores the request in the database.

Alternative Flow	1. If required information is missing, the system asks the user to complete the form. 2. If the request submission fails, an error message is displayed.
------------------	---

Table 3.2: Description of the Use Case " Submit Gift Request "

3.2.2.3 Submit Event

Request This use case diagram shows how the customer can submit an event request by providing the necessary information.

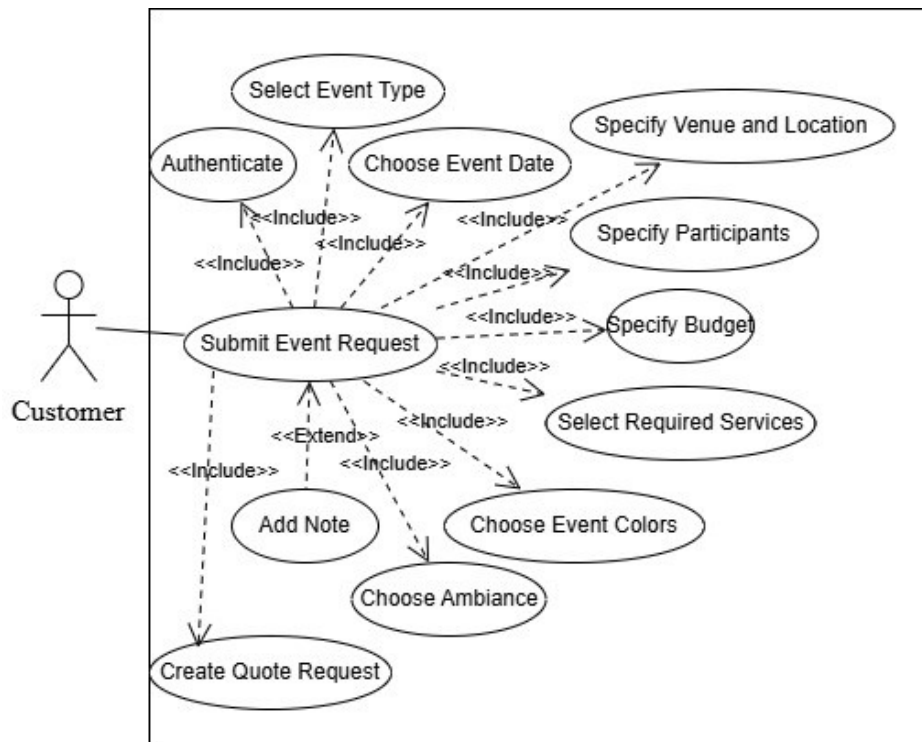


Figure 3.4: Submit Event Request Refined Use Case Diagram

Use Case Name	Access IT Trends Dashboard
Description	This use case allows the business customer to create and submit a customised gift request.
Actors	Business Customer
Pre-conditions	The customer must be authenticated.
Normal Flow	1. The customer accesses the profile section. 2. The system displays the the gifts catalogue. 3. The customer selects a category and configures the gift by choosing the model, material, quantity, logo position, and optional notes. 4. The customer submits the request 5. The system stores the request in the database.
Alternative Flow	1. If required information is missing, the system asks the user to complete the form. 2. If the request submission fails, an error message is displayed.

Table 3.3: Description of the Refined Use Case " Submit Event Request"

3.2.2.4 Track Projects

This use case diagram describes how the customer can follow the progress of submitted projects, consult request details, view attachments and quotation information, and access the related conversation.

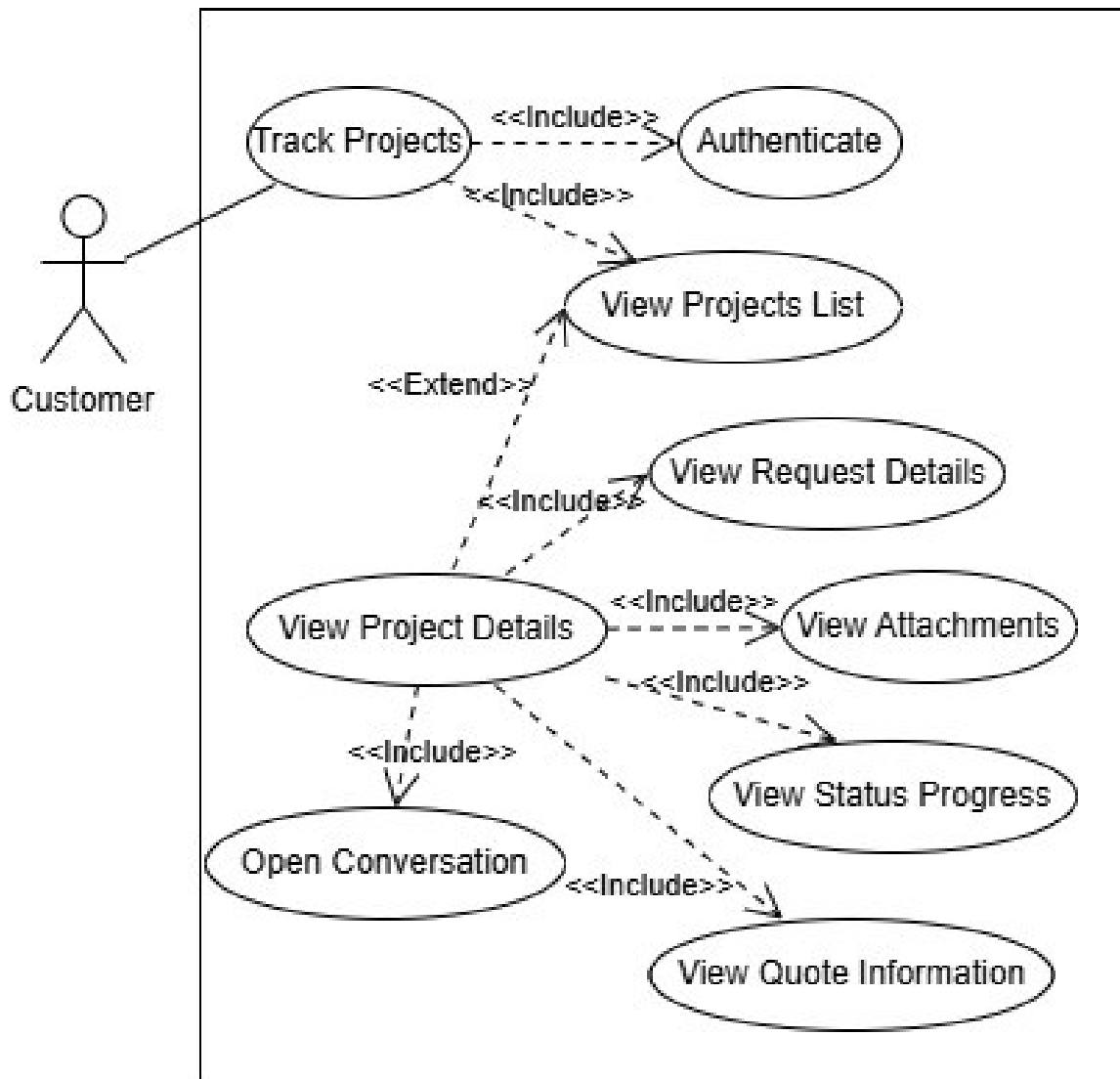


Figure 3.5: Track Projects Refined Use Case Diagram

Below is a description of the use case " Track Projects ".

Use Case Name	Track Projects
Description	This use case allows the customer to follow the progress of submitted projects and consult their details.
Actors	Customer
Pre-conditions	The customer must be authenticated.
Normal Flow	1. The customer accesses the projects section. - The system displays the list of submitted projects. 3. The customer selects a project. 4. The system displays the project details, including status, attachments, quotation information, and conversation access.
Alternative Flow	1. If no projects are found, the system displays an empty-state message. 2. If the selected project details cannot be retrieved, the system displays an error or not-found message.

Table 3.4: Description of the Refined Use Case " Submit Event Request"

3.2.2.5 Respond to Quote

This use case diagram illustrates how the customer interacts with the system to consult quotation details, read the terms and conditions, and respond by accepting or rejecting the proposed quote.

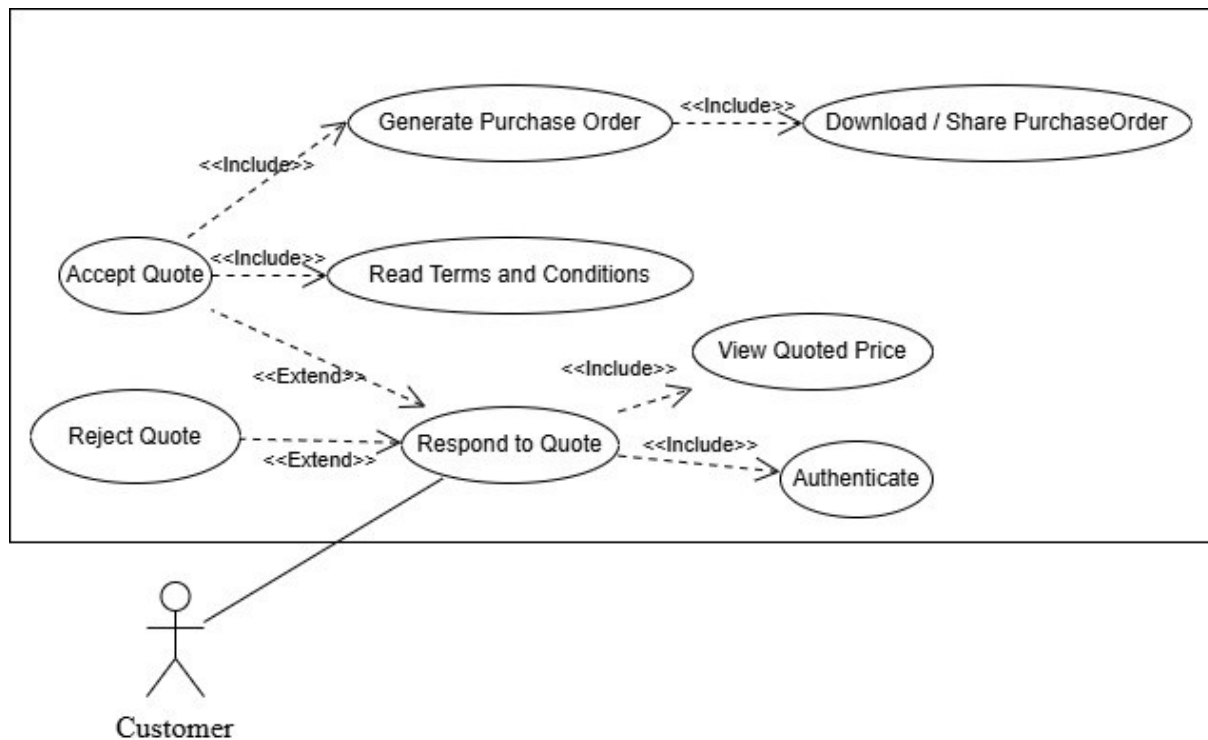


Figure 3.6: Respond to Quote Refined Use Case Diagram

Below is a description of the use case " Respond to Quote ".

Use Case Name	Respond to Quote
Description	This use case allows the customer to review a quotation and respond by accepting or rejecting it.
Actors	Customer
Pre-conditions	The customer must be authenticated and a quotation must be available for the selected request.
Normal Flow	1. The customer opens the quotation details. 2. The system displays the quoted price and the related terms and conditions. 3. The customer chooses to accept or reject the quotation. 4. The system updates the quotation status in the database. 5. If accepted, the customer may download or share the purchase order.

Alternative Flow	1. If the project or quotation data cannot be retrieved, the system displays an error or not-found message. 2. If the customer cancels the confirmation process, no response is recorded.
------------------	--

Table 3.5: Description of the Refined Use Case " Respond to Quote "

3.2.2.6 Exchange Messages

This use case diagram represents the communication process between the customer and the administrator through the project conversation space.

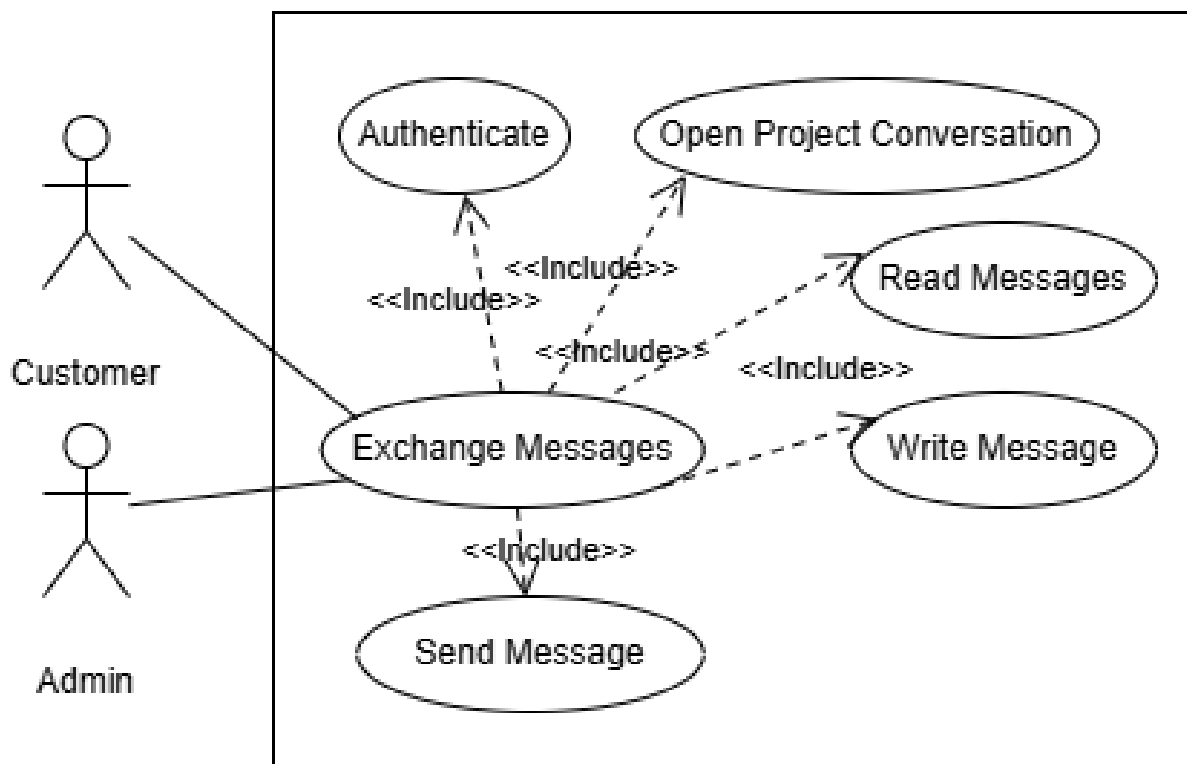


Figure 3.7: Exchange Messages Refined Use Case Diagram

Below is a description of the use case "Exchange Messages ".

Use Case Name	Exchange Messages
Description	This use case allows the customer and the admin to communicate through the project conversation space.
Actors	Customer, Admin
Pre-conditions	The user must be authenticated and the related project must exist.
Normal Flow	1. The user opens the conversation linked to a project. 2. The system displays previous messages. 3. The user writes a new message. 4. The system sends and stores the message. 5. The other party can read the message when accessing the conversation.
Alternative Flow	If no messages are available, the conversation area remains empty or displays a placeholder message.

Table 3.6: Description of the Refined Use Case " Respond to Quote "

3.2.2.7 Manage Requests

Manage Requests This use case diagram presents the actions available to the administrator for handling customer requests, including viewing requests, reviewing details, filtering requests, sending quotations, updating status, cancelling requests, and replying to customers.

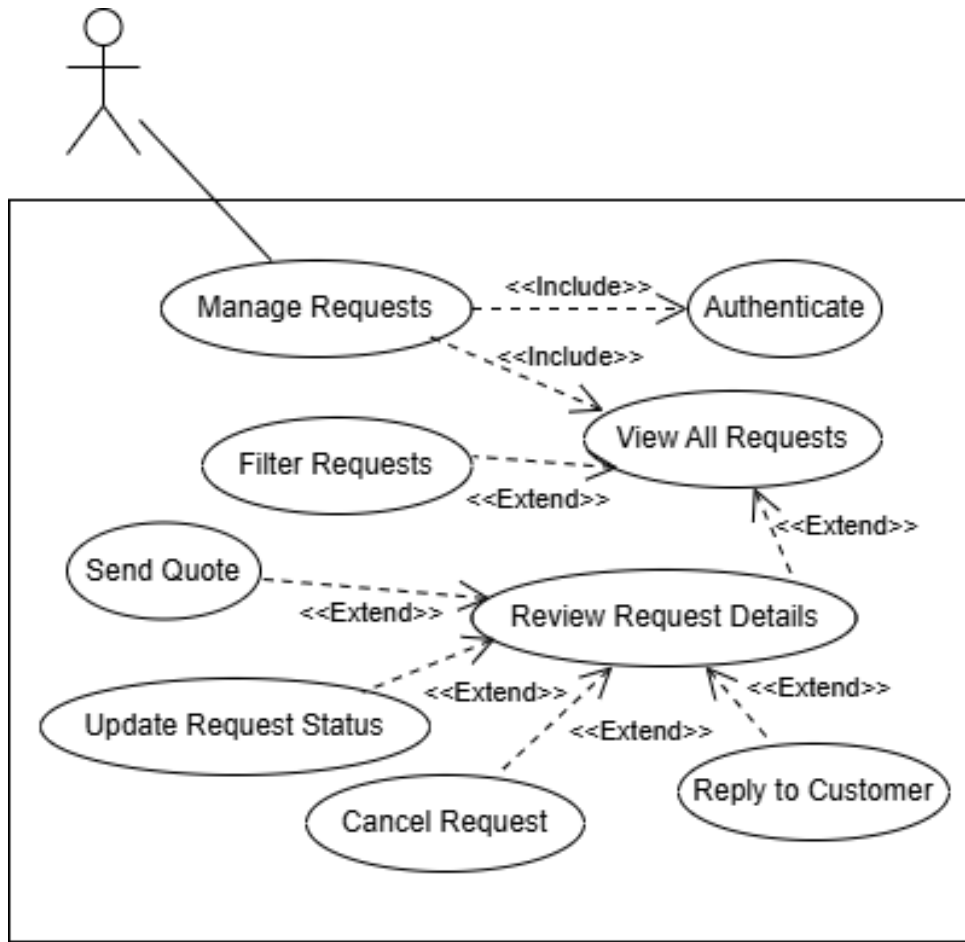


Figure 3.8: Manage Requests Refined Use Case Diagram

Below is a description of the use case " Manage Requests ".

Use Case Name	Manage Requests
Description	This use case allows the admin to handle customer requests and perform the necessary management actions.
Actors	Admin
Pre-conditions	The admin must be authenticated.

Normal Flow	1. The admin accesses the requests management section. 2. The system displays all customer requests. 3. The admin may filter requests and select one to review its details. 4. The admin performs an action such as sending a quotation, updating the request status, cancelling the request, or replying to the customer. 5. The system updates the corresponding data in the database.
Alternative Flow	1.If requests cannot be loaded, the system displays an error message. If an action fails, the system informs the administrator through an error message, and the requested update is not saved.

Table 3.7: Description of the Refined Use Case " Manage Requests"

3.3 Technical Specification

This section presents the main technologies, tools, external packages, architectural choices, and security mechanisms adopted in the development of the INWIN application.

3.3.1 Used Tools

The table below presents the used tools and technologies.

Logo	Description
For Mobile Application Development	
 [1]	Flutter is a UI framework that was created by Google to enable developers to create applications natively on different platforms using a single codebase.
 [2]	Dart is an optimized programming language for clients and is open source. This means it can be used to develop applications on different platforms efficiently. Dart has advanced features like null safety and pattern matching.
For Backend Development	
 [3]	Firebase Authentication was used to manage all aspects of identity management within the app, including registration, login, and onboarding processes.
Development Tools	
 [4]	Android Studio is the recommended software development kit for developing Android applications, which includes a wide range of tools for development, debugging, testing, and optimizing performance.
For Design and Prototyping	
 [5]	Design.com is an AI-based branding solution used in the creation of the application's logo, and it features several options for designing logos.
 [6]	Figma is a web-based graphic designing software used to design the application's user interface mockups and prototypes, and it runs exclusively on the browser.
For UML Modeling	
 [7]	Visual Paradigm Online is an online diagramming software that was utilized in designing the diagrams of the application. The tool can create various kinds of diagrams, such as UML diagrams.

Table 3.8: Technologies and Tools Used

3.3.2 Firebase Services and External Packages

This subsection presents the key Firebase services and external packages integrated into the INWIN application. Each technology was selected to address a particular requirement within the project, such as user authentication, data storage and retrieval, file management, push notifications, screen navigation, or UI development.

Service/Package	Description
For Firebase Services	
Firebase Authentication	Manages user registration, login, and password recovery, ensuring secure access to the application.
Cloud Firestore	Acts as the primary database for storing and retrieving user profiles, requests, messages, and project data.
Firebase Storage	Acts as the primary database for storing and retrieving user profiles, requests, messages, and project data.
Firebase Cloud Messaging (FCM)	Delivers notifications to users related to quotations, request updates, and new messages.
Cloud Functions for Firebase	Automates certain backend operations, particularly those involved in triggering and managing notifications.
For External Packages	
flutter-riverpod	Manages state across the application, keeping data consistent between screens.
go-router	Controls screen navigation and defines how users move through the application.
file-picker	Allows users to browse their device and select files for upload.
intl	Handles the formatting of dates and other displayed data within the application.
pdf	Generates purchase order documents in PDF format directly from the application.

Service/Package	Description
share-plus	Enables users to share generated documents straight from application.

Table 3.9: Firebase Services and External Packages

3.3.3 Software Architecture

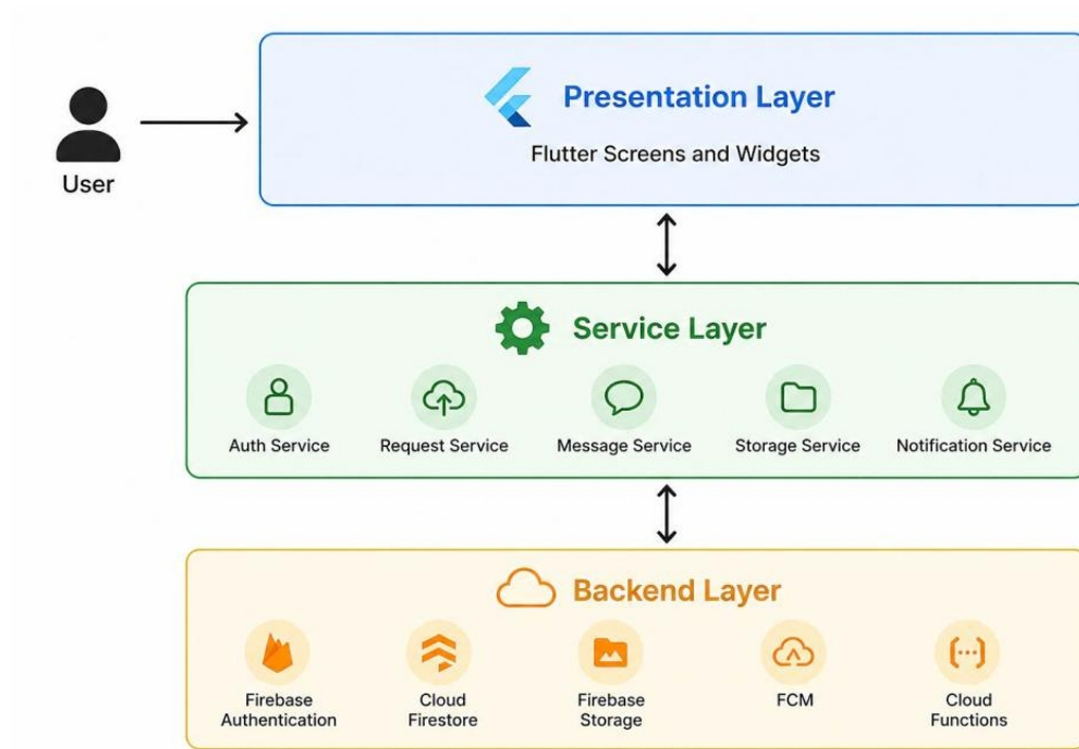


Figure 3.9: Layered Architecture of the INWIN Application

The architecture of the INWIN application represents a layered architecture that ensures the complete isolation of user interface logic from application logic, and further from backend services. The separation of logic into distinct modules allows maintaining the source code base, its scalability, and provides a well-defined communication between separate parts of the system. There are three layers in total in the INWIN application, as can be seen on the diagram provided above:

- **Presentation Layer:** This is what users actually see and interact with. It includes all the screens and widgets designed for both customers and administrators.
- **Service Layer:** the core logic of the app and processes the main operations inside.

- Backend Layer: responsible for managing internal processes including data storage, user authentication, file operations, and other things, all based on Firebase services. To put it simply, implementation of separate layers will look like this:
- presentation layer implemented using Flutter and responsible for creating the mobile user interface of INWIN app.
- The service layer is made up of dedicated Dart service classes covering authentication, request management, messaging, storage, and notifications.
- The backend layer relies on Firebase, using Firebase Authentication for access control, Cloud Firestore as the main database, Firebase Storage for uploaded files, Firebase Cloud Messaging for notifications, and Cloud Functions for backend automation.

3.3.4 App Security

Application Security is one of the core components of the INWIN framework, since the framework manages users' accounts, customers' queries, communications, and documents shared between the parties. This sub-section describes the key application security approaches used for securing user data, managing system feature access, and ensuring the integrity of data storage.

3.3.4.1 Authentication

Authentication basically involves ensuring the identity of the user accessing the application. In the case of the INWIN application, authentication was carried out through Firebase Authentication. Once logged in, Firebase authenticates the user and links him to his respective record in Cloud Firestore, ensuring that each authenticated user has a user record within the Firestore database, and the sessions stay consistent with other application data.

Authentication is achieved through logic that determines whether the authenticated user already has a record in the users collection. If there is no record, a default record will be created automatically, guaranteeing that each user has at least a record in the database. Moreover, the application listens continually to the changes in authentication status, thus being able to authenticate each user instantly upon logging in and binding him to the correct Firestore record.

The following figure illustrates the part of the authentication logic that accomplishes this task

```

Future<void> bindUserDoc(User firebaseUser) async {
  final userDoc = _db.collection('users').doc(firebaseUser.uid);
  final snapshot = await userDoc.get();

  if (!snapshot.exists) {
    await userDoc.set(_buildFallbackUserMap(firebaseUser));
  }

  userDocSubscription = userDoc.snapshots().listen(
    (doc) {
      controller.add(doc.exists ? AppUser.fromFirestore(doc) : null);
    },
    onError: controller.addError,
  );
}

```

Figure 3.10: Code snippet showing how the authenticated user is linked to the corresponding user record

The following figure presents a code snippet where the application listens to authentication state changes and handles the authenticated session.

```

authSubscription = _auth.authStateChanges().listen(
  (firebaseUser) async {
    await userDocSubscription?.cancel();
    userDocSubscription = null;

    if (firebaseUser == null) {
      controller.add(null);
      return;
    }

    try {
      await bindUserDoc(firebaseUser);
    } catch (error, stackTrace) {
      controller.addError(error, stackTrace);
    }
  }
)

```

Figure 3.11: Code snippet demonstrating how authentication state changes are monitored in the application

3.3.4.2 Authorisation and Role Management

Authorisation determines what authenticated user is allowed to access or modify in the application. In this project, authorisation was implemented using role-based access control. The system mainly distinguishes between two roles: customer and admin. After authentication, the role of the user is checked before access to data is granted.

In our application, access is controlled using three main conditions: the user must be authenticated, the user may access only his or her own data, and an admin has broader permissions across the system. This means that ordinary users are restricted to their own information, while administrators can manage a wider range of records. Some actions, such as deletion, are also denied to better protect stored data.

The figure on the next page presents the part of the authorisation logic responsible for checking authentication, ownership, and administrator privileges.

```
function isAuth() {  
  return request.auth != null;  
}  
function isOwner(uid) {  
  return request.auth.uid == uid;  
}  
function isAdmin() {  
  return isAuth() &&  
    get(/databases/$(database)/documents/users/$(request.auth.uid)).data.role == 'admin';  
}
```

Figure 3.12: Code snippet showing how authentication, ownership, and administrator role are checked

The following figure presents a code snippet where access to user records is controlled according to ownership and administrator privileges.

```
match /users/{userId} {  
  allow read: if isOwner(userId) || isAdmin();  
  allow create: if isOwner(userId);  
  allow update: if isOwner(userId) || isAdmin();  
  allow delete: if false;  
}
```

Figure 3.13: Code snippet demonstrating how access to user records is restricted

The following figure presents a code snippet where access to requests is controlled according to the role of the authenticated user.

```
match /requests/{requestId} {  
  allow read: if isAdmin() ||  
    (isAuth() && resource.data.customerId == request.auth.uid);  
  allow create: if isAuth() &&  
    request.resource.data.customerId == request.auth.uid;  
  allow update: if isAdmin() ||  
    (isAuth() &&  
      resource.data.customerId == request.auth.uid &&  
      request.resource.data.diff(resource.data).affectedKeys()  
        .hasOnly(['status', 'customerNote', 'updatedAt']));  
}
```

Figure 3.14: Code snippet demonstrating how request permissions are managed according to user role

3.3.4.3 Data Validation and Protection

Data validation refers to the process by which the program checks whether the information provided by the user has been validated before any action takes place. In the current project, validation has been done on various fields in order to identify errors and incomplete information. For example, the program validates that all required fields have been filled and that phone numbers are in the correct format.

Similar attention was paid to uploading files. Only those who have logged in can upload files, and there is also control over the size of files and their types. In this way, no invalid data is passed to the system and creates problems later on.

The figure on the next page presents a code snippet showing how user input is validated before being accepted by the application.

```
static String? phone(String? value) {  
  if (value == null || value.trim().isEmpty) return 'Numéro requis';  
  if (!TunisiaPhone.isValidLocalPart(value)) {  
    return 'Entrez un numéro à 8 chiffres';  
  }  
  return null;  
}
```

Figure 3.15: Code snippet showing how phone number input is validated before submission

```

match /markings/{userId}/{allPaths=**} {
  allow read: if request.auth != null &&
    (request.auth.uid == userId ||
    | firestore.get(/databases/(default)/documents/users/{request.auth.uid}).data.role == 'admin');
  allow write: if request.auth != null &&
    request.auth.uid == userId &&
    request.resource.size < 50 * 1024 * 1024 &&
    request.resource.contentType.matches('image/.*|application/pdf');
}

```

Figure 3.16: Code snippet showing how user input is validated before submission

3.3.4.4 Error Handling

Error handling involves the manner in which an application reacts when presented with exceptional circumstances or failure scenarios. Error handling was applied in this project using the try and catch blocks so as to ensure that errors are captured without leading to the crash of the application. In case of any error encountered by the application, the system manages the situation effectively through either logging of errors or simply informing the user about the error. This helps in stabilizing the running application.

The following is a diagram of the error capturing process done by the application.

```

Future<void> _initBackgroundServices() async {
  try {
    SystemChrome.setSystemUIOverlayStyle(const SystemUiOverlayStyle(
      statusBarColor: Colors.transparent,
      statusBarIconBrightness: Brightness.dark,
    ));

    await Future.wait([
      initializeDateFormatting('fr_FR', null),
      NotificationService.initialize(),
      SystemChrome.setPreferredOrientations([
        DeviceOrientation.portraitUp,
        DeviceOrientation.portraitDown,
      ]),
    ]);
  } catch (error, stackTrace) {
    debugPrint('Background initialization failed: $error');
    debugPrintStack(stackTrace: stackTrace);
  }
}

```

Figure 3.17: Code snippet showing how initialization errors are caught and handled in the application

3.3.4.5 Database Integrity

Database integrity will ensure that the information present within the database is accurate and consistent. For our system, integrity will be enhanced by ensuring that whenever a user signs up for our application, the user information that will be added to the database will be unique. This is achieved because upon signing up, the application registers the new user using their email address and password through Firebase Authentication. The authentication tool will not create more than one user with the same email address. The figure below illustrates how a new account is created before the user record is stored in the database.

```
final cred = await _auth.createUserWithEmailAndPassword(  
    email: email, password: password);  
  
await cred.user?.sendEmailVerification();  
  
final user = AppUser(  
    uid: cred.user!.uid,  
    fullName: fullName,  
    companyName: companyName,  
    clientType: clientType,  
    email: email,  
    phone: phone,  
    role: UserRole.customer,  
    createdAt: DateTime.now(),  
); // AppUser  
await _db.collection('users').doc(user.uid).set(user.toMap());  
await cred.user!.updateDisplayName(fullName);  
return user;  
}
```

Figure 3.18: Code snippet showing how user registration is performed before storing consistent user data

Conclusion

This chapter covered the development requirements of the INWIN application, from the functional and non-functional requirements to the use case diagrams and the technical specification. The following chapter moves into the structural and behavioural design of the system, presenting the class diagram, sequence diagrams, and activity diagrams.

Chapter 4

System Architecture and Workflow Designs

Introduction

This chapter offers insight into the design of the INWIN application through the use of UML diagrams. The intention behind this approach is to ensure that the users can easily understand how the software system functions through both static and dynamic representations of the system. On this note, the use of class diagrams illustrates how the different classes interact within the software application.

4.1 Class Diagram

The class diagram represents the static structure of the INWIN system. In other words, it represents the fundamental classes making up the system alongside their attributes, methods, and associations. The following is a brief explanation of some of the fundamental classes represented in the class diagram presented on the following page.

- **AuthService:** Handles authentication operations such as registration, login, email verification, password reset, profile update, and logout.
- **AppUser:** An object representing an end user with personal attributes including full name, organization name, email, phone number, role, category of the customer, and creation date of the account.
- **QuoteRequest:** Represents the quote request from the customer who used the application including type, status, description, file attachment, quotation data, and additional remarks.
- **RequestService:** Handles request processing including quotation generation and change of statuses.

- **ChatMessage:** Represents the message exchanged between the customer and administrator while discussing projects.
- **MessageService:** Deals with sending and retrieving messages regarding the particular request.
- **StorageService:** Handles the upload and deletion of files in regard to the particular request.
- **NotificationService:** Handles notification-related operations including token handling and request status notifications.

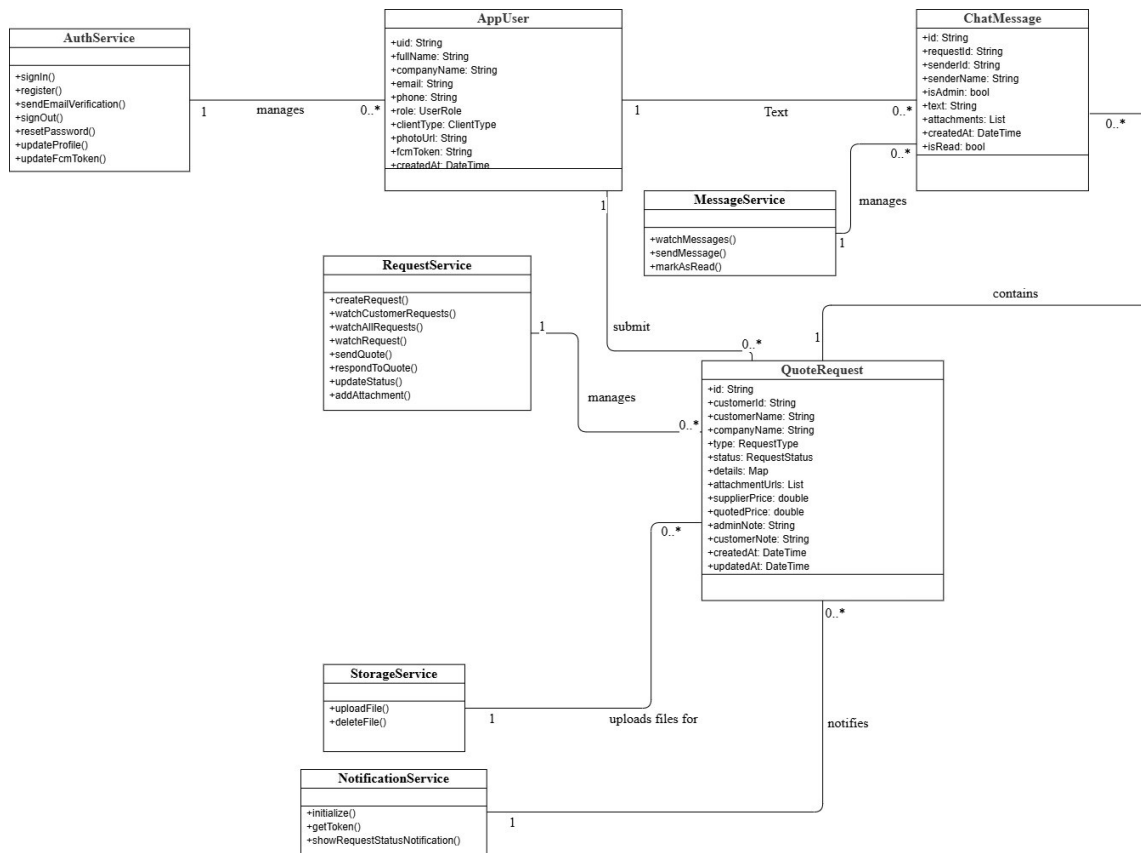


Figure 4.1: Global Class diagram

4.2 Sequence Diagrams

Sequence diagrams illustrate the dynamic aspect of the INWIN application based on the interaction among all the different actors and components within the system. The process enables one to understand how the request is handled and how data is passed from the user interface, through the service layer to the back end services during execution of a certain function.

4.2.1 Log In Sequence Diagram

Below is the sequence diagram for authentication process for the user within the INWIN application. The sequence diagram below demonstrates how the user gets authenticated using the interactions between the login UI, authentication service, Firebase Authentication, Cloud Firestore, and the router component used for routing purposes according to user roles.

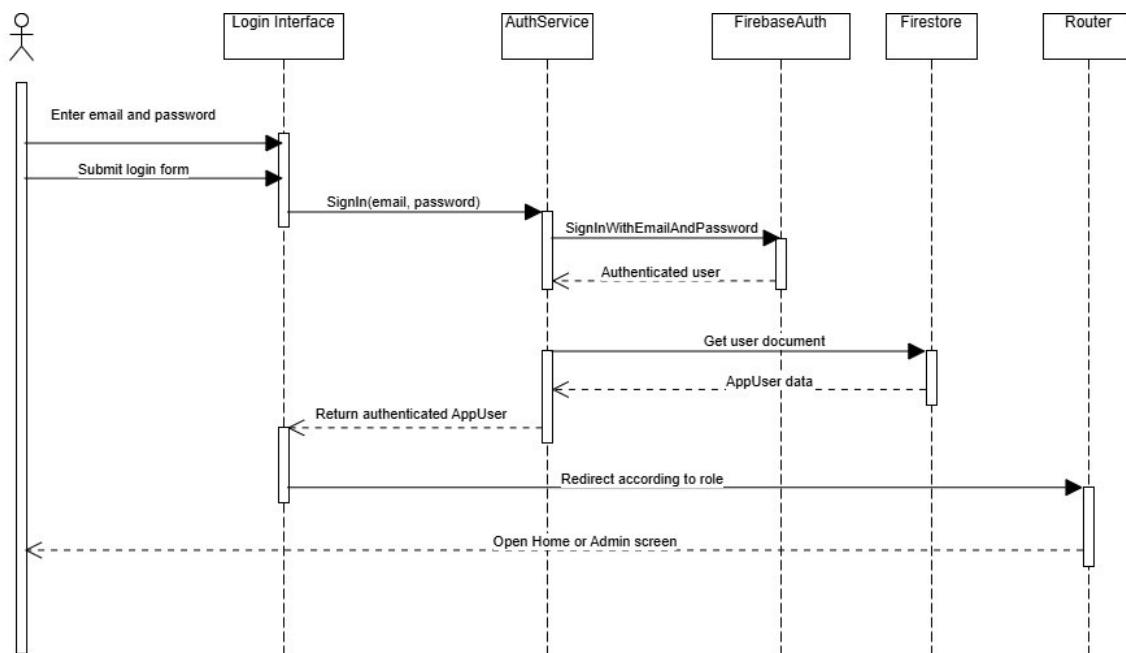


Figure 4.2: Log In Sequence Diagram

- The user provides their email and password via the login screen.
- The application ensures that all required input fields are not left blank.
- In case an input field is left blank, a validation message pops up.
- Otherwise, the login credentials are submitted to the authentication service.
- The authentication service sends the user's mail and password to Firebase Authentication. In the case of an invalid username and password combination, the application will return an error message in the login window.
- If the entered credentials are valid, the authentication process will fetch the respective user data from the Cloud Firestore database.
- After that, the role of the user is established based on the authenticated identity.
- If the user is an administrator, then the administrator area is loaded.

- In case the user is a customer, then the application redirects the customer to the customer area.
- For customer accounts, the router will also validate the status of the email address.
- If the email is not verified, the application will redirect the user to the email validation page first.

4.2.2 Submit Request Sequence Diagram

Below is the sequence diagram of "Submit Request", together with its step description, illustrating how the customer submits a request and how it is processed by the system.

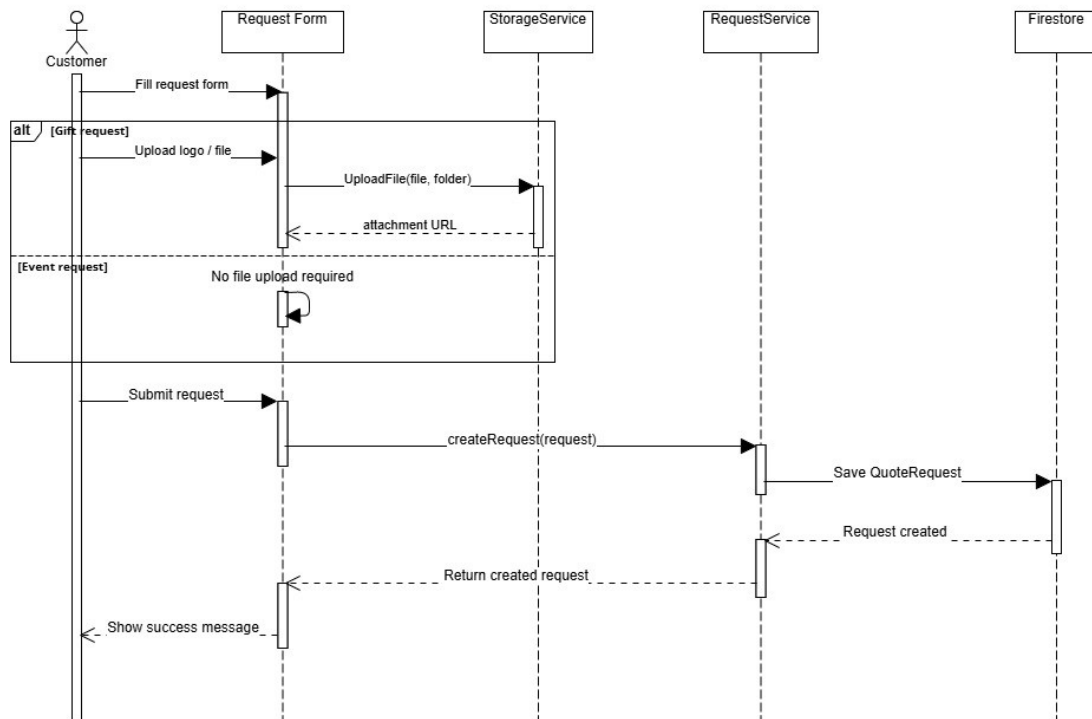


Figure 4.3: Submit Request Sequence Diagram

- If it is related to a gift, then the customer uploads a logo/file through the request form.
- The request form transfers the uploaded file to the storage service.
- The storage service uploads the file and provides the URL back to the request form.
- If it is related to an event, then no need for uploading a file.
- After filling out all the details, the customer sends the request.

- The request form transfers the request data to the request service.
- The request service creates a QuoteRequest object and saves it into Cloud Firestore.
- Cloud Firestore saves the request into the database.
- When the process finishes, then it gives the saved request back.
- Lastly, the user interface shows a success message indicating that the request was sent successfully.

4.2.3 Admin Manages Request Sequence Diagram

The sequence diagram for the scenario "Admin Manages Request" is depicted below, along with its step description, illustrating how an administrator reviews a customer request and prepares a quote for it.

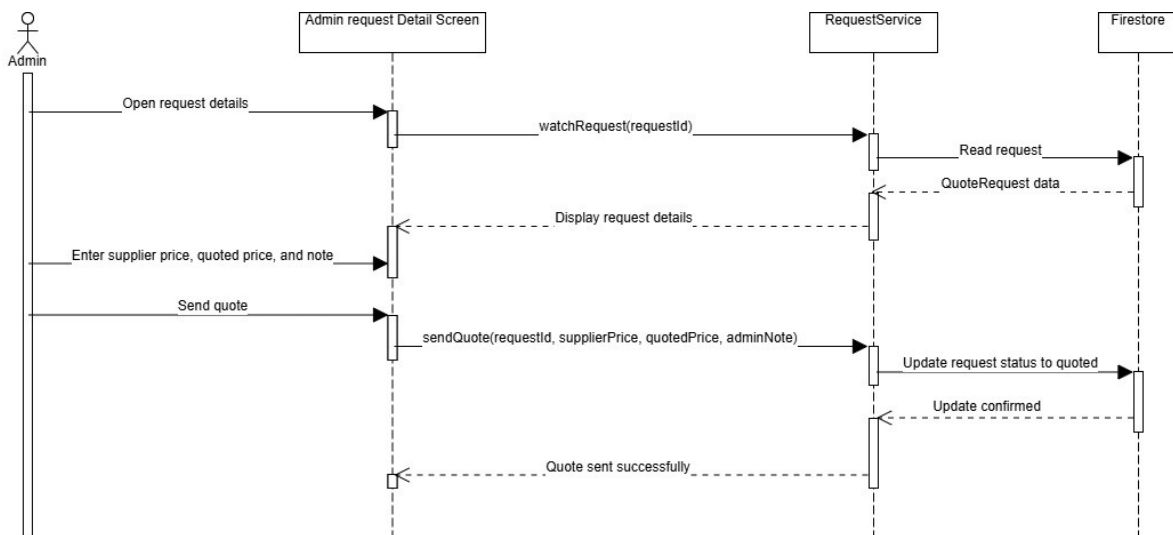


Figure 4.4: Admin Manages Request Sequence Diagram

- The administrator clicks on the details of a selected request.
- The request detail screen passes the request id to the request service.
- The request service fetches the corresponding request data from Cloud Firestore.
- Cloud Firestore provides the request data back to the request service.
- Next, the request details will be shown to the administrator.
- The administrator fills in the supplier quote, quote, and note if needed.
- The administrator proceeds to make the quotation.

- The request detail screen transfers the quotation details to the request service.
- The request service updates the request by updating the quotation and requesting state to quoted in Cloud Firestore.
- Cloud Firestore acknowledges the operation of updating.
- Later, the system shows that the quote is made successfully. message indicating that the request was sent successfully.

4.2.4 Customer Responds to Quote Sequence Diagram

The sequence diagram for "Customer Responds to Quote" can be seen on the following page, along with the steps involved in this activity as described below.

The customer views the details of the quote request.

The project detail screen sends the request id to the request service.

The request service retrieves the corresponding request data from Cloud Firestore.

The request details are received from Cloud Firestore to the request service.

The quoted request is then shown to the customer. **If the customer accepts the quote:**

- The customer chooses the accept option.
- The terms and conditions will be shown.
- The customer confirms acceptance.
- The project detail screen sends the response to the request service with the value `accepted = true`.
- The request service updates the status of the request in Cloud Firestore to accepted.
- Following confirmation, the purchase order service prepares the purchase order document.
- The generated PDF is sent back to the user interface.
- The customer can then download or share the purchase order document.

If the customer rejects the quote:

- The customer selects the rejection option.
- The project detail screen sends the response to the request service with the value `accepted = false`.

- The request service changes the request status to rejected on Cloud Firestore.
- On confirmation, the rejection confirmation message pops up.

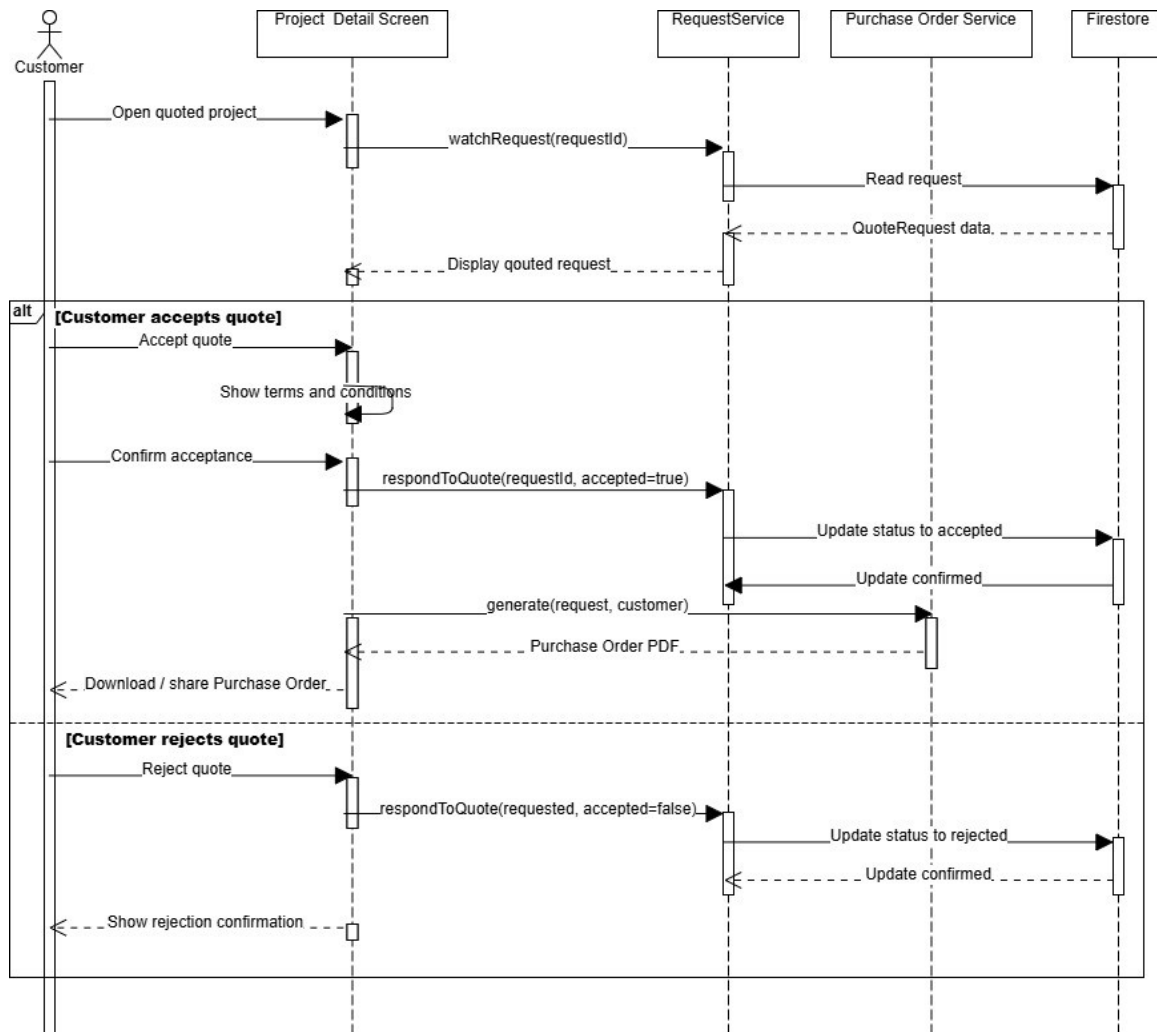


Figure 4.5: Customer Responds to Quote Sequence Diagram

4.2.5 Project Messaging Sequence Diagram

Below is the sequence diagram of "Project Messaging," together with its step description below to demonstrate how messages are sent and received in a project discussion.

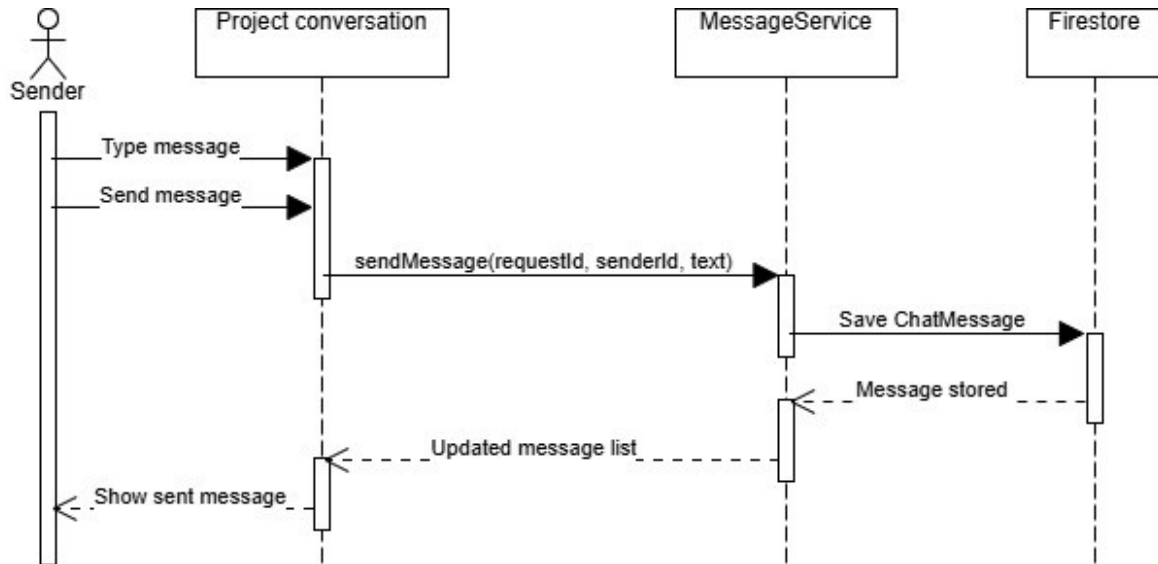


Figure 4.6: Project Messaging Sequence Diagram

- The sender opens the interface for the project conversation.
- The sender types message in the conversation panel.
- The sender sends the message.
- Message data from conversation interface gets sent to messaging service.
- The Messaging service saves the message in Cloud Firestore.
- Confirmation of successful saving of message in Cloud Firestore is sent back.
- Saved message list gets returned back to conversation interface.
- Final step, the sent message gets displayed in the conversation window.

4.3 Activity Diagrams

Activity diagrams show the sequence of activities and decisions in the process, which makes it simpler to comprehend how the system works. This part will discuss the major processes that were developed in the INWIN application.

4.3.1 User Authentication Workflow Activity Diagram

The activity diagram of "User Authentication Workflow" is presented below, and its description is provided in the next page.

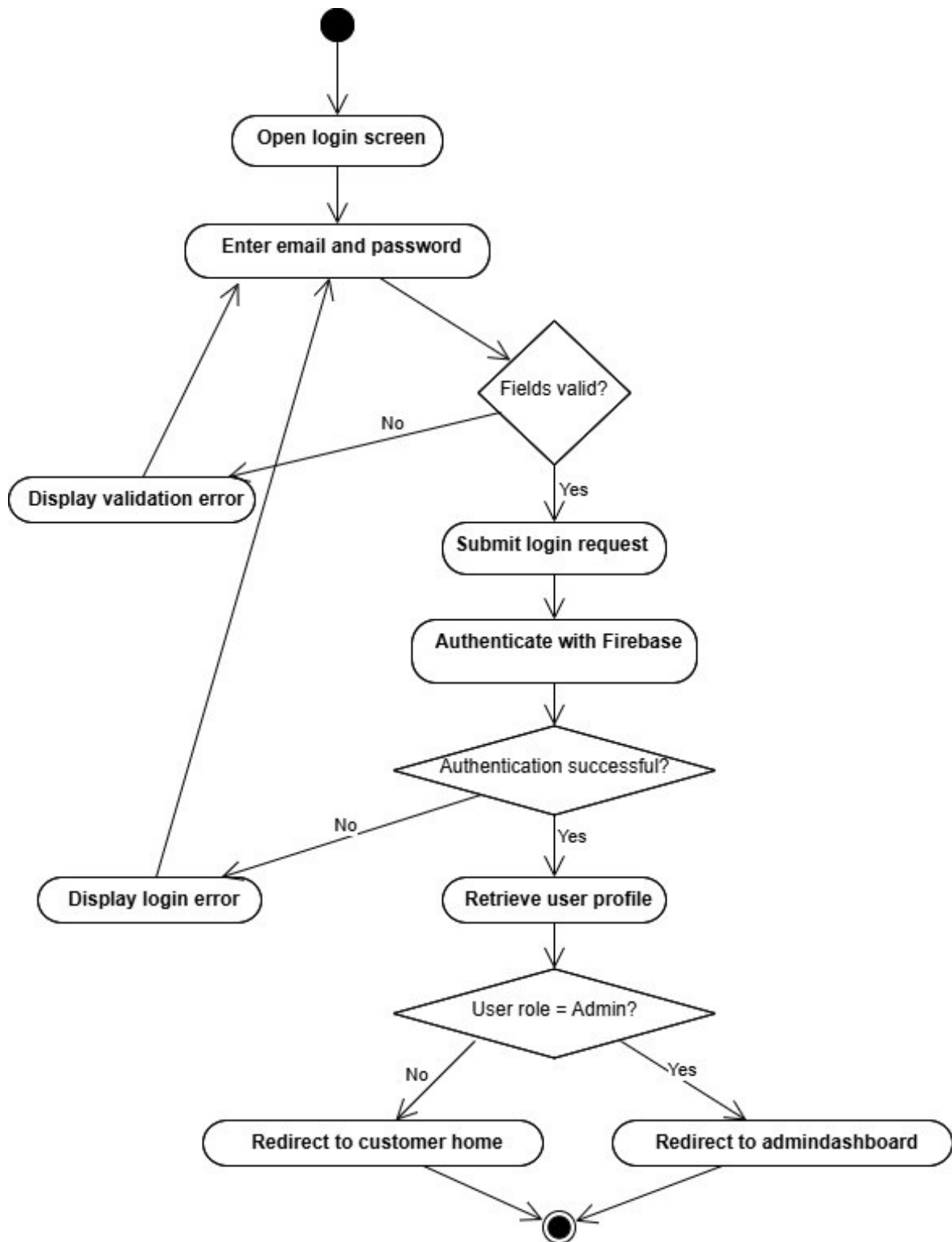


Figure 4.7: User Authentication Workflow Activity Diagram

- The user navigates to the login interface.
- The user inputs their email ID and password.

- The system validates whether the inputs entered by the user are correct.
- In case of incorrect data inputs, a validation error message appears on the screen.
- In case of correct data inputs, the system sends out the login request.
- The user is validated using Firebase Authentication service.
- In case of login failure, the user sees an error message.
- In case of successful login attempt, the user's profile is fetched.
- The role of the user account is validated.
- If the user is an administrator, the system redirects them to the admin dashboard.
- Otherwise, the system redirects the user to the customer home page.

4.3.2 Gift Request Submission Workflow Activity Diagram

The activity diagram of "Gift Request Submission Workflow" is presented on the next page, and its description is provided below.

- The user navigates to the gift catalog.
- The user chooses a gift category.
- The user completes the gift request form.
- The user uploads a file containing the logo/marketing.
- The system checks whether the uploaded file is an image file.
- If the file is an image, the user adjusts the position of the logo.
- Otherwise, the positioning step is skipped.
- The system checks the form for validity.
- In case the form is not valid or if there is no file attached, the system shows an error message.
- In case the form is valid and the file is attached, the file is uploaded.
- The gift request is created.
- The gift request is stored in the database.
- The system shows the success message.

- Lastly, the user is redirected to the projects page.

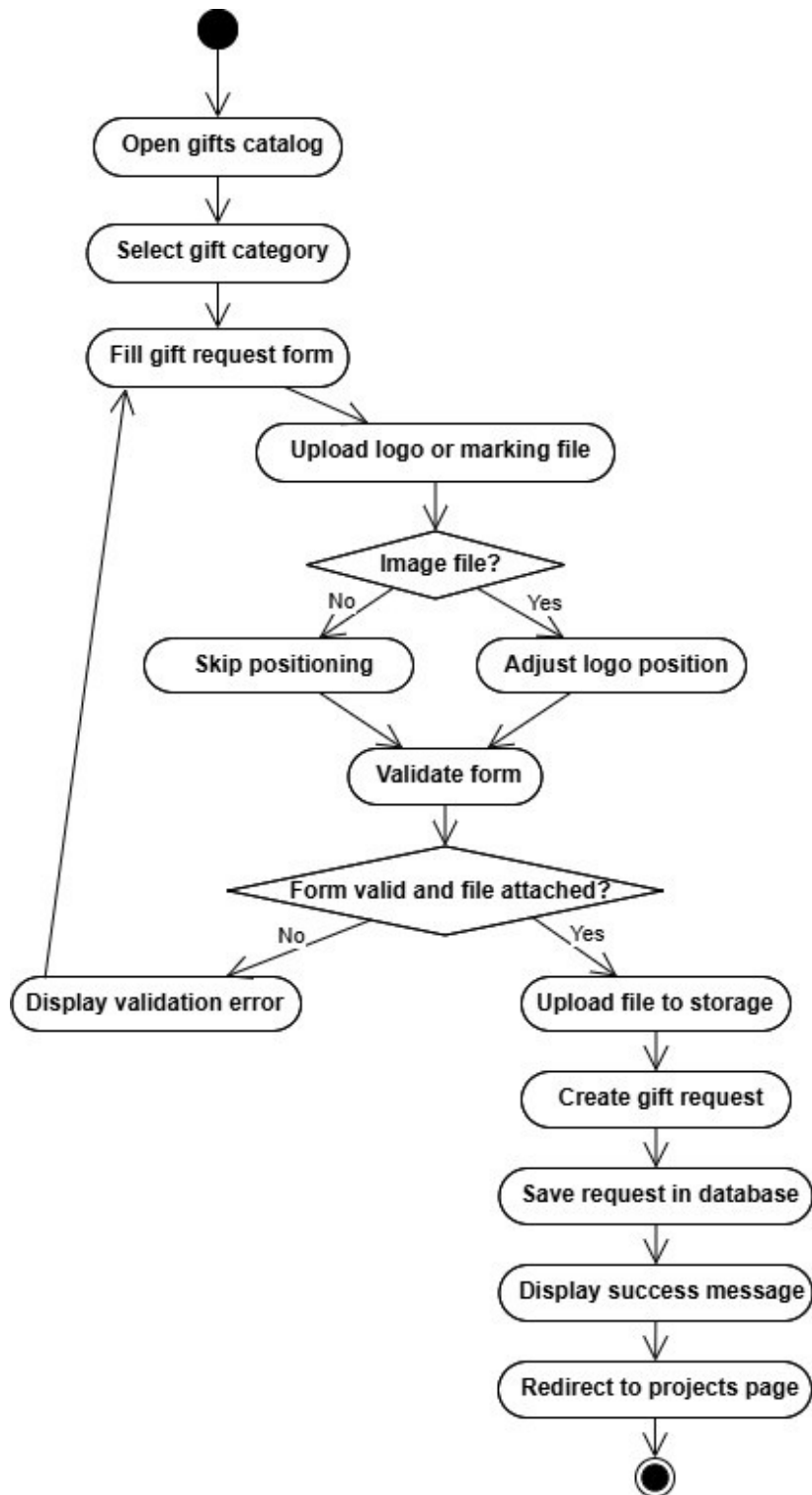


Figure 4.8: Gift Request Submission Workflow Activity Diagram

4.3.3 Event Request Submission Workflow Activity Diagram

Below are the description and the activity diagram " Event Request Submission Workflow Activity Diagram ".

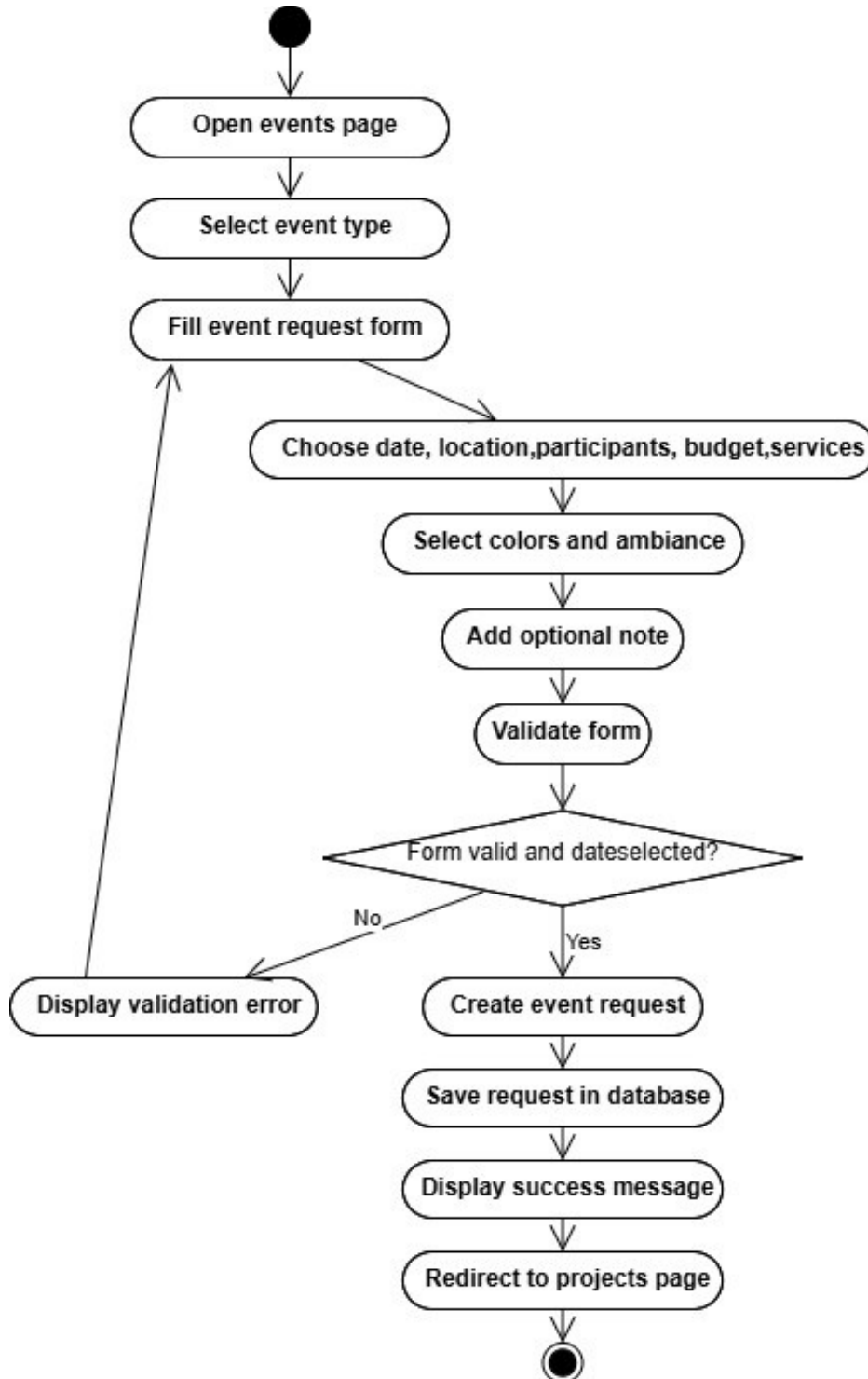


Figure 4.9: Event Request Submission Workflow Activity Diagram

- The user clicks on the events page.

- The user chooses the type of event.
- The user fills out the event request form.
- The user provides the required event information, including the date, location, number of participants, budget, and requested services.
- The user chooses the colours and atmosphere of the event.
- The user can even write a note if they want to.
- The application validates the form.
- In case the form is invalid and the necessary details are not provided, then the application shows validation errors.
- When the form is validated successfully and the date of the event is chosen, the application generates the event request.
- The event request is stored in the database.
- The application shows a success message.
- Later, the user is forwarded to the projects page.

4.3.4 Quote Response Workflow Activity Diagram

The activity diagram of "Quote Response Workflow" is presented on the next page, and its description is provided below.

- The customer opens the project details page.
- The customer views the quotation information.
- The system waits for the customer's decision.
- When the customer accepts the quotation, the system shows the terms and conditions.
- The system verifies if the terms are accepted by the user.
- When the terms are not accepted, the acceptance process is stopped.
- When the terms are accepted, the system marks the request status as accepted.
- The system creates the purchase order in PDF format.
- The system shows the purchase order to the user.

- The customer can then download or share the purchase order.
- When the user rejects the quotation, the system confirms the rejection of the quotation.
- The system changes the request status to rejected.
- Finally, the system shows a rejection confirmation message.

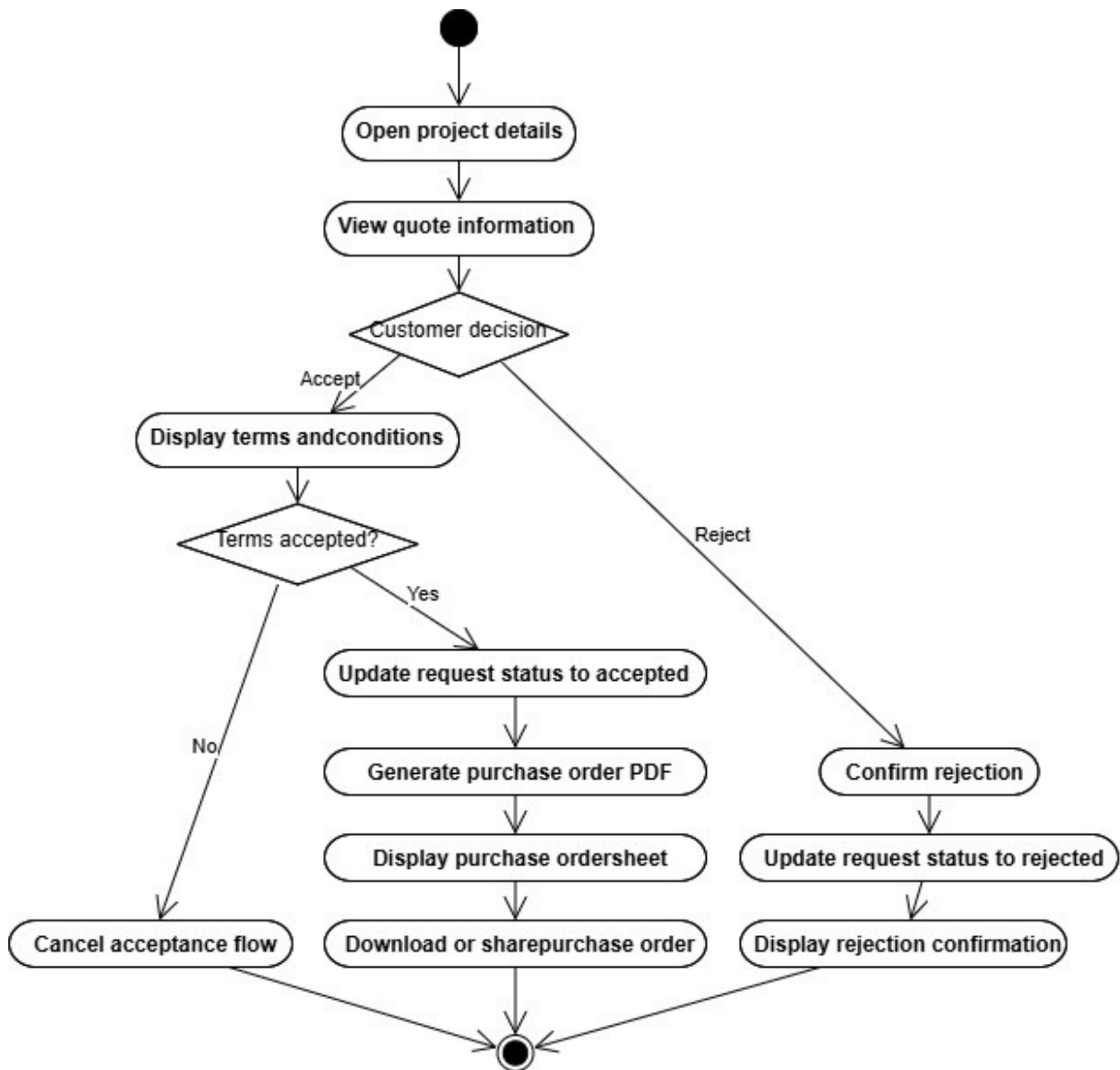


Figure 4.10: Quote Response Workflow Activity Diagram

4.3.5 Admin Request Management Workflow Activity Diagram

The activity diagram of "Admin Request Management Workflow" is presented on the next page, and its description is provided below.

- The administrator opens the admin requests page.
- The application displays all customer requests.
- The administrator clicks on one of the customer requests.
- The selected request details are opened.
- The administrator decides which action to take next.
- In case the administrator opts for reviewing action, then the request status is set to reviewing.
- If the administrator decides to send a quotation, then he/she adds the supplier price, quoted price, and a note.
- The request status is then changed to quoted.
- If the administrator decides to start production, the request status is updated to in production.
- If the administrator decides to mark the request as delivered, then the request status is changed to delivered.
- If the administrator decides to cancel the request, then the request status is changed to cancelled.
- The request status is finally saved by the system.

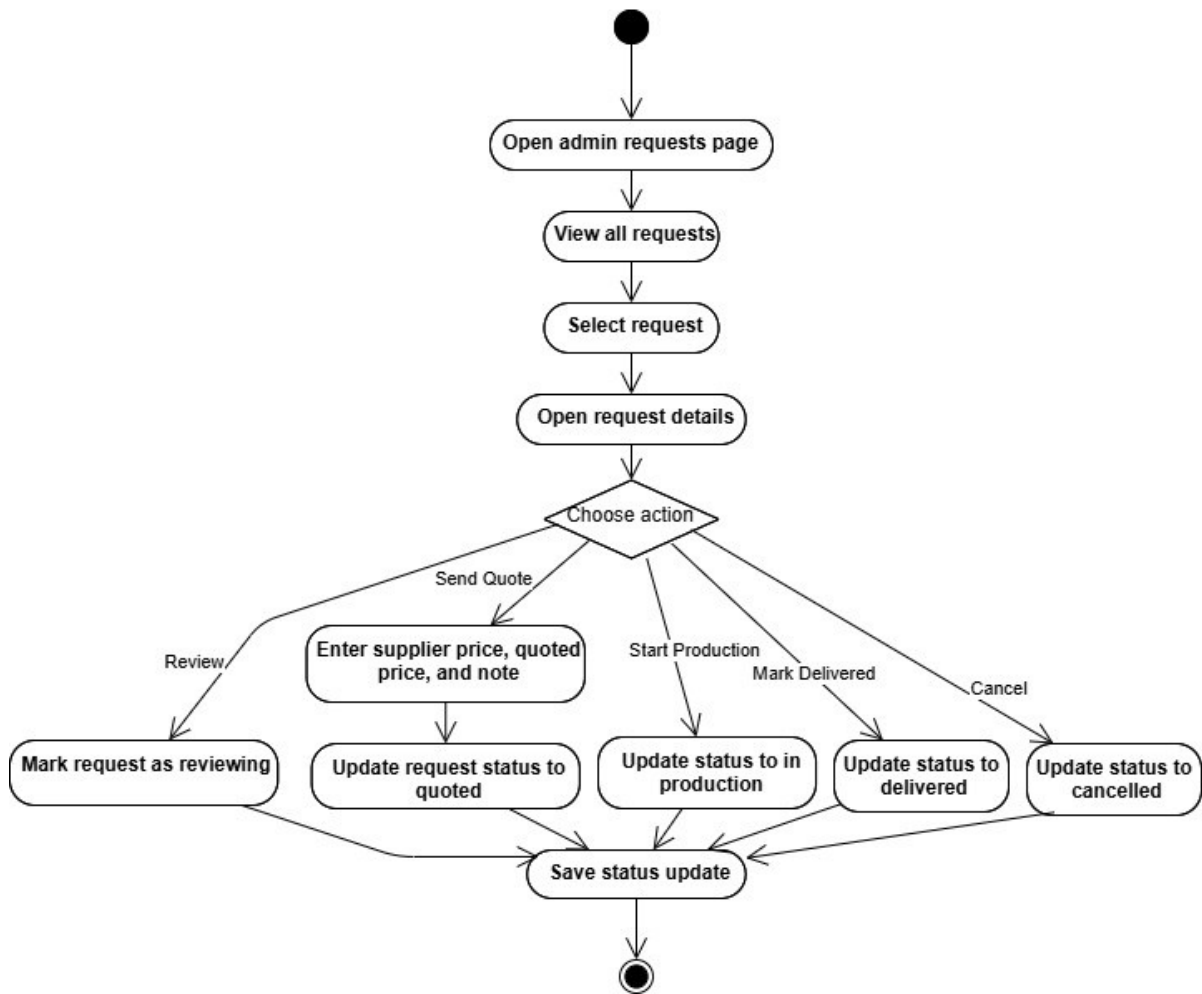


Figure 4.11: Admin Request Management Workflow Activity Diagram

4.3.6 Project Messaging Workflow Activity Diagram

Below are the activity diagram of "Project Messaging Workflow" and its description.

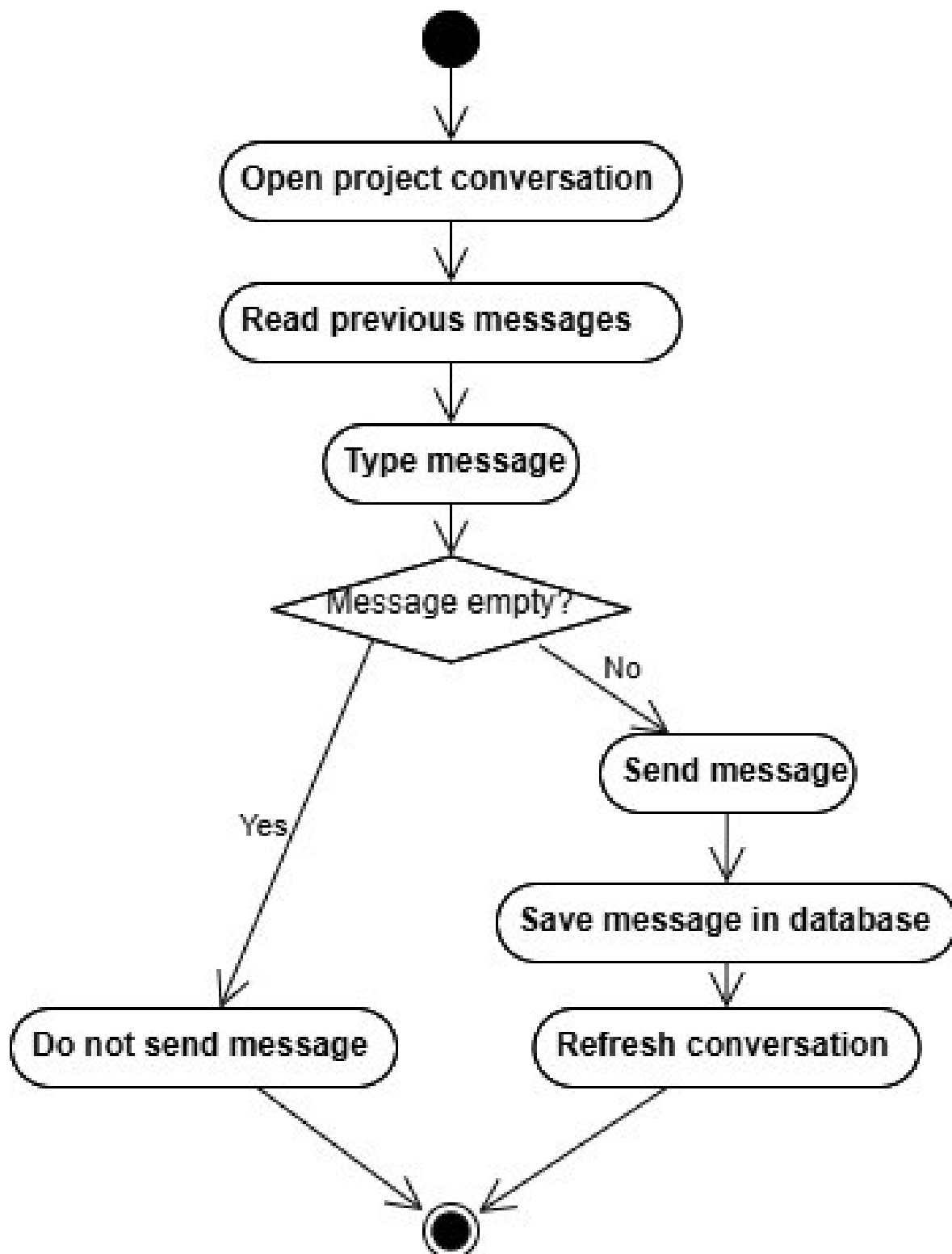


Figure 4.12: Project Messaging Workflow Activity Diagram

- The user begins the conversation for the project.
- The application shows the old messages.

- The user enters his/her message.
- The system examines if the message is blank.
- If the message is blank, it won't be transmitted.
- In case that the message is not blank, it will be sent out.
- Next, the message gets stored in the database.
- At last, the conversation gets refreshed.

Conclusion

This chapter walked through the class diagram, sequence diagrams, and activity diagrams of the application. The following chapter shifts focus to the execution.

Chapter 5

Execution

Introduction

In this chapter, we discuss the interfaces used in the INWIN application and the functionalities implemented in it. It provides an overview of the different user interfaces, explains their role within the system, and highlights the backend services and processes that support their operation.

5.1 Authentication Module

This section describes the authentication characteristics of the INWIN application. It covers the login and registration interface, email verification and other methods that can be applied to manage users' access.

5.1.1 Registration and Login Interfaces

The INWIN application uses a self-registration model, where users create their own accounts by providing their personal information and login credentials. After registration, the user logs in and is taken to the relevant part of the system depending on their position.

5.1.1.1 Registration Interface

The figure below represents the registration pages.

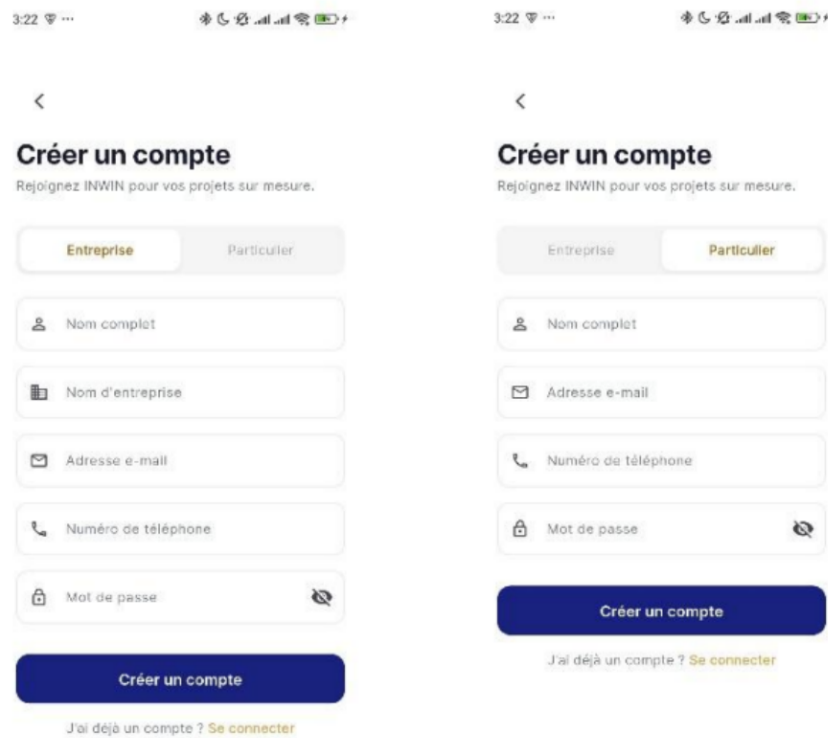


Figure 5.1: Registration Interface

The Registration Interface serves as the first phase of the authentication process in the INWIN application. Here, users can register themselves by providing their personal information and the credentials that will be used to login to the system. In this phase, users select their client type and fill out all the required fields before submitting the registration form.

5.1.1.2 Login Interface

The login screen is the starting point that will enable registered users to log into the INWIN software. The login screen comprises a very straightforward form where users enter their email address and password. Once they have been authenticated, the system gets their profiles and then redirects them to the respective part of the application depending on their role. Therefore, the login screen plays a critical role in ensuring secure access to the application. The diagram below shows the login screen.

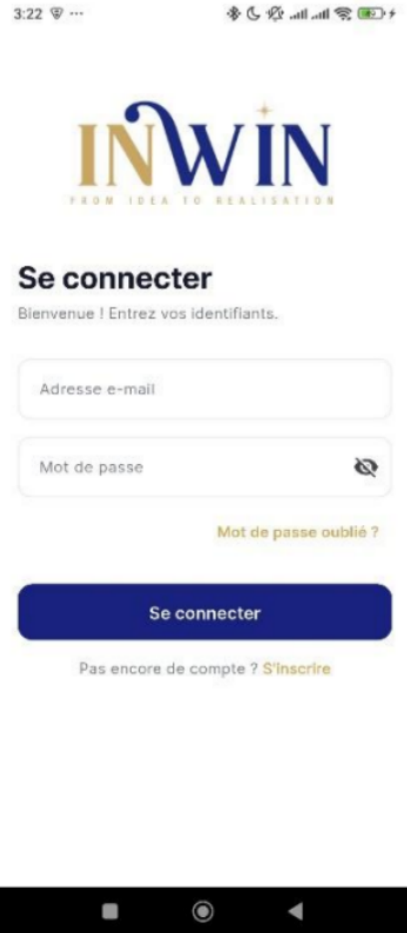


Figure 5.2: Login interface

5.1.1.3 Logout

The logout function helps the authenticated user to terminate their session securely. In the INWIN application, this feature can be accessed via the user interface and will allow the user to log out from the application at any point in time. After the logout process has been completed, the user's session will be terminated, and they will be directed back to the authentication process. This feature enhances the security of the application since unauthorized access will not be possible after terminating the user's session.

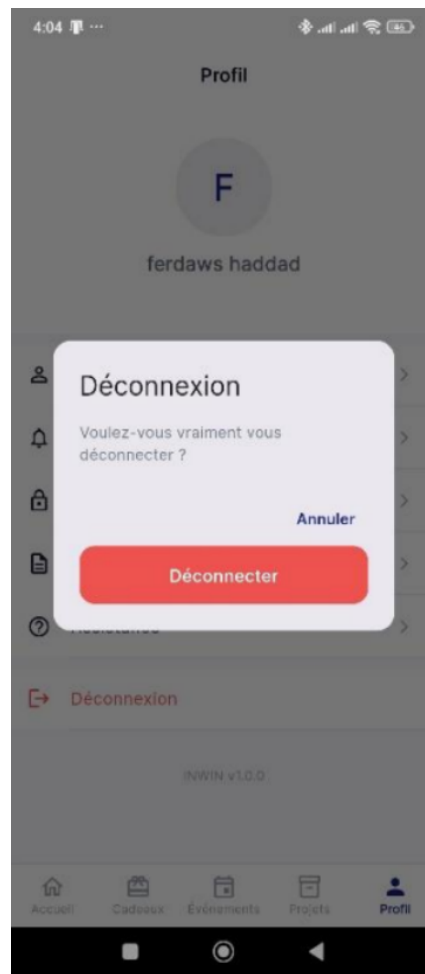


Figure 5.3: Logout Button

5.1.2 Email Verification

The figure below represents the email verification interface.

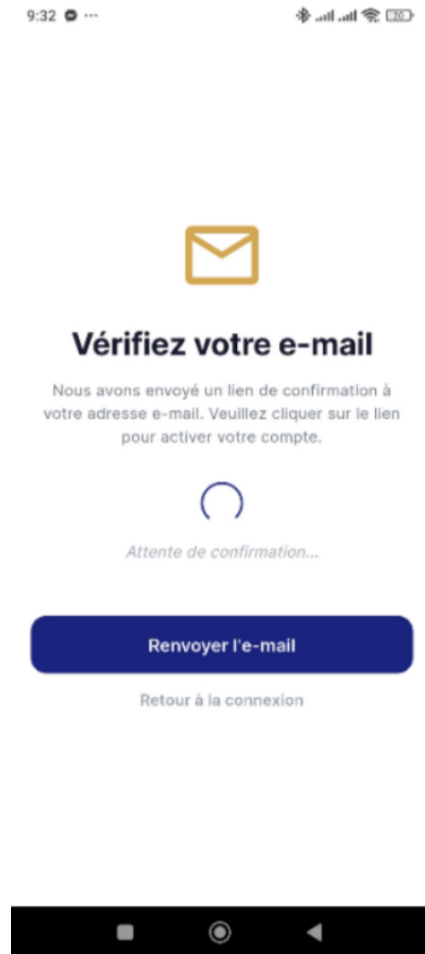


Figure 5.4: Email Verification Interface

The email verification interface is an important part of the authentication procedure within the INWIN application. After creating an account, the user needs to verify their email address in order to have full access to the customer zone of the application. Such a function is helpful for improving account security and ensuring that only authorized users have access to restricted features.

The verification screen displays information about the sent email and suggests to the user that they should confirm their account using the link contained in that email. In addition, the user is provided with the opportunity to resend the verification email and return to the login page. Upon confirming the email address, the user gains access to the main interface of the application.

5.1.3 Authentication Security and Access Logic

The authentication security of the INWIN application is guaranteed by using Firebase authentication, email verification, and routing protection. After the user logs into the system, their profile is fetched, and the role assigned to them is identified. They are then redirected to the corresponding interface. Access control is strengthened by limit-

ing access to protected routes to only authenticated users, while customers need to go through email verification before accessing the application. Code examples can be found in Annexes 1 to 4.

5.2 Customer Features

This section presents the primary functions accessible to users within the INWIN application. It highlights the user interfaces and interactions that allow customers to browse services, submit requests, track project progress, communicate with the administrator, and manage their profile information.

5.2.1 Home Interface

This section highlights the home interface of the INWIN application which acts as the main portal for users upon authentication. This enables easy navigation to the key features of the application. In addition, it gives the user an idea of the various services offered by the system. The interface has been designed in a manner that makes it easy to navigate through the system. It is mainly composed of the following parts: • **Navigation Area:** allows the user to access the main sections of the application, such as home, gifts, events, projects, profile, and notifications; • **Service Shortcuts:** provide direct access to important customer actions, including browsing the gifts catalogue and exploring the events section; • **General Interface Elements:** ensure a smooth and intuitive user experience by structuring the content of the screen clearly. The figure next page represents the main home interface of the INWIN application.

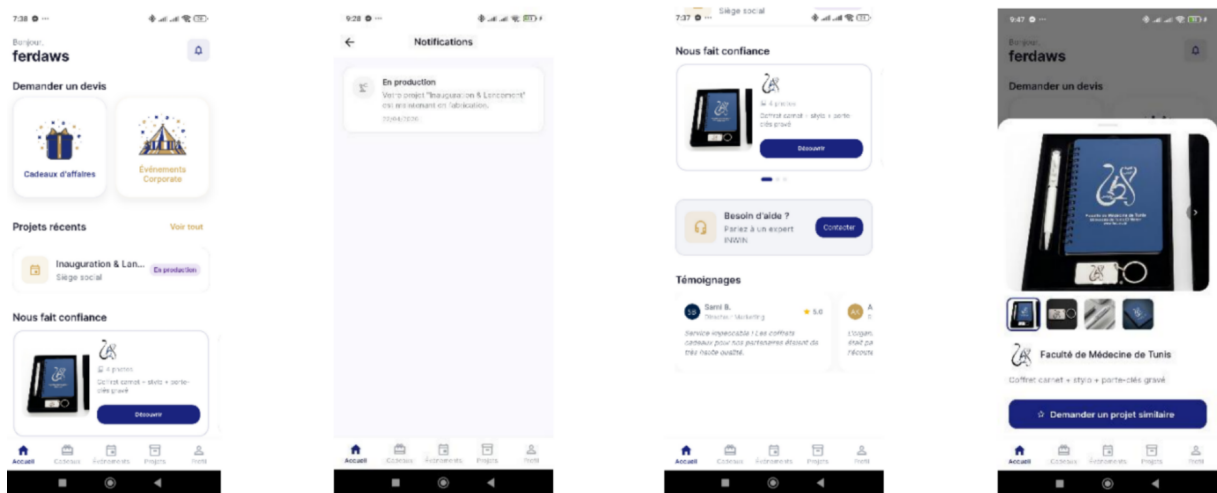


Figure 5.5: Home Interface

5.2.2 Gifts Catalogue Interface

This section presents the gifts catalogue interface of the INWIN application. It allows business customers to browse the different gift categories offered by the platform and to access the corresponding request configuration screens. The catalogue is designed to provide a clear and organised presentation of the available products, making it easier for the user to select the most suitable category before submitting a gift request.

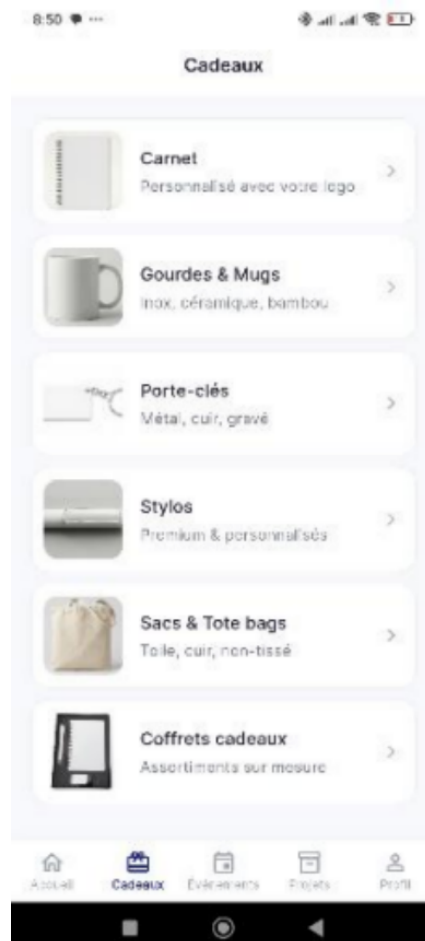


Figure 5.6: Gift Catalogue Interface

The interface shown above allows business customers to browse the different gift categories offered by the platform. Each category is presented with a title, a short description, and a representative image, enabling the user to quickly identify the appropriate product type before accessing the corresponding gift request configuration interface.

5.2.3 Gift Request Interface

This section presents the gift request interface of the INWIN application. It allows business customers to configure and submit a personalised gift request by specifying the product characteristics, uploading the required file, and providing additional information

related to the request. Depending on the uploaded file type, the interface may also provide an interactive logo positioning feature. The figure below represents the initial state of the gift request interface before a file is uploaded



Figure 5.7:

The following figure illustrates the gift request interface after an image file has been uploaded, enabling the logo positioning functionality.

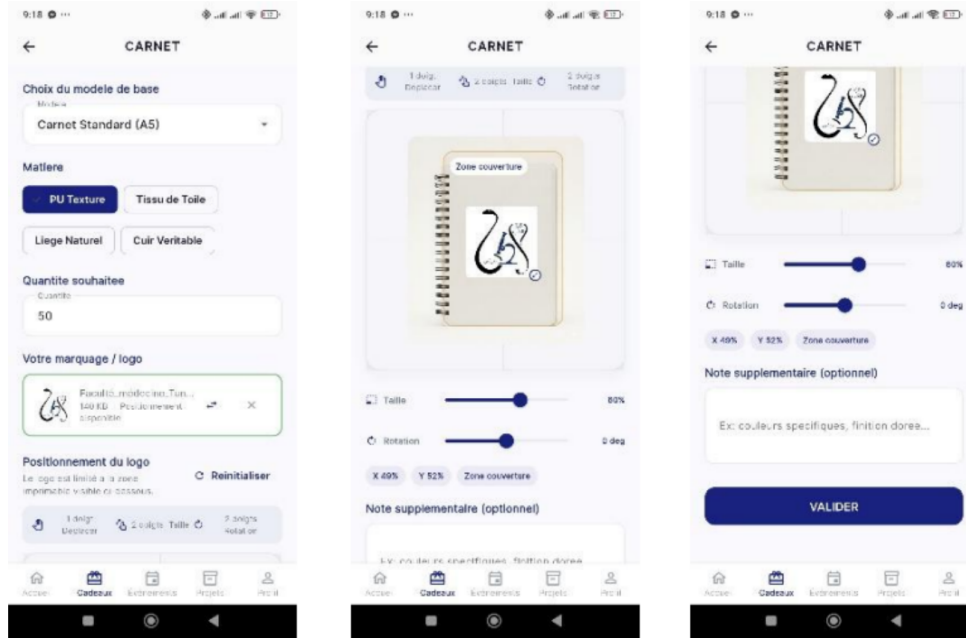


Figure 5.8: Gift Request Interface with Logo Positioning

The interface shown above is supported by the request submission process implemented in the application. When the customer submits the form, the uploaded file is first stored in Firebase Storage, after which the request data is structured and saved in Cloud Firestore as a new request entry. This process ensures that both the request information and the related attachment are properly associated and stored for later processing by the administrator.

5.2.4 Event Catalogue Interface

This section presents the events catalogue interface of the INWIN application. It allows users to browse the different event categories proposed by the platform before accessing the corresponding request form. The interface is designed to provide a clear overview of the available event types and to help the user select the category that best matches their needs. The figure below represents the events catalogue interface of the INWIN application.



Figure 5.9: Event Catalogue Interface

5.2.5 Event Request Interface

This section presents the event request interface of the INWIN application. It allows customers to submit event-related requests by providing the information required for the preparation and organisation of the event. Through this interface, the user can define the main characteristics of the request, including the event type, location, date, number of participants, budget, required services, preferred colours, ambiance, and an optional note. The figure next page represents the event request interface of the INWIN application.

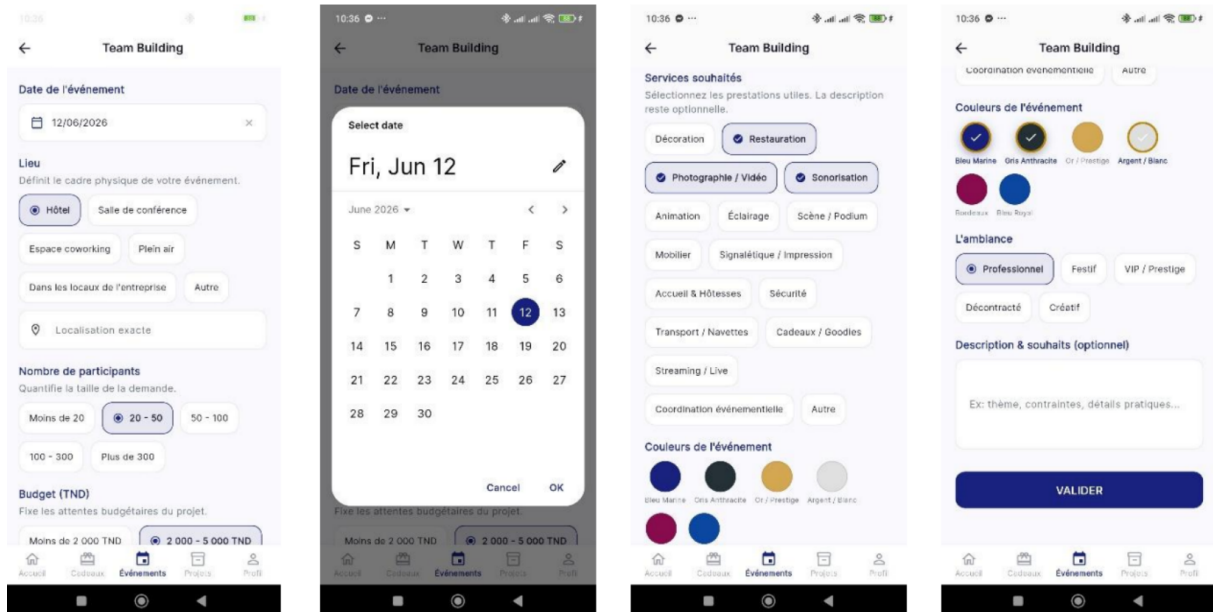


Figure 5.10: Event Request Interface

The interface shown above is supported by the event request creation process implemented in the application. When the customer submits the form, the system validates the entered information, structures the request data, and stores it in Cloud Firestore as a new request entry. This process ensures that the submitted event details are properly recorded and made available for later processing by the administrator.

5.2.6 Project Tracking and Messaging Interfaces

This section presents the project tracking and messaging interface of the INWIN application. It allows customers to follow the progress of their submitted requests and to communicate directly with the administrator within the same project space. Through this interface, the user can consult request details, view status updates, access quotation information, and exchange messages related to the project. It is mainly composed of the following parts: • **Project Details Area:** displays the main information related to the selected request; • **Status Tracking Area:** shows the progress of the request through its different stages; • **Quotation Area:** allows the customer to consult and respond to the quotation when available; • **Messaging Area:** allows the customer to read previous messages and send new ones to the administrator. The following figure represents the project tracking and messaging interface of the INWIN application.

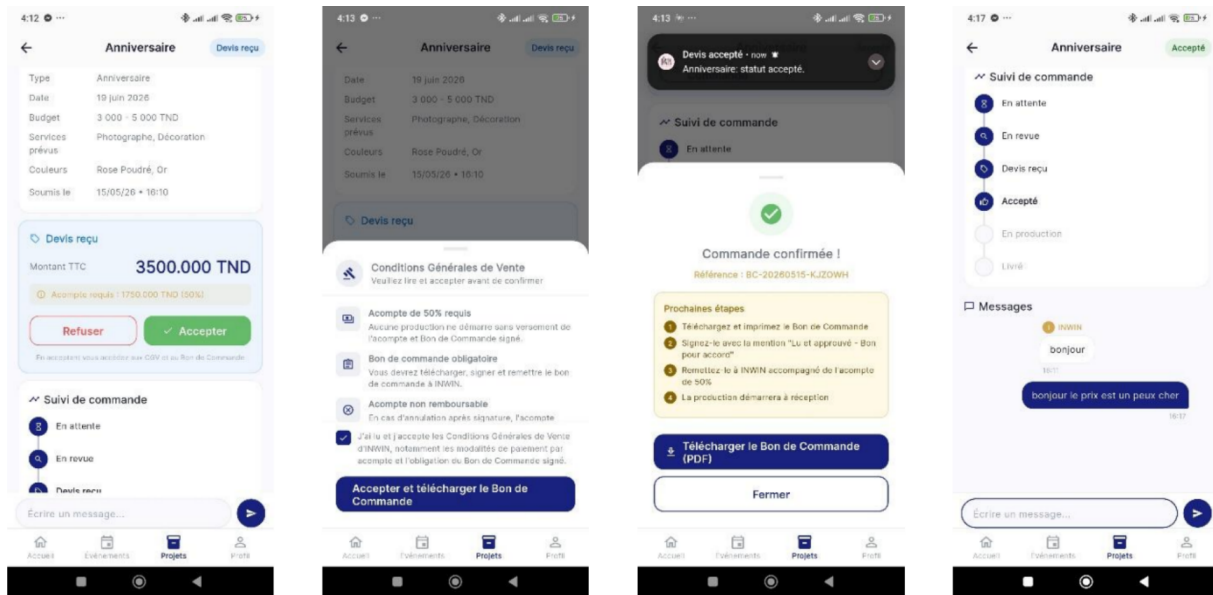


Figure 5.11: Project Tracking Interface

This interface, which is seen above, is facilitated by both the quotation response process and messaging process that have been integrated into the application. When there is a quotation provided, the customer can either accept or reject the same, and then the status of the request will be updated accordingly. Any messages sent via the conversation area will be saved in Cloud Firestore, making it accessible by both the client and administrator. The following figure illustrates the purchase order generated after the customer accepts the quotation.

CLIENT

ferdaws
Particulier

ferdawshaddad1@outlook.com
+21653438654

COMMANDE

Référence : BC-20260515-KJZOWH
Date : 15/05/2026
Type : Événement
Statut : Devis accepté

DÉTAIL DE LA COMMANDE

Désignation	Détail
Type d'événement	Anniversaire
Date de l'événement	19/06/2026
Lieu	Salle
Budget	3 000 - 5 000 TND
Services	Photographe, Décoration
Ambiance	Chaleureux
Couleurs Thème	Rose Poudré, Or

Montant HT 2941.176 TND

TVA (19%) 558.824 TND

TOTAL TTC 3500.000 TND

Acompte dû (50%) 1750.000 TND

Solde à la livraison (50%) 1750.000 TND

MODALITÉS DE PAIEMENT

Paiement en espèces (remise en main propre)

L'acompte de 50% est exigé avant tout démarrage de production. Aucune commande ne sera traitée sans réception du présent bon de commande signé accompagné de l'acompte.

Figure 5.12: Purchase Order Generated After Quotation Acceptance

5.2.7 Profile Management Interface

This section presents the profile management interface of the INWIN application. It allows users to view and manage their personal account information, access account-related options, and perform actions related to session control. Through this interface, the user can update profile details, request a password reset, consult additional application pages, and log out securely. It is mainly composed of the following parts: • **Profile Information Area:** displays the user's main personal information, such as name and account type; • **Account Management Area:** allows the user to modify profile details and request password reset; • **Navigation and Support Area:** provides access to

additional pages such as terms and conditions and assistance; • **Logout Option:** allows the user to securely end their session. The figure below represents the profile management interface of the INWIN application.

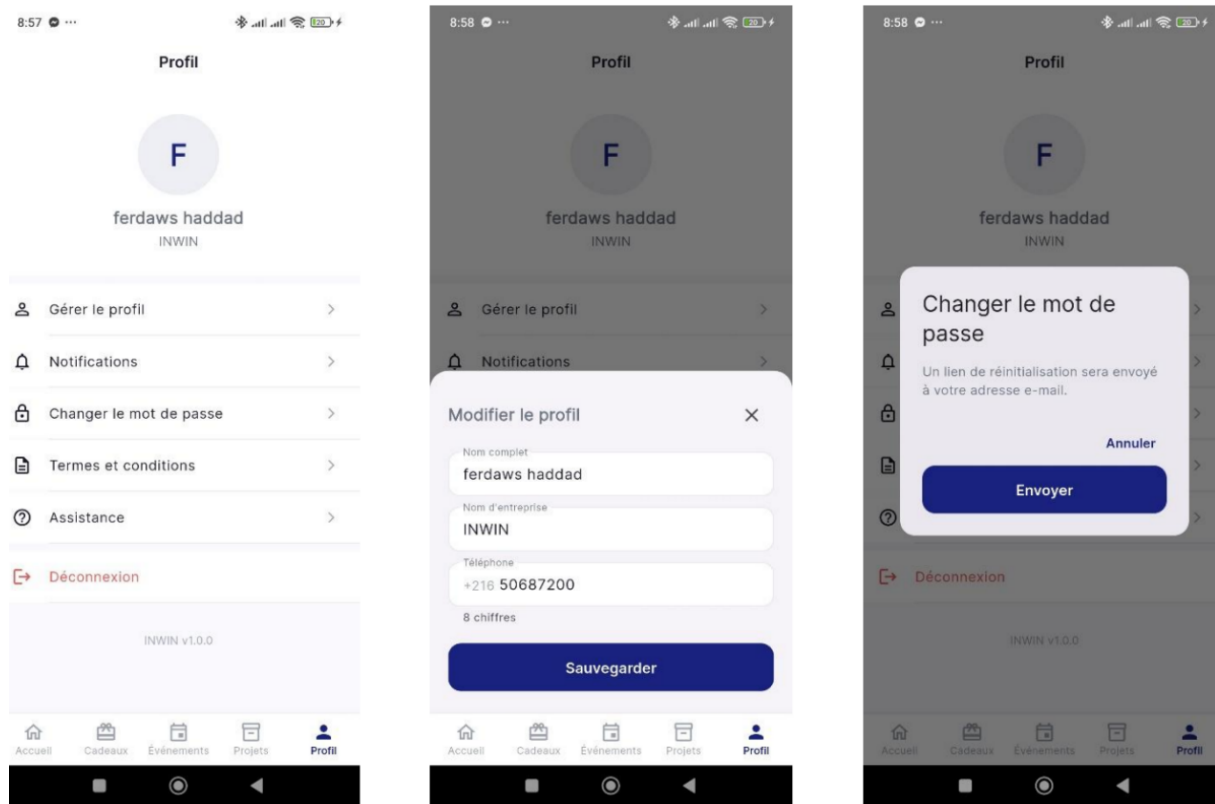


Figure 5.13: Profile Management Interface

5.3 Administrator Features

This section presents the different functions of the INWIN application which are at the disposal of the administrator. It highlights the different interfaces that the administrator uses to manage the requests, quotations, project statuses, and communication with the customers.

5.3.1 Main Interface

This section presents the requests dashboard of the INWIN application. It acts as the primary administration dashboard from where the administrator can get an insight into all the requests that have been made by customers via the application. The dashboard gives a global perspective on the requests and makes it easy to manage them. It is mainly composed of the following parts: • **Requests Overview Area:** displays the list of submitted customer requests; • **Filtering Area:** allows the administrator to

filter requests according to specific criteria such as status, type, or client category; • **Navigation Function:** allows the administrator to open the detail interface of a selected request for further processing. The figure below represents the requests dashboard of the INWIN application.

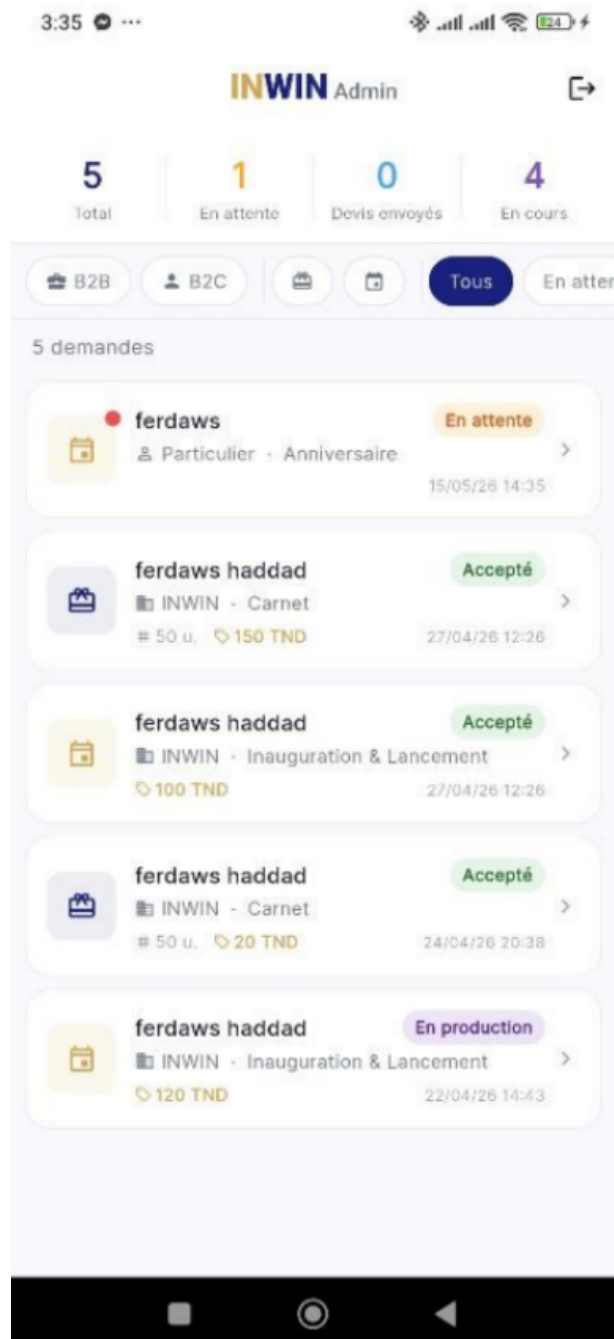


Figure 5.14: Admin Main Interface

5.3.2 Details Interface

This section presents the request detail management interface of the INWIN application. It enables the administrator to look at all the details of the customer request

selected and view all the key steps necessary for handling the request. Through this interface, the administrator will be able to view request details, check the current status of the request, and plan for the next step. It is mainly composed of the following parts:

- **Request Information Area:** displays the complete details of the selected customer request.
- **Status Monitoring Area:** shows the current status of the request within its processing lifecycle.
- **Action Access Area:** provides access to the main management actions available to the administrator.
- **Conversation Area:** displays the messages exchanged between the administrator and the customer.

The figure below represents the request detail management interface of the INWIN application.

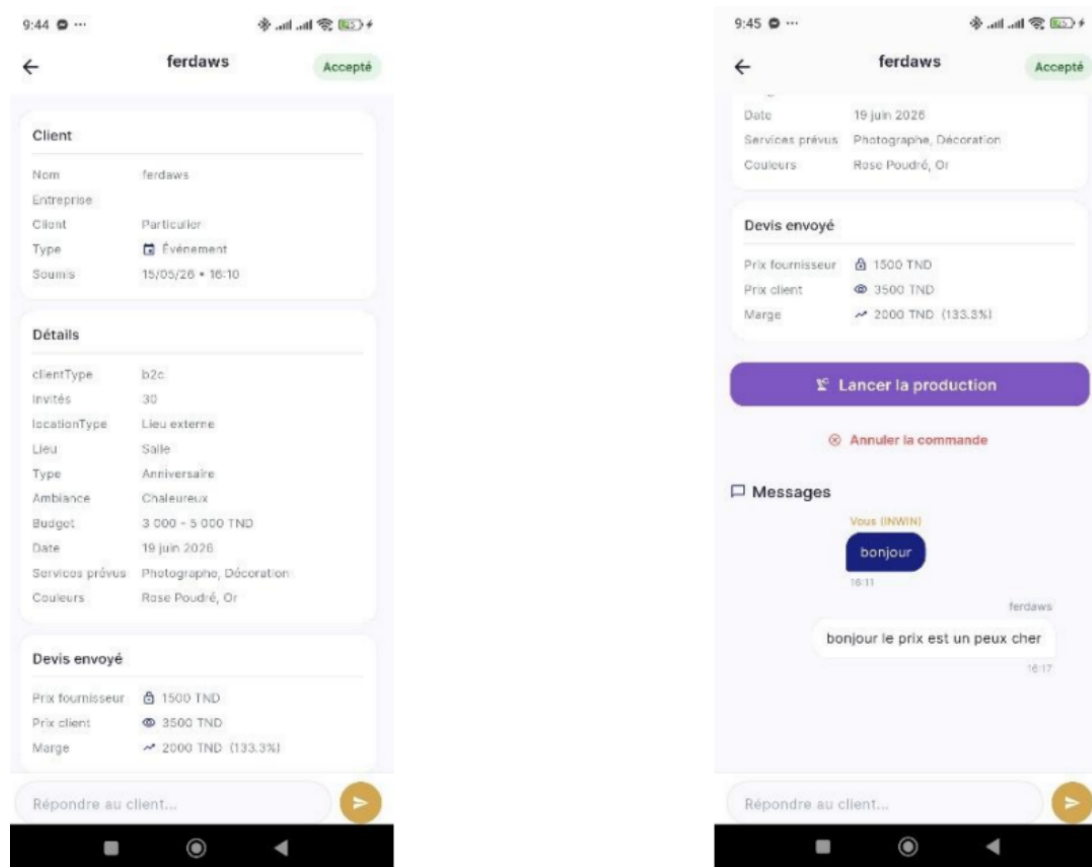


Figure 5.15: Request Detail Management Interface

5.3.3 Quote Preparation and Status Updates

This section presents the functionalities that allow the administrator to prepare quotations and manage the evolution of customer requests. Through the request detail interface, the administrator can input the data regarding quotations, keep track of the request status based on its development, and follow up on every project. The figure below represents the quotation and status management interface of the application.

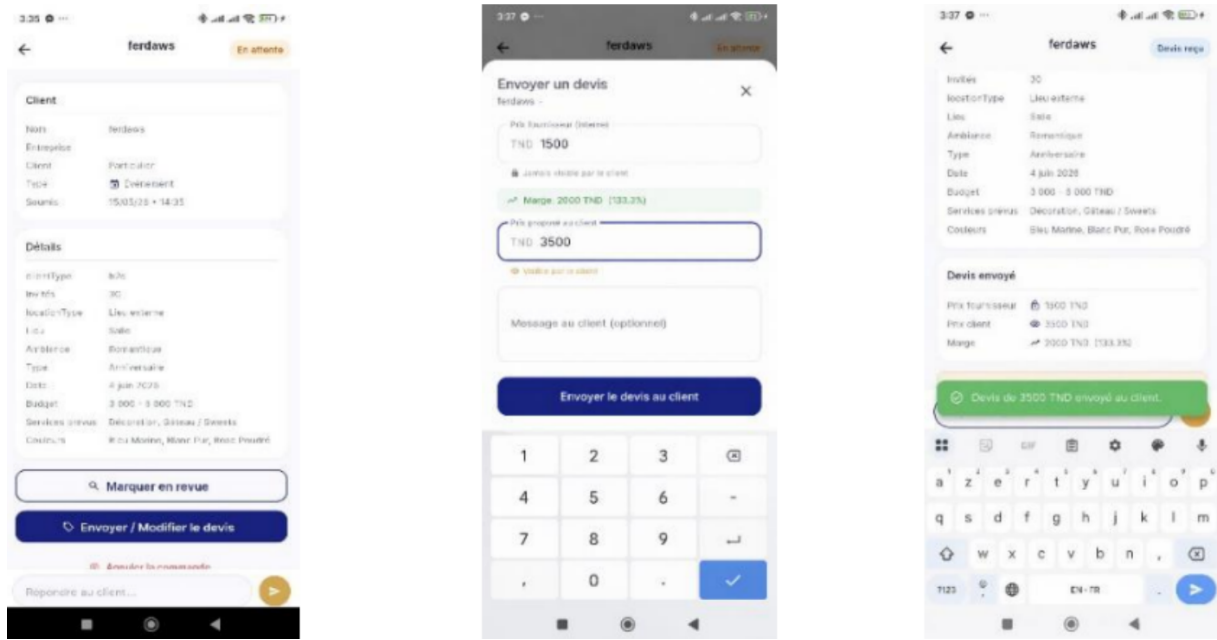


Figure 5.16: Quotation Preparation Interface

5.4 Backend and Database Integration

This section presents the main backend services integrated into the INWIN application, highlighting the role of Firebase in authentication, data storage, file management, and notifications.

5.4.1 Firebase Authentication

Firebase Authentication was integrated into the INWIN application to manage account creation, login, email verification, password reset, and secure session handling. The figure below illustrates the Firebase Authentication console used in the INWIN application to manage registered users and support the authentication process.

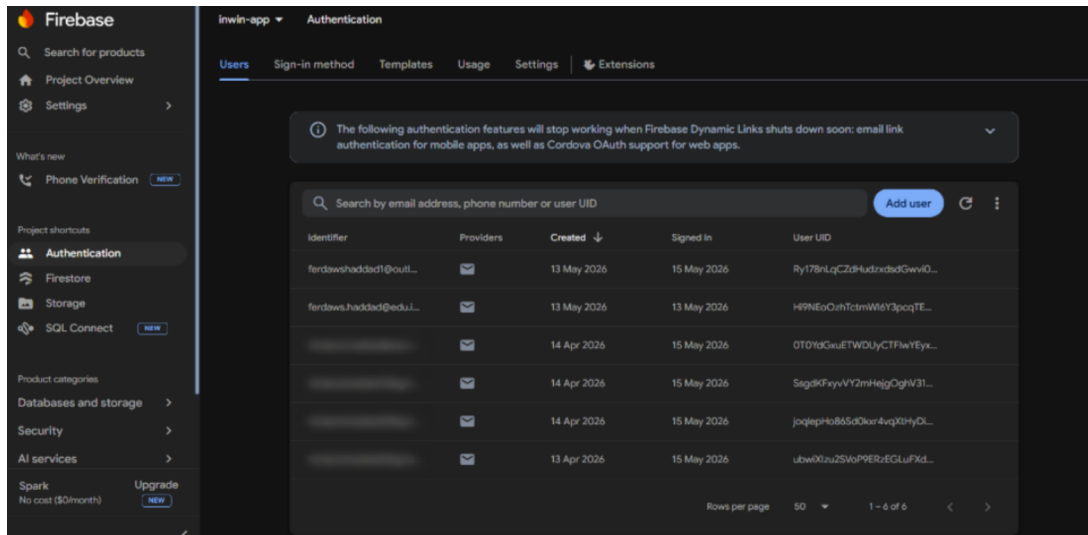


Figure 5.17: Firebase Authentication Console Containing Registered Users

5.4.2 Cloud Firestore

The integration of Cloud Firestore within the INWIN application was done for purposes of making it the major database management software for storing structured data within the application. This has been instrumental in enabling the storage of user profiles, customer requests, projects, and messages. The figure below illustrates the Cloud Firestore database structure used in the INWIN application.

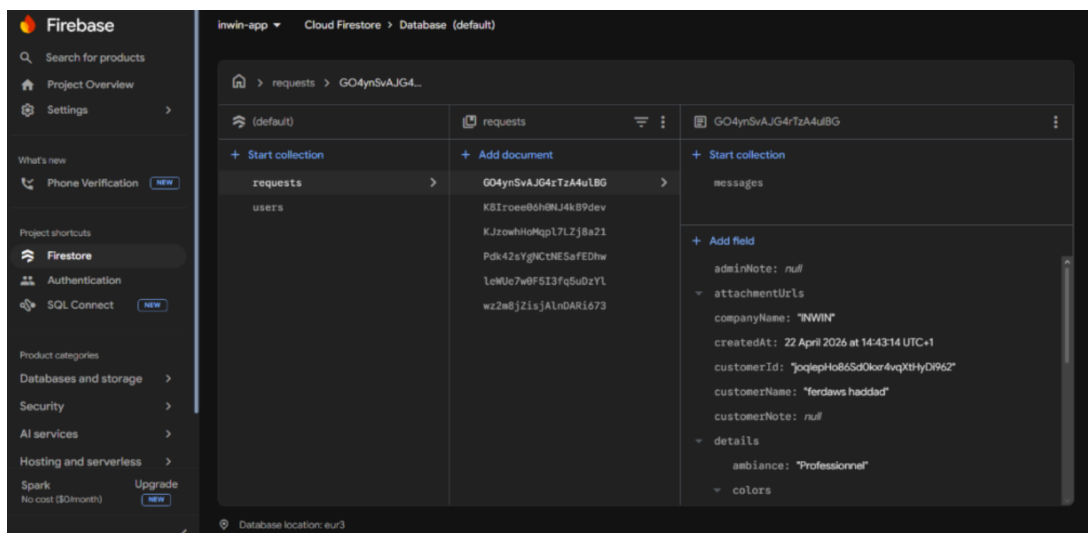


Figure 5.18: Cloud Firestore Database Structure

5.4.3 Firebase Storage

The integration of Cloud Firestore within the INWIN application was done for purposes of making it the major database management software for storing structured

data within the application. This has been instrumental in enabling the storage of user profiles, customer requests, projects, and messages. The following code snippet illustrates how the uploaded file is sent to the storage service during the gift request submission process.

```
try {
    final user = ref.read(currentUserProvider).valueOrNull!;
    final url = await ref.read(storageServiceProvider).uploadFile(
        file: File(_file!.path!),
        folder: 'markings/${user.uid}',
        onProgress: (progress) {
            if (!mounted) return;
            setState(() => _uploadProgress = progress);
        },
    );
}
```

Figure 5.19: File Upload Through the Storage Service

The following code snippet shows how the returned file URL is attached to the request data before the request is saved in the database.

```
final req = QuoteRequest(
    id: '',
    customerId: user.uid,
    customerName: user.fullName,
    companyName: user.companyName,
    type: RequestType.gift,
    status: RequestStatus.pending,
    details: details,
    attachmentUrls: [url],
    createdAt: DateTime.now(),
    updatedAt: DateTime.now(),
); // QuoteRequest
```

Figure 5.20: Attachment URL Integration in Request Creation

5.4.4 Firebase Cloud Messaging and Cloud Functions

Firebase Cloud Messaging and Cloud Functions were integrated into the INWIN application to support notification delivery and automated backend event handling, particularly for request status updates and project messages. The following figure illustrates the beginning of the Cloud Function that detects a change in request status and retrieves the customer notification token.

```
export const onRequestUpdated = functions
  .region("europe-west1")
  .firestore.document("requests/{requestId}")
  .onUpdate(async (change) => {
    const before = change.before.data();
    const after = change.after.data();

    if (before.status === after.status) return;

    const customerId: string = after.customerId;
    const customerToken = await getCustomerToken(customerId);
    if (!customerToken) return;
```

Figure 5.21: Detection of Request Status Change

The following figure illustrates the part of the Cloud Function that sends the notification after the new request status has been identified.

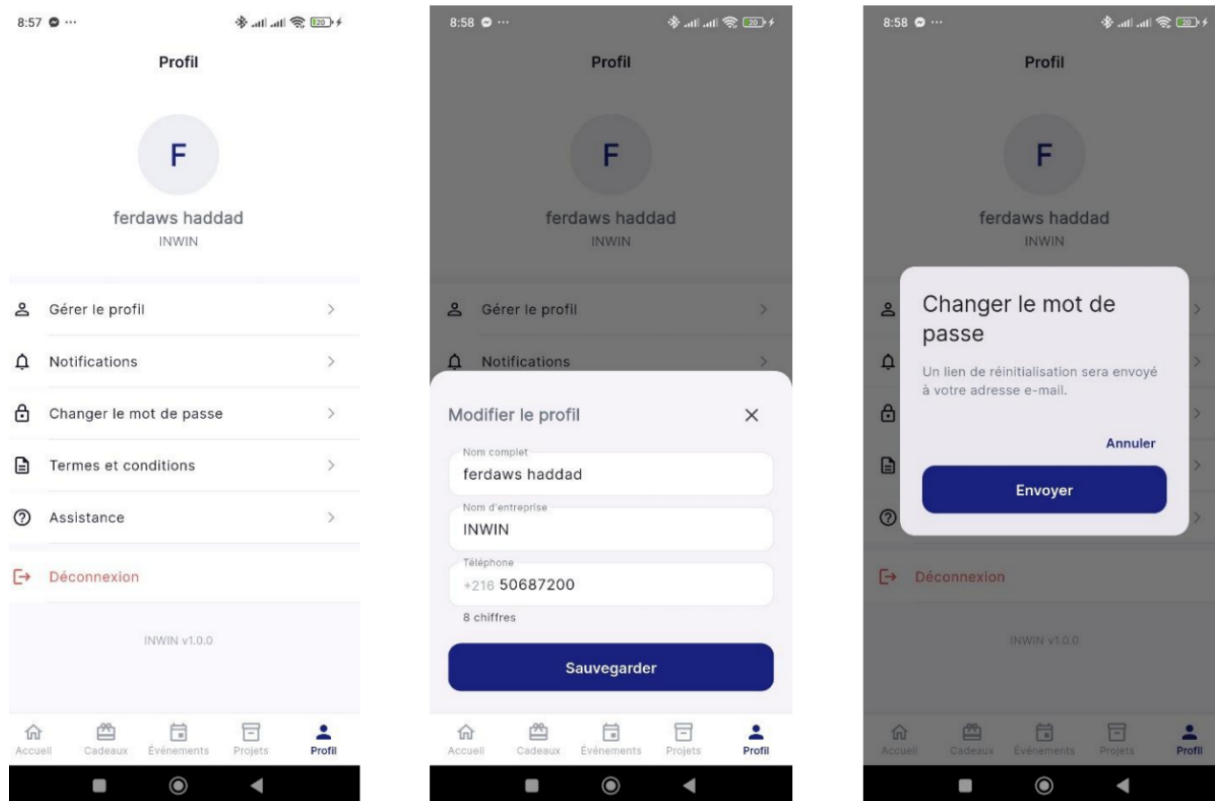


Figure 5.22: Sending a Request Status Notification

5.5 Conclusion

This chapter presented a detailed overview of the main interfaces of the INWIN application, highlighting their roles within the system, accompanied by short explanations on the backend processes that enable their functionality.

General Conclusion

Digital technologies are continuously transforming the way individuals and businesses organize events and manage corporate services, and INWIN aims to contribute effectively to this evolution. This report explored how a centralized mobile platform can simplify event organization, improve communication between clients and service providers, and facilitate corporate gift customization through a modern and user-friendly digital solution.

The development process presented throughout this report highlights the importance of combining technical implementation, market analysis, user needs, and modern project management approaches in order to design an efficient and innovative application. INWIN was developed with flexibility, scalability, and usability in mind to ensure that it responds to the evolving expectations of both individuals and businesses in the digital era.

Furthermore, the project demonstrated the significance of integrating multiple services within a single platform to improve user experience and simplify management processes. Features such as reservation management, real-time messaging, budget tracking, review systems, and corporate gift customization contribute to creating a complete and centralized ecosystem adapted to modern event management needs.

Looking ahead, several improvements and future enhancements can further strengthen the platform. Future developments may include online payment integration, advanced recommendation systems, AI-based personalization, event analytics, and expanded corporate management functionalities. These additions would help improve efficiency, increase user engagement, and support the continuous evolution of the platform.

This report has demonstrated the practical viability of INWIN as an innovative solution in the Tunisian market. By focusing on accessibility, personalization, communication, and centralized management, the application provides a strong foundation for a

smarter and more modern approach to event organization and corporate gifting services.

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Annexes

Annex 1 : Questionnaire for Customer

Formulaire particuliers

Description du formulaire

Âge ? *

- ☐ 18 – 25 ans
- ☐ 26 – 35 ans
- ☐ 36 – 50 ans
- ☐ Plus de 50 ans

Avez-vous déjà utilisé une application d'organisation d'événements ? *

- ☐ Oui
- ☐ Non

Si non, pourquoi ?

- ☐ Trop compliqué
- ☐ Pas utile
- ☐ Je ne connais pas

À quelle fréquence organisez-vous des événements ? *

- ☐ Rarement
- ☐ 1 à 2 fois par an
- ☐ Plusieurs fois par an

Quels types d'événements vous intéressent ? *

- ☐ Mariage
- ☐ Anniversaire
- ☐ Fête privée
- ☐ Baby shower
- ☐ Autre

Qu'est-ce qui est le plus important pour vous ? *

- ☐ Prix
- ☐ Qualité
- ☐ Créativité
- ☐ Rapidité
- ☐ Réputation

Quel budget consacrez-vous généralement ? *

- ☐ Petit budget
- ☐ Budget moyen
- ☐ Budget élevé

Quelles fonctionnalités souhaitez-vous dans l'application ? *

- ☐ Demande de devis rapide
- ☐ Chat avec organisateur
- ☐ Galerie d'exemples
- ☐ Paiement en ligne

Quelles difficultés avez-vous rencontrées ? *

- ☐ Trouver un lieu
- ☐ Gérer le budget
- ☐ Trouver des prestataires
- ☐ Organisation générale
- ☐ Manque de temps

Quelles difficultés avez-vous rencontrées ? *

- ☐ Trouver un lieu
- ☐ Gérer le budget
- ☐ Trouver des prestataires
- ☐ Organisation générale
- ☐ Manque de temps

Qu'est-ce qui vous encouragerait à utiliser cette application ? *

- ☐ Gain de temps
- ☐ Prix attractif
- ☐ Simplicité
- ☐ Qualité du service

Annex 2 : Questionnaire for Business

Formulaire inwin entreprise

Description du formulaire

Secteur d'activité

Réponse courte

Taille de l'entreprise : *

- ☐ Petite
- ☐ Moyenne
- ☐ Grande

Organisez-vous des événements professionnels ? *

- ☐ Oui régulièrement
- ☐ Occasionnellement
- ☐ Jamais

Types d'événements *

- ☐ Séminaire
- ☐ Conférence
- ☐ Team building
- ☐ Lancement de produit
- ☐ Réunion
- ☐ Autre :

Quels sont vos principaux problèmes ? *

- ☐ Organisation complexe
- ☐ Manque de prestataires fiables
- ☐ Budget difficile à gérer
- ☐ Manque de temps

Utilisez-vous des cadeaux d'affaires ? *

- ☐ Oui
- ☐ Non

Si oui :

- ☐ Goodies personnalisés
- ☐ Produits personnalisés
- ☐ Coffrets personnalisés

Une application vous serait utile pour : *

- ☐ Gérer vos événements
- ☐ Commander des cadeaux personnalisés

Fonctionnalités attendues *

- ☐ Demande de devis rapide
- ☐ Personnalisation produits
- ☐ Support client

Quel budget consacrez-vous ? *

- ☐ Moins de 2 000 TND
- ☐ 2 000 – 5 000 TND
- ☐ Plus de 5 000 TND

Qu'est-ce qui est le plus important ? *

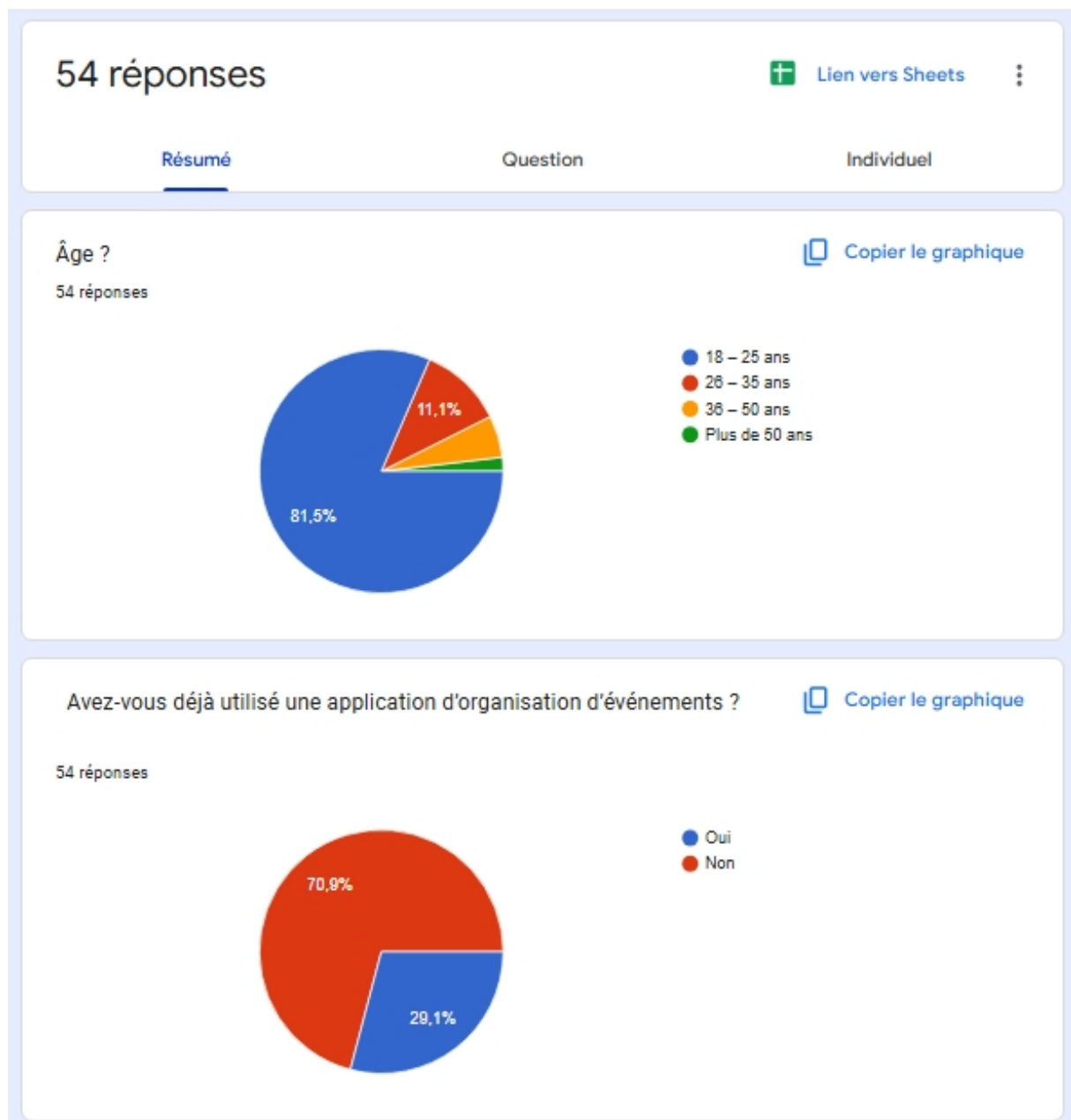
- ☐ Prix
- ☐ Qualité
- ☐ Rapidité
- ☐ Expérience

Vos recommandations pour une application idéale :

Réponse courte

.....

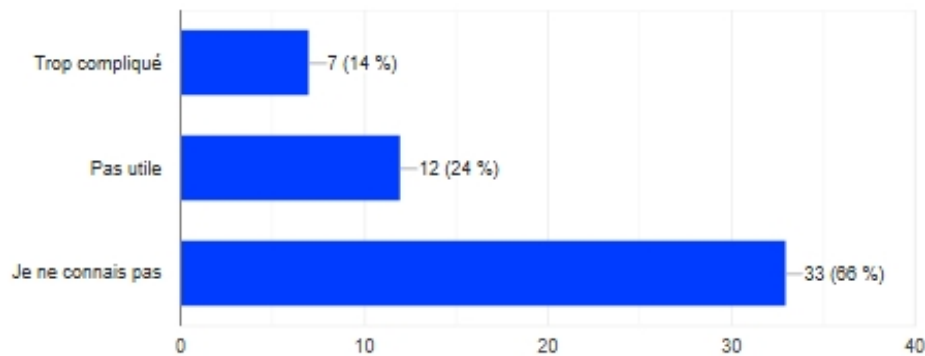
Annex 3: Questionnaire for Customer Responses



Si non, pourquoi ?

 Copier le graphique

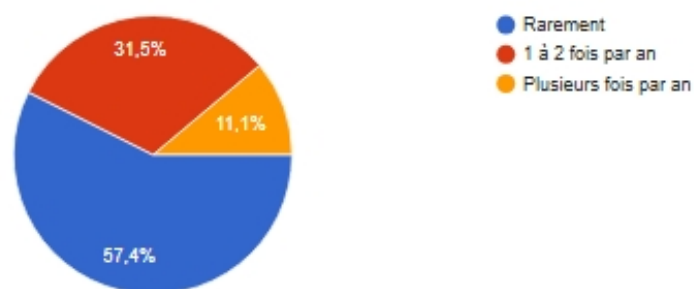
50 réponses



À quelle fréquence organisez-vous des événements ?

 Copier le graphique

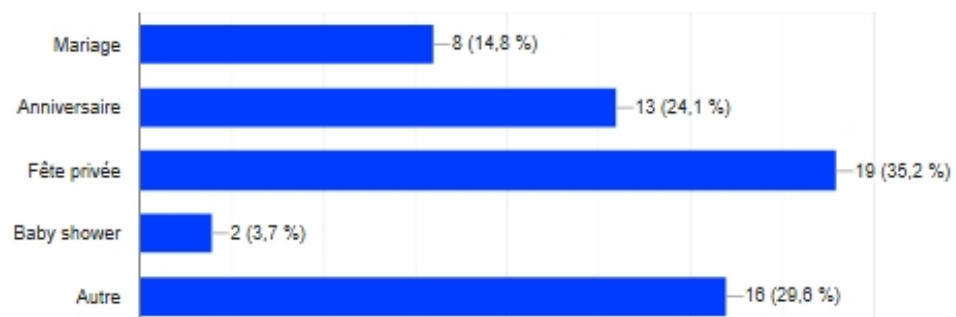
54 réponses



Quels types d'événements vous intéressent ?

 Copier le graphique

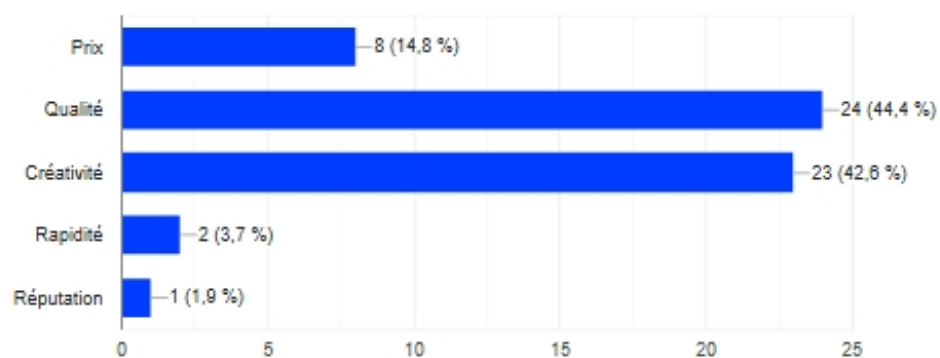
54 réponses



Qu'est-ce qui est le plus important pour vous ?

 Copier le graphique

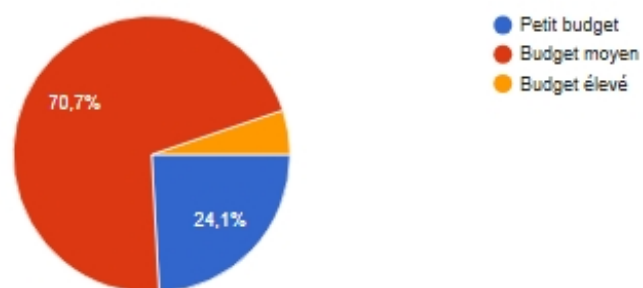
54 réponses



Quel budget consacrez-vous généralement ?

 Copier le graphique

54 réponses



Quelles fonctionnalités souhaitez-vous dans l'application ?

 Copier le graphique

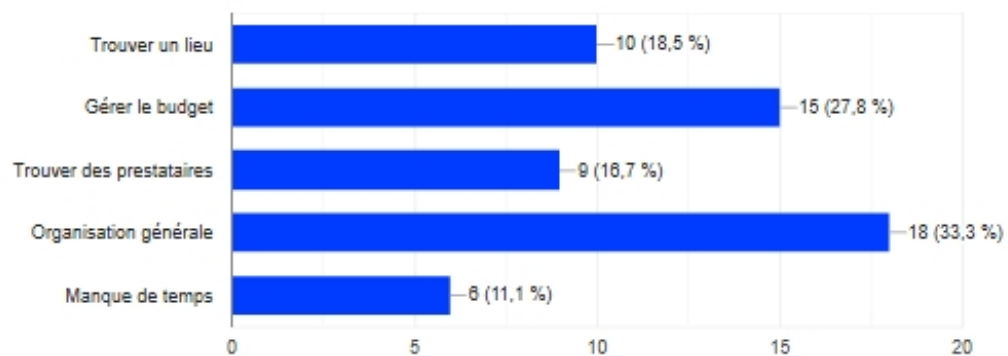
54 réponses



Quelles difficultés avez-vous rencontrées ?

 Copier le graphique

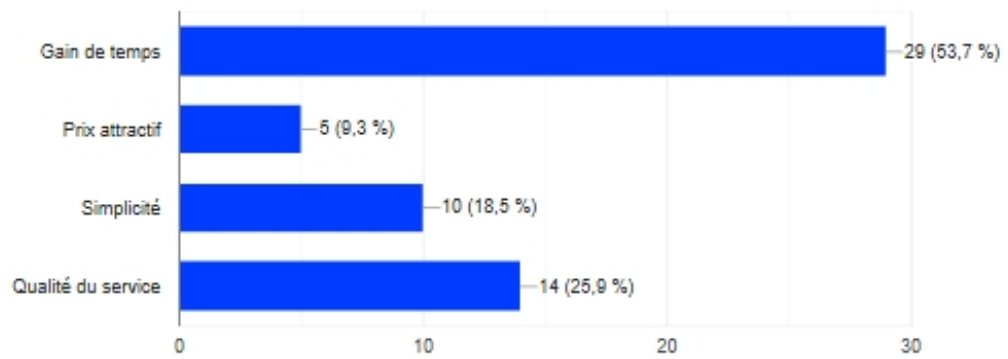
54 réponses



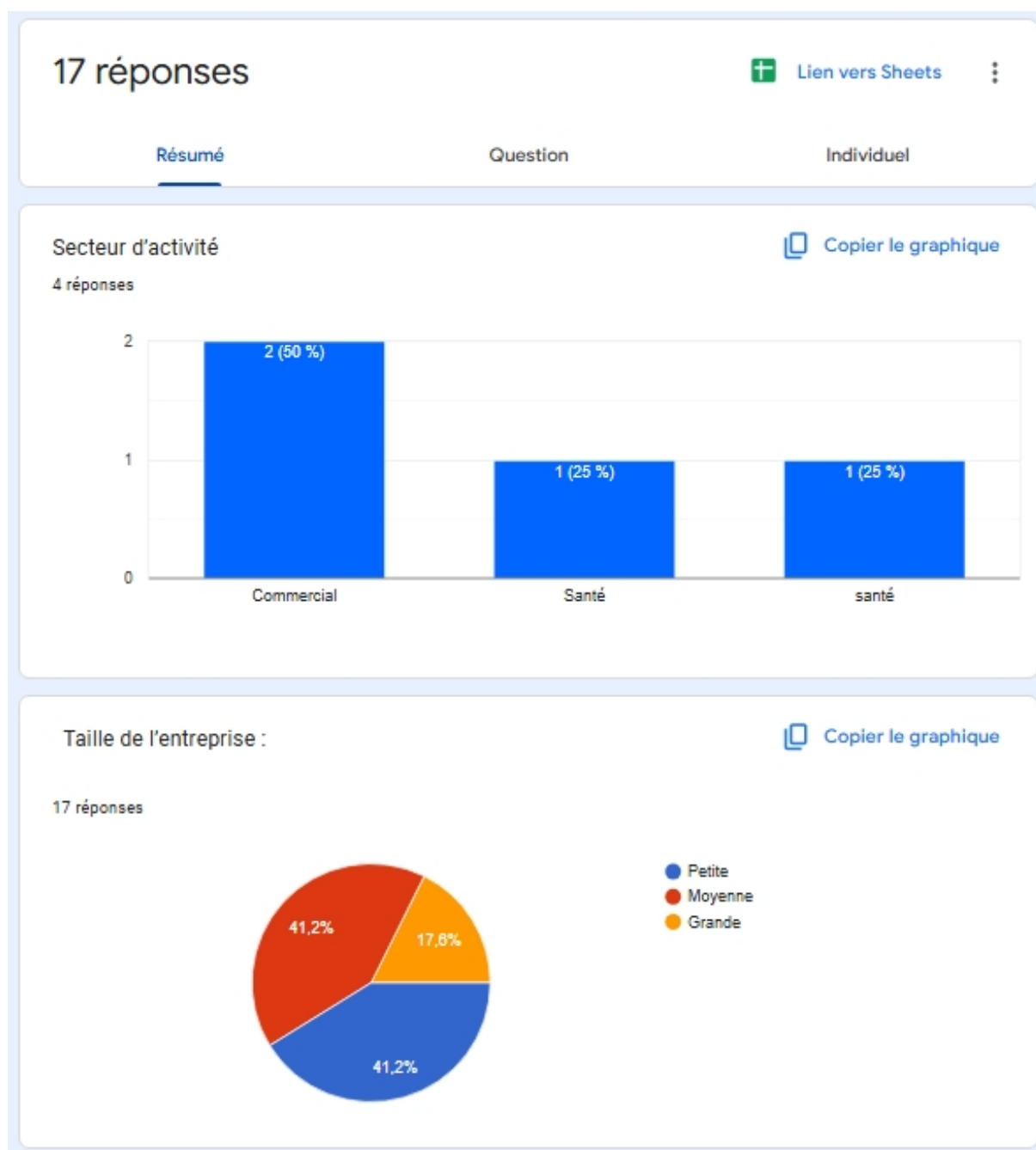
Qu'est-ce qui vous encouragerait à utiliser cette application ?

 Copier le graphique

54 réponses



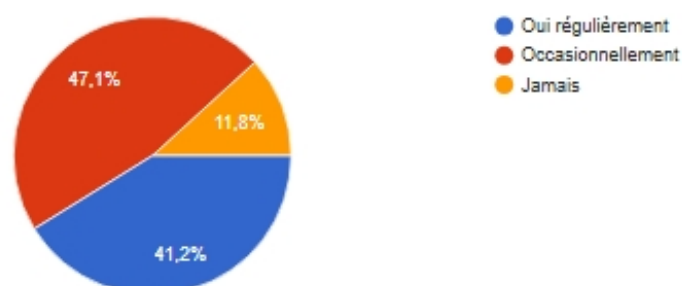
Annex 4 : Questionnaire for business Responses



Organisez-vous des événements professionnels ?

[Copier le graphique](#)

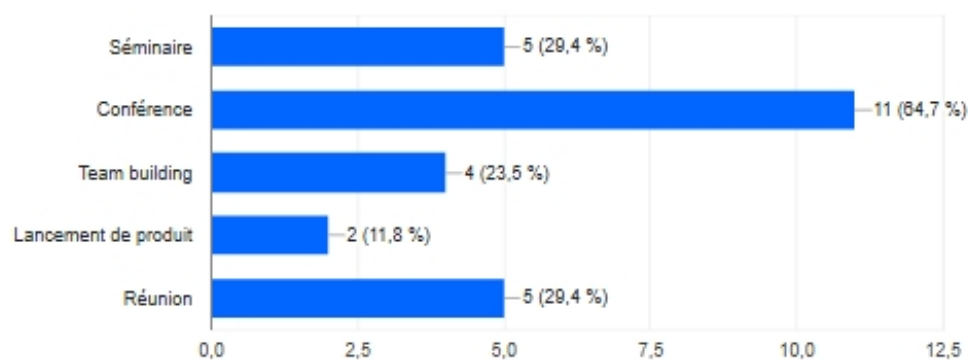
17 réponses



Types d'événements

[Copier le graphique](#)

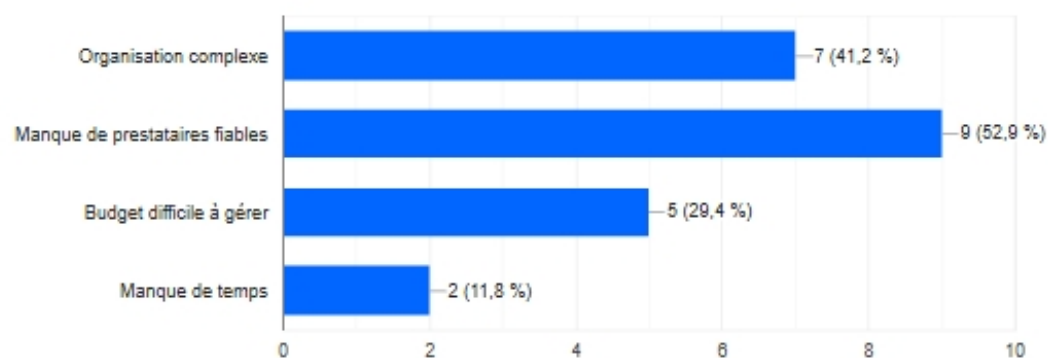
17 réponses



Quels sont vos principaux problèmes ?

[Copier le graphique](#)

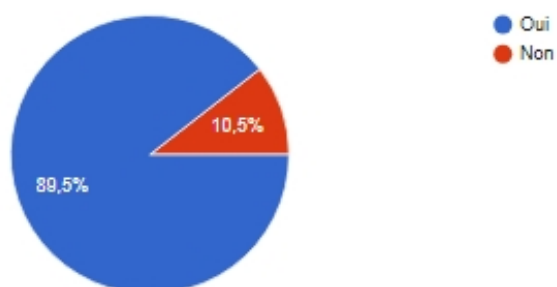
17 réponses



Utilisez-vous des cadeaux d'affaires ?

[Copier le graphique](#)

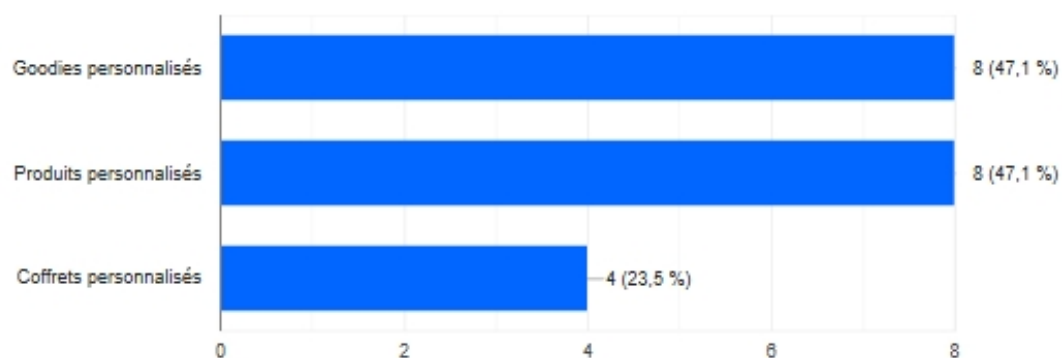
17 réponses



Si oui :

[Copier le graphique](#)

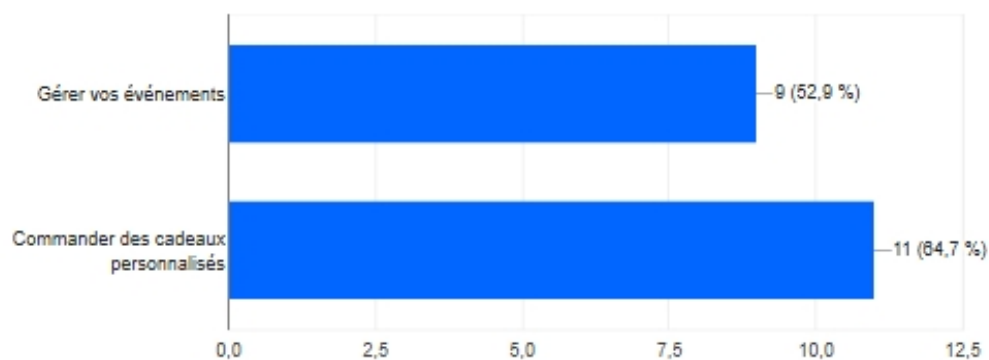
17 réponses



Une application vous serait utile pour :

[Copier le graphique](#)

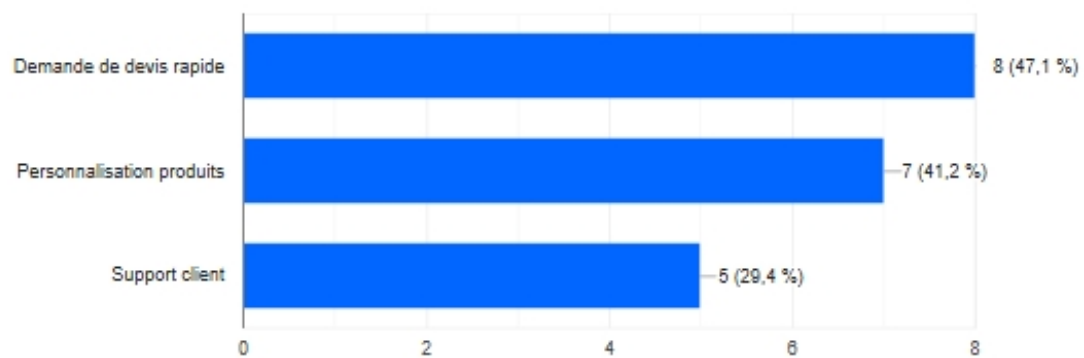
17 réponses



Fonctionnalités attendues

 Copier le graphique

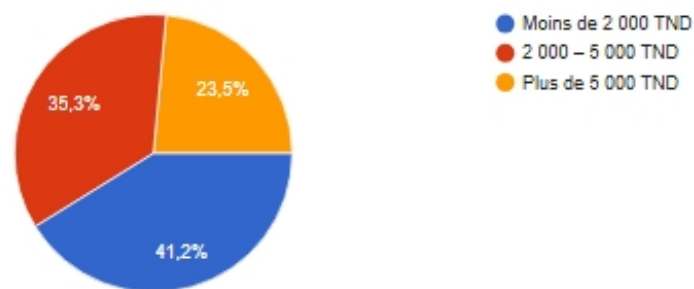
17 réponses



Quel budget consacrez-vous ?

 Copier le graphique

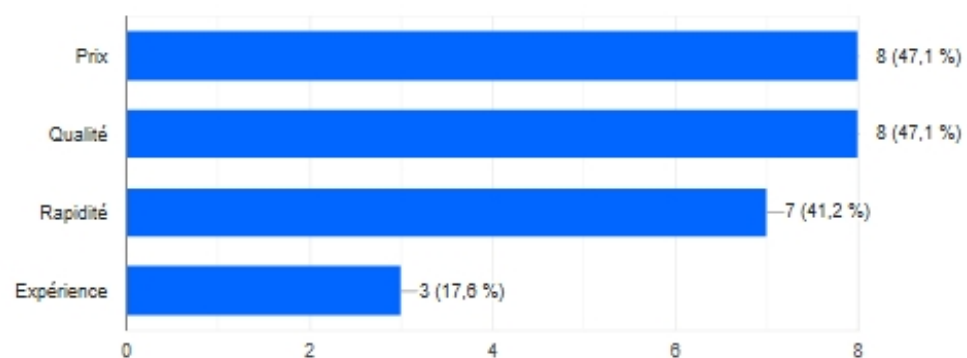
17 réponses



Qu'est-ce qui est le plus important ?

 Copier le graphique

17 réponses



Annexe 5 : La charte graphique de l'application



Sommaire

1. Idées directrices

2. LOGO

- description LOGO
- plateforme utilise pour crée le Logo

3. TYPOGRAPHIES

- Typographie principale
- Typographie secondaire

4. les couleur

5. le forme étoile

6. La version blanche

7. Propositions de logos non retenues

- logos non retenues avec font bleu
- logos non retenues avec font blanc
- logos non retenues plus complique
- logos non retenues plus coloré

8. Gestion des images



Idées directrices

L'application InWin incarne des valeurs de créativité, d'élégance et de personnalisation. Elle se positionne comme un partenaire stratégique qui accompagne les entreprises dans la conception d'expériences uniques, que ce soit à travers l'organisation d'événements ou la création de cadeaux d'affaires sur mesure.

Ces principes se reflètent dans son identité visuelle, pensée pour transmettre à la fois professionnalisme et innovation. Le design de l'application repose sur des éléments graphiques fluides et modernes, symbolisant l'accompagnement et la transformation d'une idée en une expérience concrète et mémorable.



description LOGO

Le logo InWin reflète une identité visuelle élégante, moderne et haut de gamme, en parfaite cohérence avec le positionnement de l'entreprise dans l'organisation d'événements et la création de cadeaux d'affaires personnalisés.



description LOGO

voici le logo sans background



J'ai utilisé le logiciel Photoshop pour supprimer l'arrière-plan du logo, afin d'obtenir une version propre et adaptable à différents supports.



plateforme utilise pour cree le Logo

Le logo de l'application INWIN a été conçu à l'aide de la plateforme Design.com, un outil en ligne qui permet de créer des logos modernes et personnalisés grâce à des modèles variés et une interface intuitive.



typographie principale

Il se distingue par une typographie raffinée, où le contraste entre les lettres dorées et blanches crée une harmonie visuelle équilibrée.



Le nom police utilisé est :

Iridescent BC

typographie secondaire

La lettre centrale, stylisée de manière fluide et originale, apporte une touche de créativité et symbolise la transformation d'une idée en une réalisation concrète, - en lien direct avec le slogan "From Idea to Realisation".

FROM IDEA TO REALISATION

Le nom police utilisé est :

Carrois Gothic SC

les couleur

Le bleu foncé représente la confiance, le professionnalisme et la stabilité. Il traduit le sérieux de l'entreprise et sa capacité à accompagner ses clients dans des projets importants avec rigueur et fiabilité.



les couleur

Le doré symbolise le prestige, la qualité et l'excellence. Il met en valeur le caractère haut de gamme des services proposés, notamment dans l'organisation d'événements et la création de cadeaux d'affaires personnalisés.



les couleur

Le blanc apporte une touche de simplicité, de clarté et d'équilibre. Il améliore la lisibilité du logo tout en renforçant l'aspect moderne et épuré de l'identité visuelle.





le forme

L'élément étoilé situé au-dessus du logo représente l'excellence, la réussite et la touche finale qui rend chaque projet unique. Il évoque également la capacité d'InWin à apporter une valeur ajoutée et un effet "wow" à chaque événement ou produit.



La version blanche

La version blanche du logo InWin met en avant une identité visuelle épurée et moderne, parfaitement adaptée aux supports visuels chargés ou aux arrière-plans photographiques.

L'utilisation du fond blanc apporte une sensation de clarté, de simplicité et de légèreté, permettant au logo de ressortir de manière nette et élégante. Cette version renforce la lisibilité tout en conservant le caractère professionnel de la marque.



à quoi pourrait ressembler le logo



à quoi pourrait ressembler le logo



à quoi pourrait ressembler le logo





logos avec font bleu



logos avec font blanc







Gestion des images



Evitez l'utilisation de logo sur un arrière-plan sombre ou chargé

Annex 6: Login Logic

```
Future<void> _login() async {
  if (!_formKey.currentState!.validate()) return;
  setState(() { _loading = true; _error = null; });
  try {
    final user = await ref.read(authServiceProvider).signIn(
      email: _emailCtrl.text.trim(),
      password: _passCtrl.text,
    );
    if (user != null && mounted) {
      context.go(user.role == UserRole.admin ? '/admin' : '/home');
    } else if (mounted) {
      setState(() => _error = "Compte non trouvé.");
    }
  } on Exception catch (e) {
    setState(() => _error = e.toString().replaceAll('Exception: ', ''));
  } finally {
    if (mounted) setState(() => _loading = false);
  }
}
```

Annex 7: Email Verification Logic

```
_timer = Timer.periodic(const Duration(seconds: 5), (timer) async {
  await ref.read(authServiceProvider).reloadUser();
  if (!context.mounted) return;
  if (ref.read(authServiceProvider).isEmailVerified()) {
    timer.cancel();
    _goHome();
  }
});
```

```
Future<void> _resendEmail() async {
  if (!_canResend) return;
  final messenger = ScaffoldMessenger.of(context);
  try {
    await ref.read(authServiceProvider).sendEmailVerification();
    if (!mounted) return;
    setState(() {
      _canResend = false;
      _resendTimer = 60;
    });
    _startResendTimer();
  }
}
```

Annex 8: Route Protection and Role-Based Redirection Logic

```
if (loc == '/splash') {
  final prefs = ref.read(sharedPreferencesProvider);
  final onboardingDone = prefs.getBool('onboarding_done') ?? false;

  if (!onboardingDone) return '/onboarding';
  if (!isAuth) return '/login';

  if (user.role == UserRole.customer) {
    final isEmailVerified =
      ref.read(authServiceProvider).isEmailVerified();
    if (!isEmailVerified) return '/verify-email';
  }

  return user.role == UserRole.admin ? '/admin' : '/home';
}
```

```

// 1. If not authenticated, force redirect to login unless already on a public route
if (!isAuth) {
    return publicRoutes.contains(loc) ? null : '/login';
}

// 2. If authenticated, check email verification (customers only)
if (user.role == UserRole.customer) {
    final isEmailVerified = ref.read(authServiceProvider).isEmailVerified();
    if (!isEmailVerified) {
        // If not verified, only allow /verify-email
        return loc == '/verify-email' ? null : '/verify-email';
    }
    // If verified, don't allow staying on /verify-email
    if (loc == '/verify-email') return '/home';
}

// 3. Prevent authenticated users from accessing public routes (except splash)
if (publicRoutes.contains(loc) &&
    loc != '/splash' &&
    loc != '/verify-email') {
    return user.role == UserRole.admin ? '/admin' : '/home';
}

// 4. B2B Restricted Routes
if (loc.startsWith('/gifts') && user.clientType.isB2C) {
    return '/home';
}

```

Annex 9: Firestore Security Rules

```

function isAdmin() {
    return isAuth() &&
        get(/databases/(database)/documents/users/${request.auth.uid}).data.role == 'admin';
}

// Users: can read/write own doc; admin can read all
match /users/{userId} {
    allow read: if isOwner(userId) || isAdmin();
    allow create: if isOwner(userId);
    allow update: if isOwner(userId) || isAdmin();
    allow delete: if false;
}

```

```

// Requests: customers own their requests; admin can read/write all
match /requests/{requestId} {
  allow read: if isAdmin() ||
    (isAuth() && resource.data.customerId == request.auth.uid);
  allow create: if isAuth() &&
    request.resource.data.customerId == request.auth.uid;
  // Customer can only update customerNote and status (accept/reject)
  allow update: if isAdmin() ||
    (isAuth() &&
      resource.data.customerId == request.auth.uid &&
      request.resource.data.diff(resource.data).affectedKeys()
        .hasOnly(['status', 'customerNote', 'updatedAt']));
  allow delete: if false;

  // Messages subcollection
  match /messages/{messageId} {
    allow read: if isAdmin() ||
      (isAuth() && get(/databases/${database}/documents/requests/${requestId}).data.customerId == request.auth.uid);
    allow create: if isAdmin() ||
      (isAuth() && get(/databases/${database}/documents/requests/${requestId}).data.customerId == request.auth.uid);
    allow update: if isAdmin();
    allow delete: if false;
  }
}

```